

Sparse Koopman Supports for Multibasin Forecasting and Transductive Regime Alignment

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Abstract

Koopman autoencoders (KAEs) seek a higher-dimensional latent representation in which nonlinear dynamics evolve linearly. However, systems with multiple basins need not admit an exact, state-reconstructing finite-dimensional Koopman embedding with one continuous global coordinate map, so a single-transition KAE may have to approximate across dynamically different regions. We posit that sparse encoders can expose latent supports as inspectable, model-produced regime variables without basin labels or other regime annotations during training. On 15 controlled multibasin systems, sparse-latent KAEs have lower reported aggregate forecast error than the dense-latent KAE controls, although decoder normalization differs across encoder families. In a fixed-data 128-dimensional Lorenz-96 comparison, a signed-support KAE recipe reduces NRMSE relative to an audited exact-dense tanh KAE by 6.30% and 24.87% at physical times 2.5 and 5, winning all ten paired model seeds. On a new-initial-condition dataset from a synthetic 512-dimensional four-well Allen-Cahn system, sign-split temporal-support families at $T = 20$ stratify the same-time $T = 20$ modal-well fate labels (ARI 0.767); thresholded dense support yields one family, which does not test information in continuous dense values. Conditional on a recipe whose T20 scope, Jaccard-0.40 rule, temporal weight, and whole-packet advancement were selected with validation fate metrics, the objective, representative fitting, and assignment use no labels. A single-well-dominated final-field slice prospectively frozen for this dataset after earlier mixed-domain fragmentation (130/256 trajectories) is exactly four-family aligned for all ten seeds; the all-trajectory final-state map misses its uniqueness ceiling by 0.009, and this representation-focused recipe loses to dense at long PDE horizons. Separately, a soft-thresholded global- K recipe (43.4% of evaluated latent activations at $|z| \leq 10^{-3}$) using direct rollout reduces mean field MSE over forecast states by 6.30% and 5.48% through physical times 16 and 20; both terminal intervals cross zero, and H200 terminal also misses the frozen effect-size and win-count criteria. On the same-checkpoint three-dataset forecasts, all seven subsequently frozen, outcome-aware secondary physical metrics favor sparse and five survive Holm correction, showing broad but correlated phase/interface/energy concordance within this condition rather than PDE-residual or dissipation-law validation. In a post-hoc transductive diagnostic on the 14 systems whose evaluation assignment realizes at least two labels, LISTA support families carry substantially more information about native or nearest-center-proxy labels than the dense-latent control’s thresholded support families. The remaining catalog proxy collapses to one observed label; it is excluded from primary aggregation but retained in raw and per-system evidence and in an all-15-system sensitivity. Finally, on the controlled benchmark, support families learned after the first training stage route a second-stage bank of local affine latent models and yield descriptively lower forecasting error than a same-architecture, same-budget global- K LISTA baseline; this comparison is optimistic because the staged checkpoint selector overlaps the reported evaluation starts and differs from the baseline selector. Together these results separately support two bounded conclusions: sparse recipes can improve high-dimensional forecasting, and separately trained sparse-support maps can stratify finite-time regimes. They do not establish a representation-to-forecast causal chain, a global exact linearization, or out-of-sample basin discovery.

1 Introduction

Learning representations for nonlinear dynamical systems remains a central problem in machine learning, scientific computing, and control. A particularly useful goal is to embed nonlinear dynamics into coordinates in which the evolution is linear, or at least locally linear, because linear models can be combined with mature tools for prediction, estimation, and control, including model predictive control and linear quadratic regulation (Mayne et al., 2000; Grüne and Pannek, 2017; Korda and Mezić, 2018; Kalman, 1960; Mamakoukas et al., 2019). This goal is also well motivated theoretically. Classical dynamical-systems results provide local topological linearizations near hyperbolic fixed points (Grobman, 1959; Hartman, 1960), while Koopman eigenfunction constructions can yield linearizing coordinates on basins of attraction for stable equilibria and periodic orbits (Lan and Mezić, 2013; Kvalheim and Revzen, 2021). Thus, nonlinear systems may admit linear dynamics with a specific change of coordinates.

The difficulty is that these results are primarily existential. They establish when useful coordinates can exist, but they do not by themselves provide a finite-data procedure for constructing such coordinates. Koopman operator theory formalizes the linearization idea by lifting nonlinear state evolution to a linear operator acting on observables (Koopman, 1931; Koopman and Neumann, 1932; Mezić, 2005; Brunton et al., 2022). This viewpoint has inspired data-driven approximations of Koopman dynamics, such as DMD and eDMD (Schmid, 2010; Williams et al., 2015, 2016), and more recently to Koopman autoencoders, which learn an encoder from state space to a latent representation, a linear latent transition matrix, and a decoder back to state space (Takeishi et al., 2017; Lusch et al., 2018; Otto and Rowley, 2019; Azencot et al., 2020; Shi and Meng, 2022; Fathi et al., 2024). These models provide an appealing route from existence results to practical learning: rather than hand-designing observables, they attempt to learn the embedding directly from trajectory data.

A central challenge for Koopman approximations is that many practically interesting nonlinear systems typically contain multiple equilibria or basins of attraction; for example, a damped mechanical system can settle into different wells, a biological or chemical model can approach different stable equilibria, and a controlled system can move between regions with distinct local dynamics. Theory places strong restrictions on exact finite-dimensional global linearizations with continuous, state-reconstructing encoder-decoder pairs in multi-attractor settings (Lan and Mezić, 2013; Brunton et al., 2016; Pan and Duraisamy, 2024; Liu et al., 2025), and empirical work documents practical difficulty for a single global model (Fathi et al., 2024). Local or basin-restricted linear descriptions may still be useful, but their simultaneous representation by one finite matrix is not guaranteed. This motivates testing whether a learned representation can expose local predictive structure without assuming that every basin shares one exact global linearization.

This paper proposes inducing sparse latent structure as a practical and interpretable method for Koopman learning of multi-attractor systems. By inducing the state to activate only a small subset of coordinates in lifted latent space, we can incentivize different regions of state to occupy different active coordinates, since nearby states should have similar latent embeddings, and thus enable local linearizations. In this sparse representation, nonzero coefficient values in a latent vector can provide continuous coordinates for prediction within a selected region, while the active indices provide a discrete support object that can be inspected across states and used to select local linearizations.

Sparse coding and LISTA-style encoders provide a plausible inductive bias for this purpose. They induce piecewise-linear, union-of-subspaces structure (Olshausen and Field, 1996; Chen et al., 1998; Donoho and Elad, 2003; Gregor and LeCun, 2010), which may help when useful predictive descriptions differ across state-space regions. We hypothesize that sparse-latent Koopman autoencoders will assign local dynamical regimes to distinct latent support families, yielding a learned regime variable

without basin supervision. Periodically decoding and re-encoding the model’s prediction can then refresh the inferred latent support as a rollout moves across state space (Fathi et al., 2024).

Contributions. This paper makes three empirical contributions. First, across controlled multi-basin systems, sparse-latent KAEs have lower reported aggregate rollout error than the dense-latent KAE controls; fixed-data Lorenz–96 and a separately forecast-optimized Allen–Cahn comparison extend the result to high-dimensional direct rollouts against fail-closed exact-dense tanh KAEs. Using the same 20 checkpoints without retraining or reselection, an outcome-aware H200 advantage persists on three new initial-condition datasets, while independent retraining replication and terminal superiority remain unsupported. Second, a post-hoc transductive diagnostic on 14 informative controlled systems shows that LISTA absolute-threshold support families align with evaluation-only native or nearest-center-proxy labels on high center-margin states, without using those labels to train the model or as features in the mask merge. On a new-initial-condition 512-dimensional multibasin PDE dataset, a different sign-split temporal-support extension transfers strong same-time alignment between $T = 20$ support families and $T = 20$ modal-well fates across all trajectory final states, and exactly one family per fate on a single-well-dominated final-field slice prospectively frozen after earlier mixed-domain fragmentation, while narrowly missing its stricter all-trajectory final-state uniqueness gate. Its objective, representative fitting, and assignment are label-free conditional on a recipe whose T20 scope, Jaccard-0.40 rule, temporal weight, and whole-packet advancement were selected with validation fate metrics. Third, on the controlled benchmark, support families formed after the first stage of LISTA training route a second-stage bank of local affine latent predictors and produce descriptively lower error than a same-architecture, same-budget global- K LISTA model under an asymmetric, partly evaluation-overlapping checkpoint-selection protocol. These are forecasting, alignment, and descriptive routing claims, respectively; they do not identify a representation-to-forecast causal chain. Unstructured LISTA, Sparse MLP, Dense MLP, support-family formation, assignment, and staged routing use no basin information during training or rollout; the structured-transition diagnostic rows are the exception, using a known count only to size their blocks.

2 Problem formulation

The introduction motivates a multibasin version of Koopman learning. Given sampled trajectories of a nonlinear system, the aim is to learn a state-reconstructing latent representation whose evolution is linear, without being told which basin or local dynamical regime generated each observation. We formulate the problem at the cadence of the stored benchmark observations. For each system, let $\mathcal{X} \subseteq \mathbb{R}^{d_x}$ be the observed state space and let

$$x_{k+1} = F_{\Delta t}(x_k), \quad x_k \in \mathcal{X}, \quad (1)$$

where $x_k \in \mathbb{R}^{d_x}$ is the state at the k th stored sample, $F_{\Delta t}$ denotes the transformation to the next discrete observation step based on underlying dynamics, and Δt is the system-specific observation interval. Forecasting is therefore defined by repeatedly applying the map $F_{\Delta t}$; a horizon of h means h applications of $F_{\Delta t}$; we denote k compositions as $F_{\Delta t}^{(k)}$. Because Δt may differ across systems, equal values of h may yield different physical time.

Basins of attraction. Many systems of interest are multistable: different initial conditions can approach different long-run behaviors. A fixed point is a state x^* satisfying $F_{\Delta t}(x^*) = x^*$. More generally, a trajectory may approach an attractor A , such as a stable equilibrium, limit cycle,

countable finite set, or chaotic invariant set. The basin of attraction of A is the set of initial conditions whose trajectories converge to A :

$$\mathcal{B}(A) = \left\{ x_0 \in \mathbb{R}^{d_x} : \inf_{a \in A} \|F_{\Delta t}^{(k)}(x_0) - a\|_2 \rightarrow 0 \text{ as } k \rightarrow \infty \right\}. \quad (2)$$

When multiple attractors coexist, their basins partition the state space up to basin boundaries. This defines the setup for this paper. The task is to learn a representation that can predict while preserving inspectable information about where the state lies in the multibasin geometry.

Koopman theory. The stored-step map $F_{\Delta t}$ induces a Koopman operator $\mathcal{K}$ on scalar observables $g : \mathcal{X} \rightarrow \mathbb{R}$ by composition, $(\mathcal{K}g)(x) = g(F_{\Delta t}(x))$. Although $F_{\Delta t}$ may be nonlinear, $\mathcal{K}$ is linear on the space of observables. A finite-dimensional Koopman approximation searches for a vector observable $\Phi : \mathcal{X} \rightarrow \mathbb{R}^{d_z}$ and a matrix K such that $\Phi(F_{\Delta t}(x)) \approx K\Phi(x)$, while retaining enough information to reconstruct the state. Additional background is provided in [section K](#).

Koopman autoencoders. We study this setting with Koopman autoencoders (KAEs). The model learns an encoder $E_\theta : \mathbb{R}^{d_x} \rightarrow \mathbb{R}^{d_z}$, a decoder $D_\phi : \mathbb{R}^{d_z} \rightarrow \mathbb{R}^{d_x}$, and a latent linear transition matrix $K \in \mathbb{R}^{d_z \times d_z}$. For observation x_k at arbitrary timestep k , the corresponding open-loop h -step forecast is

$$\hat{x}_{k+h} = D_\phi \left(K^h E_\theta(x_k) \right). \quad (3)$$

The same matrix K is applied at every step of the rollout, so any state-dependent structure required for prediction must be encoded in the learned representation rather than in a supplied regime label.

Training windows. Training uses windows of adjacent stored observations. For a batch of B windows of rollout length L , $\{x_{b,k}, x_{b,k+1}, \dots, x_{b,k+L}\}_{b=1}^B$, where b indexes windows and ℓ indexes offsets within a window, the model encodes each observed state, rolls out from the initial encoded state, and decodes both reconstructed observations and latent forecasts. The encoded/reconstructed quantities are defined for $\ell = 0, \dots, L$, while rollout/forecast quantities are defined for $\ell = 1, \dots, L$:

$$z_{b,k+\ell}^{\text{enc}} = E_\theta(x_{b,k+\ell}), \quad z_{b,k+\ell}^{\text{roll}} = K^\ell z_{b,k}^{\text{enc}}, \quad x_{b,k+\ell}^{\text{rec}} = D_\phi(z_{b,k+\ell}^{\text{enc}}), \quad \hat{x}_{b,k+\ell} = D_\phi(z_{b,k+\ell}^{\text{roll}}). \quad (4)$$

The training losses described later are built from these reconstructed and rolled-out quantities. Each application of K is interpreted as one stored benchmark step, matching the observation map in [equation \(1\)](#).

Label-free training and rollout. In the intended training and deployment setting, the sparse-support mechanism is not given basin labels, the number of basins, or trajectory-to-basin assignments. Apart from disclosed evaluation-only uses and recipe/architecture-selection exceptions, no such annotation enters the training objective, encoding, representative fitting, assignment, or runtime routing. Native labels and centers are generally used after training for evaluation; for the 13 catalog systems without them, the known benchmark count is used after training to estimate centers and construct nearest-center proxy labels. The Allen-Cahn temporal-support packet is a disclosed recipe-selection exception: validation fate metrics selected T20 scoring, family Jaccard 0.40, the temporal weight, and advancement of the entire fixed ten-seed packet; no seed was omitted. Conditional on those choices, the objective, representative fitting, and new-simulation-seed holdout assignment remain label-free. The block-diagonal and soft-block rows are a separate architecture-size exception: on the controlled benchmarks, their number of transition blocks is set equal to the benchmark basin count. Details are in [sections A](#) and [I](#).

3 Method

3.1 Sparse supports as local regime variables

A finite-dimensional Koopman representation for the multi-basin systems in [section 2](#) should encode two complementary quantities: (i) continuous coordinates that linearize the dynamics within a basin, and (ii) indicate which local linearization is active. We use sparsity to couple these two roles. If the state x is represented by a sparse latent code z , then the support of z can act as a regime variable, while the nonzero coefficient values provide coordinates for the corresponding local linear representation.

Given an overcomplete dictionary D , sparse coding estimates z by solving the Lasso objective ([Beck and Teboulle, 2009](#))

$$z^*(x) = \arg \min_{z \in \mathbb{R}^{dz}} \frac{1}{2} \|x - Dz\|_2^2 + \lambda_{sc} \|z\|_1, \quad \lambda_{sc} > 0. \quad (5)$$

The L_1 term is a convex surrogate that promotes sparse codes; it does not directly minimize the number of nonzero coefficients and need not recover the sparsest feasible code.

The classical solver to this problem (ISTA ([Beck and Teboulle, 2009](#))) can be unrolled into a depth- Q network of ISTA iterations, yielding learned ISTA (LISTA) ([Gregor and LeCun, 2010](#)); in our setting, LISTA is an encoder yielding the sparse code $z = E_\theta(x)$, where the *support* is the set of active z coordinates. With a linear decoder, the support determines which decoder columns participate in reconstructing x , and the associated nonzero coefficients specify coordinates within that selected reconstruction subspace. When D and z are learned jointly, as in sparse coding ([Olshausen and Field, 1996](#)), this induces a data-adaptive partition of the input state space where each region can be locally reconstructed with a code having the same support.

We therefore hypothesize that combining Koopman prediction with sparse coding may expose useful multibasin structure without explicit basin labels in training. The Koopman objective encourages continuous coefficients to support linear rollout, while the sparsity objective promotes selective activation; neither objective by itself guarantees that different regimes use different supports. If that separation emerges, a support pattern can serve as a discrete route key while its nonzero coefficients carry continuous local predictive coordinates. When native basin labels are available in benchmarks, we use them only after individual models are trained to assess support–basin alignment or construct diagnostic interventions; in the disclosed Allen–Cahn exception, validation fate metrics also select the T20 scoring scope, Jaccard-0.40 family rule, temporal weight, and advancement of the fixed ten-seed temporal-support packet. Otherwise the alignment evaluator constructs the nearest-center proxy labels described below.

3.2 Koopman autoencoder and sparse encoder

The predictor follows the KAE formulation in [section 2](#): an encoder, linear decoder, and latent transition, all of which are optimized jointly, produce the rollout in [equation \(3\)](#). The reported LISTA rows decode through a dictionary whose atoms are normalized. The reported sparse and dense MLP rows use ordinary learned linear decoders without that atom-normalization constraint. Decoder normalization is therefore not matched across encoder families and is a limitation of cross-family attribution. We experiment with K that is dense, block diagonal, or softly block-regularized.

Sparse encoding. Our main sparse encoder is LISTA (Gregor and LeCun, 2010). It first computes a dense pre-code $c_t = f_\theta(x_t)$, then applies learned shrinkage refinements for $q = 0, \dots, Q - 1$:

$$u_t^{(0)} = \text{shrink}(c_t, \tau), \quad u_t^{(q+1)} = \text{shrink}(Su_t^{(q)} + c_t, \tau), \quad z_t = u_t^{(Q)}. \quad (6)$$

Here, $\text{shrink}(a, \tau) = \text{sign}(a) \max(|a| - \tau, 0)$ is applied element-wise, S is learned, and τ controls the threshold scale for shrinkage. The thresholding step creates exact zeros, so the encoder explicitly selects active latent coordinates while learning the selection rule from the prediction objective.

Transition families. We vary the structure of K separately from the encoder. The dense transition leaves K unrestricted. The block-diagonal transition partitions latent coordinates into M fixed groups and constrains

$$K_{\text{block}} = \text{blkdiag}(K_1, \dots, K_M). \quad (7)$$

We also use a soft-block LISTA ablation that keeps K dense but penalizes cross-group entries,

$$\mathcal{L}_{\text{offblock}} = \sum_{i,j: g(i) \neq g(j)} |K_{ij}|, \quad (8)$$

where $g(i)$ is the fixed architectural group containing coordinate i .

3.3 Training objective

Training uses adjacent observation windows. For a batch of B windows with prediction horizon L , the model produces reconstructed future states $x_{b,k+\ell}^{\text{rec}}$, latent rollouts $z_{b,k+\ell}^{\text{roll}}$, encoded future latents $z_{b,k+\ell}^{\text{enc}}$, and decoded rollout predictions $\hat{x}_{b,k+\ell}$ as defined in equation (4). The rollout is seeded by the initial state, and the following sums are over offsets $\ell = 1, \dots, L$:

$$\bar{\mathcal{L}}_{\text{pred}} = \frac{1}{BL\sqrt{d_x}} \sum_{b,\ell} \|\hat{x}_{b,k+\ell} - x_{b,k+\ell}\|_2, \quad \bar{\mathcal{L}}_{\text{rec}} = \frac{1}{BL\sqrt{d_x}} \sum_{b,\ell} \|x_{b,k+\ell}^{\text{rec}} - x_{b,k+\ell}\|_2, \quad (9)$$

$$\bar{\mathcal{L}}_{\text{lin}} = \frac{1}{BL} \sum_{b,\ell} \|z_{b,k+\ell}^{\text{roll}} - z_{b,k+\ell}^{\text{enc}}\|_2, \quad \bar{\mathcal{L}}_{\text{sp}} = \frac{1}{BL} \sum_{b,\ell} \|z_{b,k+\ell}^{\text{roll}}\|_1. \quad (10)$$

The observation-space terms are normalized by $\sqrt{d_x}$ so that their scale is comparable across systems of different state dimension. The latent terms are left in their native scale because the latent dimension is fixed within each comparison.

The total objective is

$$\mathcal{L} = \lambda_{\text{pred}} \bar{\mathcal{L}}_{\text{pred}} + \lambda_{\text{rec}} \bar{\mathcal{L}}_{\text{rec}} + \lambda_{\text{lin}} \bar{\mathcal{L}}_{\text{lin}} + \lambda_{\text{sp}} \bar{\mathcal{L}}_{\text{sp}} + \mathcal{L}_{\text{struct}}. \quad (11)$$

Here $\mathcal{L}_{\text{struct}} = 0$ for the dense and block-diagonal transition variants. For the soft-block transition, $\mathcal{L}_{\text{struct}} = \lambda_{\text{off}} \mathcal{L}_{\text{offblock}}$, which probes whether a soft transition grouping is sufficient to induce useful support structure. The prediction and latent-linearity terms make the representation useful for Koopman rollout, the reconstruction term keeps the code tied to the observed state, and the sparsity term encourages selective activation.

3.4 Support objects

Let $z = E_\theta(x) \in \mathbb{R}^{d_z}$. A *support rule* converts z into a Boolean vector $m(x) \in \{0, 1\}^{d_z}$, where $m_i(x) = \mathbf{1}\{|z_i| > 0\}$. Equivalently, the exact support is $S(x) = \{i : m_i(x) = 1\}$. Exact supports defined by an arbitrary threshold can be unstable: a small perturbation in x -space can change the support, so one basin may be covered by several nearby or partially overlapping active-coordinate vectors. We therefore distinguish exact supports from *support families*, which group overlapping Boolean masks to reduce fragmentation. This construction does not by itself guarantee robustness or basin-scale meaning.

Both support-family experiments begin with $m_{\text{abs},i}(x) = \mathbf{1}\{|z_i| > 10^{-3}\}$ and $S_{\text{abs}}(x) = \{i : m_{\text{abs},i}(x) = 1\}$, but they use distinct fixed merging protocols. For the alignment diagnostic, denoted $F_{\text{abs}}^{\text{align}}$, we count distinct masks over all states in the evaluation trajectory collection and visit them from most to least frequent. Each family f stores one representative mask r_f ; a new mask m joins the family with the largest Jaccard score

$$J(m, r_f) = \frac{\sum_i \mathbf{1}\{m_i = 1 \text{ and } r_{f,i} = 1\}}{\sum_i \mathbf{1}\{m_i = 1 \text{ or } r_{f,i} = 1\}},$$

with $J(0, 0) = 1$, if that score is at least $\tau = 0.50$; otherwise it starts a new family. Evaluation-only labels and their native or endpoint-estimated centers then select a per-label high center-margin subset only for scoring $H(B \mid F_{\text{abs}}^{\text{align}})$; they do not select the masks used to form families or equalize label sample counts. The diagnostic is transductive because the partition is fit on the same evaluation trajectory collection containing the scored states, rather than on a disjoint fitting collection. The staged forecasting experiment instead fits $F_{\text{abs}}^{\text{route}}$ on training-distribution trajectories with $\tau = 0.40$. The two family partitions must not be treated as interchangeable. Pseudocode and implementation details are provided in [section D](#).

3.5 Periodic re-encoding

If a support indicates the currently relevant local dynamics, a long rollout should be able to update that support. We therefore evaluate periodic re-encoding, following [Fathi et al. \(2024\)](#). Starting from $z_k = E_\theta(x_k)$ and a re-encoding period m , the model advances for m Koopman steps, decodes the predicted state, and encodes that prediction again:

$$\tilde{z}_{k+m} = K^m z_k, \quad \tilde{x}_{k+m} = D_\phi(\tilde{z}_{k+m}), \quad z_{k+m} = E_\theta(\tilde{x}_{k+m}). \quad (12)$$

The same procedure is repeated over the rollout horizon. Each rollout uses only the model’s predicted state; it does not use the true future state, a basin label, or an oracle for basin transitions. However, the reported “best-periodic” forecasting value is selected separately for each system, model, seed, and horizon by minimizing MSE on the same reported evaluation trajectories over a fixed cadence grid. This is oracle-over-grid test selection, not a validation-frozen deployment rule, and it makes the reported values optimistic.

4 Experiments

The experiments evaluate whether sparse-latent KAEs reduce forecasting error, whether their post-hoc support families align with evaluation-only native or proxy basin labels, whether those effects survive high-dimensional chaotic and physics-based stress tests, and whether a separately fitted training-distribution support-family route has lower descriptive controlled-forecasting error. A one-checkpoint coordinate intervention is retained as a scoped descriptive diagnostic.

Benchmark systems. The main benchmark contains 15 heterogeneous, author-defined analytic or piecewise-analytic two-dimensional vector fields with multiple attractors. It includes potential-based, locally gated, oscillatory, and bifurcation-derived constructions; exact equations and retained metadata are in Appendix B. The two gated environments expose native labels and centers for evaluation. The other 13 catalog environments expose neither, so the evaluator advances trajectory endpoints, estimates the manifest-specified number of centers by deterministic k -means, and assigns every evaluation state to its nearest estimated center. Native and proxy labels and centers never train any row or form support families. The known count is used only to size structured-transition diagnostic blocks and, after training, to estimate catalog-system centers; it is not a runtime routing feature. A post-submission cadence audit found that the historical Dysts KAE cache generator retained each system’s native integration interval instead of realizing the requested $30dt$ interval, whereas the standalone controls used the intended cadence. We therefore quarantine all historical Dysts forecasts, cross-model comparisons, and significance tests; they are not evidence in this draft. A corrected matched rerun of all six KAE rows is in progress and no corrected outcome is reported here. The independent high-dimensional tests use fully observed 128-dimensional Lorenz–96 and a two-channel, periodic 16×16 four-well vector Allen–Cahn system with $d_x = 512$. Both KAEs use four-times overcomplete lifts in these experiments. For Allen–Cahn, validation fate metrics selected the Jaccard-0.40 evaluation rule and the weakest temporal-group objective, then advanced one fixed ten-seed packet; no model seed was omitted. Labels enter these hyperparameter and packet-level gates, but not training, representative fitting, or assignment conditional on the selected rules. Only after the whole packet passed was a new dataset generated, and support representatives remained fit to the original validation data. Vector-field expressions and more details are in Appendices B, C, and I.

Training models. We compare the six KAE models shown in the upper block of table 1 by varying the encoder and latent transition structure. We test three LISTA sparse encoders (where LISTA-BD denotes block diagonal latent transitions and LISTA-SB denotes soft-block regularization), two sparse-latent MLP controls applying the sparsity penalty $\bar{\mathcal{L}}_{\text{sp}}$ (10) with dense (Fathi et al., 2024) or block-diagonal (BD) transitions, and a dense-latent MLP control baseline with no sparsity incentive. Each model is trained for 15 random seeds on each retained controlled system. The corrected Dysts protocol matches LISTA, LISTA-BD, and LISTA-SB at one learned shrinkage refinement and retrains every comparator; no corrected Dysts result is reported yet. Exact details are in Appendix A.

Statistical details. Point estimates use `scipy.stats.trim_mean` with proportion 0.25 over finite seeds within each system, then arithmetic means across systems. This implementation sorts n values and removes $\lfloor n/4 \rfloor$ from each tail; a complete 15-seed cell therefore averages its central nine values. We call this the interquartile mean (IQM) (Agarwal et al., 2021). The table annotations use paired tests and Holm corrections with the inferential units specified in Appendix F.

4.1 Forecasting experiments

Sparse-latent KAEs reduce forecasting error. We evaluate trained sparse and dense-latent KAEs on the 15-system controlled benchmark. Forecasting is evaluated on held-out trajectories at the stored observation cadence. For a start i and step t , the raw error is the squared Euclidean state error, $e_{i,t} = \sum_{j=1}^{d_x} (\hat{x}_{i,t,j} - x_{i,t,j})^2$. For the KAE rows, reported MSE through horizon H averages this state-summed error over steps $t = 1, \dots, H$ within each start and then over starts. Nonfinite step errors are omitted; a KAE rollout with a finite prefix can therefore contribute even if it diverges

Table 1: Forecasting performance on the 15-system controlled multibasin benchmark; lower is better. Each KAE cell is an arithmetic mean across systems after a 25%-per-tail trimmed mean over seeds within each system; complete 15-seed cells average the central nine values. Superscripts mark within-system paired Wilcoxon tests against Dense MLP after Holm correction across systems (*, $p < 0.05$). Standalone controls use three configured seeds and independently generated evaluation trajectories, so they are unmatched descriptive references rather than formal comparisons. Classical DMD/EDMD require each contributing start to have a finite mean through H , whereas KAE and mixture-local rows permit finite-prefix contribution. **Bold** marks the best KAE entry. Historical Dysts values are excluded after a timestep-provenance failure.

Model	H100	H500	H1000
LISTA	0.0413*	0.154*	0.176*
LISTA-BD	0.0414*	0.126*	0.147*
LISTA-SB	0.0387*	0.125*	0.150*
Sparse MLP, BD	0.0477*	0.150*	0.178*
Sparse MLP (Fathi et al., 2024)	0.0400*	0.0955*	0.109*
Dense MLP (Lusch et al., 2018) [<i>baseline</i>]	0.832	2.68	2.91
DMD	2.58	2.14	1.70
Polynomial EDMD	0.864	0.968	0.887
RBF-dictionary EDMD	0.660	0.808	0.751
k -means local linear	1.40	3.04	138
GMM local linear	1.65	3.57	3.93
Soft-gated GMM local linear	1.66	3.71	4.07

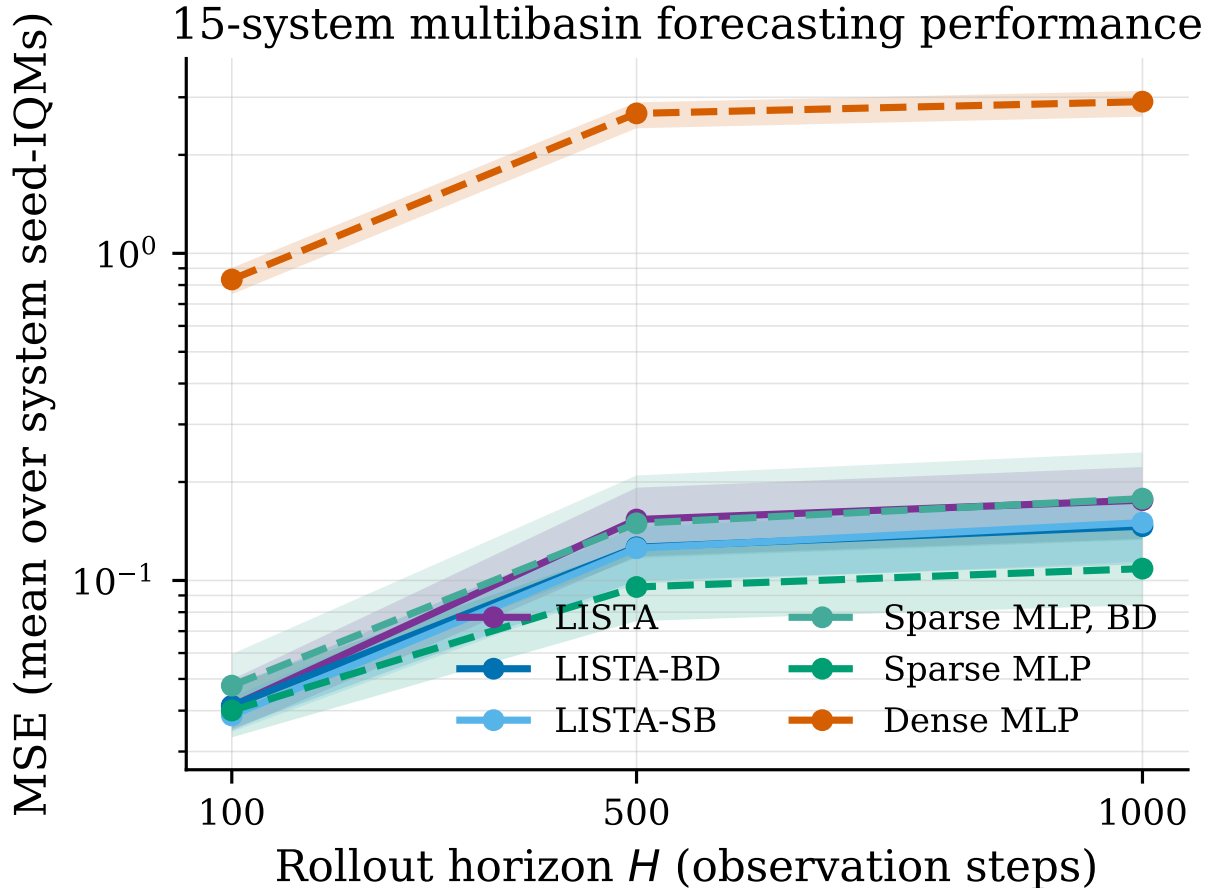


Figure 1: Forecasting horizon trends on the 15-system controlled benchmark. Lines show arithmetic means across all retained systems after summarizing finite seeds within each system by IQM; no system is trimmed. Bands show fixed-system seed-bootstrap 95% intervals after system-wise log-relative normalization, anchored to the displayed all-system mean. They summarize finite-seed uncertainty for an equally weighted average system, not cross-system heterogeneity.

before H . The unmatched standalone reducers are qualified in Appendix A. The results in table 1 and figure 1 show lower aggregate MSE for every sparse KAE row than for Dense MLP at all displayed controlled-benchmark horizons. Every sparse row carries corrected significance annotations at all three horizons. At H1000, Dense MLP has roughly 17–27 times the MSE of the sparse rows. The unmatched standalone controls show that conventional state-space and local-linear fits do not reproduce the best sparse-KAE errors in this packet, but their three seeds and independently generated trajectories preclude a matched causal attribution.

4.2 High-dimensional stress tests

Sparse forecasting survives a genuinely dense high-dimensional control. On fixed $d_x = 128$ Lorenz-96 data with $d_z = 512$, the signed-support KAE recipe reduces NRMSE relative to the exact-dense tanh KAE by 6.30% (95% interval [5.09%, 7.32%]) at physical time 2.5 and 24.87% ([22.98%, 26.51%]) at time 5. It wins all ten paired seeds at both horizons; exact two-sided sign-flip

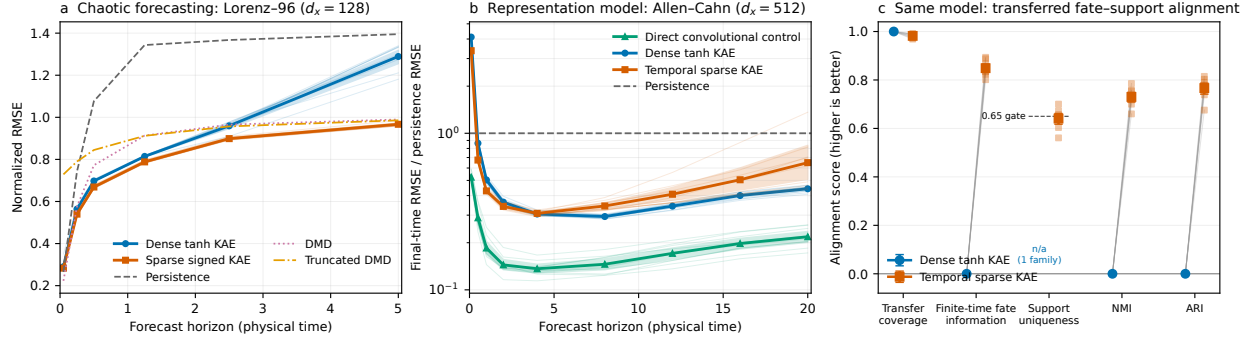


Figure 2: Separate high-dimensional evidence; faint lines or paired traces are individual model seeds and opaque marks show means with 95% model-seed bootstrap intervals. **(a)** Fixed-data Lorenz-96 forecasting through physical time 5. **(b)** Final-time Allen-Cahn RMSE relative to persistence through physical time 20, on a logarithmic ordinate. **(c)** Final-state support transfer to a new-simulation-seed Allen-Cahn holdout. Panels (b,c) use the representation-focused temporal-support model, not the separate forecast-optimized model in figure 3. Finite-time fate information is $1 - H(B | F)/H(B)$, and support uniqueness is $1 - H(F | B)/H(F)$; the latter is 0.641, narrowly below its internally frozen 0.65 gate. The single-well slice was prospectively frozen for this new holdout after earlier mixed-domain fragmentation. Thresholded dense support yields one family and therefore zero support-family NMI and ARI; this does not test information in dense latent values.

tests followed by Holm correction give $p = 0.003906$ for each. Both rows use direct repeated- K rollout without re-encoding. Dense hidden layers use tanh, the latent output is linear, and sparsity, weight decay, transition regularization, and every zero-inducing module are absent; all ten audits pass and active density is 0.9996, versus 0.4884 for signed LISTA. This is a complete-recipe comparison: sign splitting mechanically permits at most one active coefficient per pair, and LISTA versus MLP parameterization and decoder normalization differ. The time-5 endpoint is five times the 20-step training window, and sparse also has lower NRMSE than persistence, ordinary DMD, and rank-32 DMD at both declared long horizons.

PDE fate transfer in single-well-dominated fields. On the new $d_x = 512, d_z = 2,048$ four-well Allen-Cahn test, the validation-fitted sign-split temporal-support family map has transfer coverage 0.982, ARI 0.767 ([0.742, 0.787]), NMI 0.730, and purity 0.923; thresholded dense support transfers one family and has ARI zero, which does not test information in dense latent values. Sparse wins all ten paired support-ARI comparisons ($p = 0.001953$). Across all 256 trajectory final states, its normalized $H(F | B)/H(F) = 0.359$, or support-uniqueness score 0.641, misses the frozen 0.65 gate by 0.009 and its mean 10.8 families exceed the four finite-time fates. Both support families and modal-well fate labels are evaluated from the $T = 20$ final field; this is neither fate prediction from x_0 nor trajectory-wide alignment. On the evaluation-only slice prospectively frozen for this new holdout after earlier mixed-domain fragmentation, whose final fields occupy their modal well over at least 90% of the grid, all ten model seeds yield exactly four families and perfect ARI, NMI, purity, and both conditional-entropy scores across 130/256 trajectories. Thus this single-well-dominated final-field slice has exactly one transferred family per finite-time modal-well fate; it does not prove geometric basin-interior membership. The remaining fragmentation is associated with spatially mixed, interface-rich final fields. This model has near-complete pre-split group activity and is not evidence that coefficient sparsity causes alignment: it improves over dense through physical time 4, but loses from time 8 onward. At time 20, direct, dense, and sign-support final RMSE relative to

persistence are 0.218, 0.442, and 0.649.

A separate soft-thresholded sparse global model improves through-horizon PDE forecasting. With direct repeated- K rollout and no periodic re-encoding, the forecast-optimized soft-thresholded sparse recipe has mean field MSE 0.04220 versus 0.04503 for audited dense tanh through physical time 16, and 0.04472 versus 0.04731 through time 20. These are ratio-of-arm-means reductions of 6.30% and 5.48%, each with 8/10 paired-seed wins and marginal paired-seed bootstrap intervals [2.60%, 9.54%] and [1.75%, 8.76%]. Their MSE ratios to persistence are 0.0958/0.1023 at time 16 and 0.0938/0.0993 at time 20 for sparse/dense. Terminal reductions are smaller—2.38% and 3.22%—with intervals crossing zero. Both arms have the same trainable tensor shapes/count and shared convolutional, decoder, and K backbone, 12,698,690 trainable parameter entries, and 8,504,386 effective forward-path parameters; the joint treatment is soft-thresholding at 0.15 plus an L_1 weight of 0.01. The internally frozen rule required all four through-horizon and terminal cells to pass; it failed, and the conditional holdout was not generated or opened. We subsequently froze the H200 through-horizon endpoint and crossed the same 20 checkpoints, without retraining or reselection, with three prospectively generated 256-trajectory datasets. Sparse/dense MSE is 0.04616/0.04903, a 5.86% full-trajectory mean reduction with 8/10 paired-seed wins, a paired-seed bootstrap interval [2.40%, 9.21%], exact one-sided sign-flip $p = 0.00879$, and positive effects of 4.68%/5.88%/7.03% on the three datasets. A frozen secondary translation on these exact forecasts finds favorable H200 cumulative direction on all seven phase/interface/energy metrics; pixel phase, well-area, interface, full-energy, and gradient-energy errors fall by 4.61%/6.07%/2.25%/6.20%/8.58% and survive seven-way Holm correction, while modal accuracy and potential-energy error are positive but nonsignificant. This outcome-aware same-checkpoint evidence establishes new-initial-condition robustness and broad physical concordance in the same condition, not independent retraining replication, a repair of the original four-cell gate, terminal superiority, or cross-condition physics generalization. The model is soft-thresholded with mean evaluated latent near-zero fraction 0.434 at $|z| \leq 10^{-3}$, versus 0.0025 for dense, but it is trained separately from the temporal-support model, so no alignment-to-forecast mediation or single-component attribution is claimed.

The same checkpoints do not validate support-selected Allen–Cahn local laws. An internally metric-blind audit used score-initial supports only, locked every projector before future encoding, and rolled out the same global K without re-encoding. Provenance, information-firewall, and exact initial-capture checks pass, but the sparse masks remain broad (mean active fraction 0.696, or 1,425/2,048 coordinates) and every one of ten seeds yields a single training-fitted support family. Thus zero seeds meet the predeclared two-family gate, making family-specific signature and routing estimands undefined. Exact-support raw- K leakage is 3.29 and 3.25 times the coordinate-permutation null at H160/H200, the global- K -over-identity residual is $1.052 > 1$, and every matched-dense specificity family fails; the predeclared x_0 -support operationalization therefore fails on these checkpoints. Repeated fixed-support sparse forecasts nevertheless have 5.81% and 6.27% lower mean field MSE than dense full forecasts (8/10 wins; intervals [1.58%, 9.55%] and [2.20%, 9.87%]). This is a same-checkpoint, same-score-set decomposition of the preceding forecast comparison, not an independent validation: the result supports the bounded sparse-versus-dense forecast statement but not support-mediated prediction, approximate invariant coordinate subspaces, or distinct local laws. The conditional new-dataset mechanism bridge was ineligible and was not launched.

Support projections reduce one-step leakage without establishing invariance. In a separate held-out one-step diagnostic across all 15 controlled systems and three seeds, a no-next-

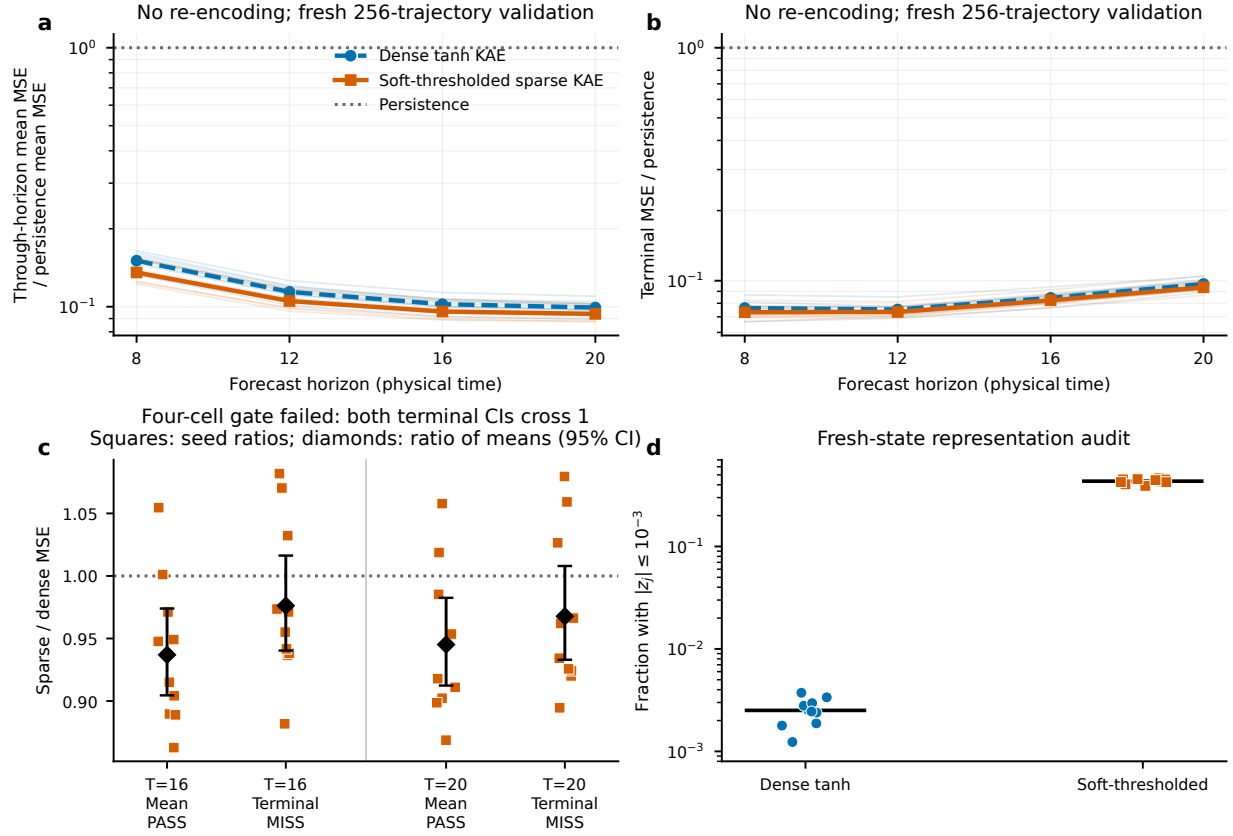


Figure 3: Separate forecast-optimized Allen-Cahn comparison on 256 fresh validation trajectories. Both KAEs use direct rollout without re-encoding. Panels report mean and terminal field MSE relative to persistence, all four frozen H160/H200 mean and terminal cells, and fresh-state near-zero fractions. In panel (c), orange squares are individual paired-seed sparse/dense ratios; black diamonds are ratios of arm means with marginal paired-seed bootstrap intervals, so a diamond need not center the seed ratios. The through-horizon sparse effects are positive, while both terminal marginal intervals include one. H200 terminal has a 3.22% reduction and 7/10 wins, below the frozen 5% and 8/10 criteria, so the four-cell gate fails.

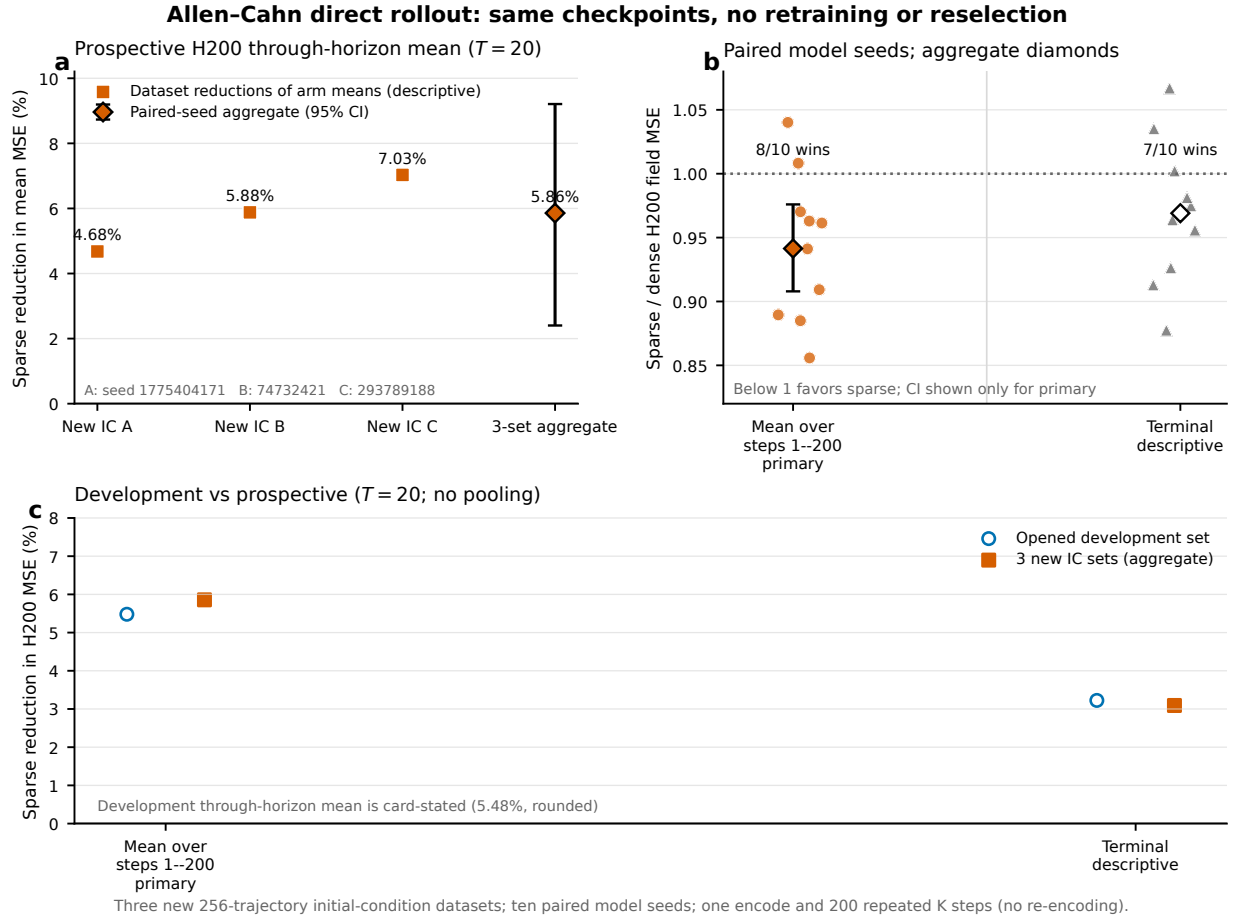
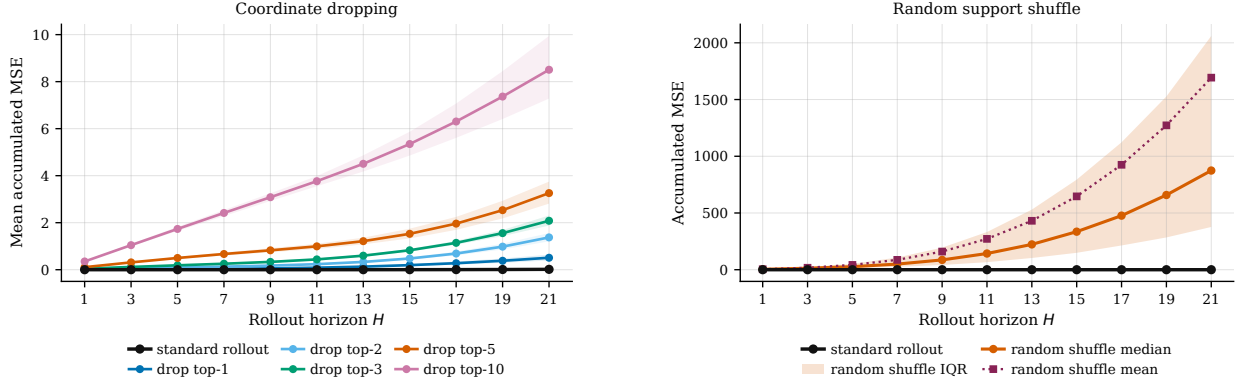


Figure 4: Same-checkpoint Allen-Cahn new-initial-condition robustness check at H200 (physical time 20). The endpoint was frozen after the original four-cell gate failed. The three prospectively generated dataset effects are descriptive; inference averages datasets within paired model seed. Opened development and new-IC results are displayed separately and never pooled. Checkpoints were neither retrained nor reselected.



(a) Dropping the largest active coordinates.

(b) Moving active values to inactive coordinates.

Figure 5: Descriptive support-coordinate interventions for one Local-linear gates LISTA checkpoint (seed 0). **(a)** The standard rollout is compared with cumulative dropping of the largest active coordinates of the initial sparse code; bands are bootstrap 95% intervals over 100 starts. **(b)** Active values are reassigned to random inactive coordinates; the band is the interquartile range of the pooled 2,000 dependent point-shuffle outcomes (100 starts times 20 shuffles) at each horizon. No confirmatory test is attached.

state guard emitted by the evaluator before execution has support-projected leakage 0.0047, versus 0.2065 under a sign-pair-preserving coordinate-permutation null. Leakage normalized by actual $K - I$ change is 0.338 versus 0.783. Whole-matrix Frobenius leakage is 0.0538 versus 0.2080, or 0.554 versus 0.783 after the same change normalization; every system favors the observed masks on all four diagnostics. Their ratios are numerically close to the internally frozen future-persistent primary. The guard adds only 0.96 percentage points because persistence is high, so this shows robustness on a highly persistent sample; it neither removes the future-conditioning concern broadly nor tests boundary-heavy switching. Observed support-projected one-step leakage is lower than under matched permutations, but does not establish invariant coordinate subspaces or distinct local laws: operator differentiation is null-like (ratio 0.993) and fails. The test remains two-dimensional and latent rather than decoded or multistep. A triggered zero-weight-decay dense tanh control uses the same physical states and per-state sparse cardinalities: activity observed/null is 0.022 sparse versus 0.567 dense (ratio 0.039), while restricted-residual observed/null is 0.053 versus 0.486 (ratio 0.108); all 14 common eligible systems favor sparse. This supports complete-recipe specificity relative to matched dense top- k coordinates, not a sparsity-only causal effect; see [section J](#).

Representative support-coordinate intervention. As a scoped mechanism diagnostic, we perturb the initial active set of one Local-linear gates LISTA checkpoint at seed 0 and evaluate 21-step forecasts. The evaluator first restricts candidates to the tie-inclusive high center-margin slice whose native label remains equal to its initial label throughout the 21-step true future. It balances the retained 100 starts across the three native labels (34/33/33); coordinate dropping and random support use exactly the same starts. The interventions are:

1. **Coordinate dropping:** Force the coordinates of z with the $k \in \{1, 2, 3, 5, 10\}$ largest magnitude coefficients to zero before forecasting.
2. **Random support:** the active coefficient values are preserved, but reassigned to randomly chosen inactive latent coordinates.

At $H = 21$, the standard rollout has mean accumulated MSE 0.0158. Dropping $k = 1, 2, 3, 5$, and 10 leading coordinates raises this to 0.508, 1.37, 2.08, 3.26, and 8.51, respectively; random support shuffling is also destructive in [figure 5](#) and [section E](#). This is descriptive evidence that this checkpoint uses coordinate identity. Because it covers one system, model, and seed and has no confirmatory test, it does not establish a general causal effect across architectures or systems.

4.3 Support-basin alignment experiments

This post-hoc experiment examines whether sparse support families fit after training are associated with evaluation-only native or proxy basin labels. For the two gated environments, the evaluator uses native labels and centers. For the 13 catalog environments, it rolls the 128 trajectory endpoints forward 5,000 steps, clusters the converged endpoints into the known benchmark count by deterministic k -means, and labels every state by its nearest estimated center. Fourteen systems realize at least two observed labels; the Triple-well Duffing proxy assigns all 16,512 states to one estimated center. Native and proxy labels and centers do not enter model training or support-family formation. The known count enters structured-transition block sizing as disclosed above, but its use here is post-hoc center estimation rather than a feature in the learned support or route.

For a state x , let $d_1(x)$ and $d_2(x)$ be its distances to the nearest and second-nearest native or estimated center. We call $d_2(x) - d_1(x)$ the *center-margin proxy*. A large value indicates separation under this nearest-center construction; it need not equal distance to a true basin boundary or separatrix, especially when basin geometry is not Voronoi. The diagnostic first fits $F_{\text{abs}}^{\text{align}}$ using the absolute threshold 10^{-3} and Jaccard threshold 0.50 over all states in each evaluation trajectory collection. It then retains states at or above the within-label 75th percentile of center margin, including all ties, and scores the already fitted partition against label B . This threshold does not resample labels to equal sizes and does not guarantee that exactly 25% of each label is retained. Conditional entropy $H(B | F_{\text{abs}}^{\text{align}})$, computed with natural logarithms and reported in nats, is the main metric; the number of family IDs observed on the high-margin slice is a descriptive companion. Most system–seed slices contain 4,129–7,167 states. Triple-well Duffing retains all 16,512 because its center margins tie, but its one-label proxy makes $H(B | F) = 0$ for every model and therefore cannot measure alignment. Primary point estimates and tests use the 14 systems with at least two observed labels; all raw Duffing rows remain in the evidence packet and an all-15-system sensitivity sidecar. Because the scored states belong to the same evaluation trajectory collection used both to estimate catalog-system centers and to fit the partition, this measures transductive alignment to native or proxy labels, not generalization of a frozen family codebook.

LISTA families align with evaluation labels on high center-margin states. In [table 2](#), LISTA has the lowest primary transductive conditional entropy, 0.138 nats, compared with 1.37

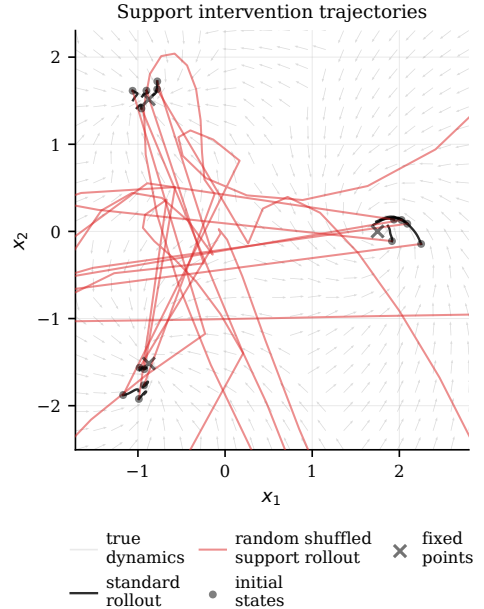


Figure 6: Randomly shuffled support trajectories (red) and standard rollouts (black) for the same representative checkpoint.

Table 2: Transductive basin-support diagnostics on the 14 controlled systems whose frozen evaluation assignment realizes at least two labels. F_{abs} is fitted over all states in each evaluation trajectory collection with absolute threshold 10^{-3} and Jaccard threshold 0.50, then scored at or above the within-label 75th center-margin percentile, retaining ties. Labels and centers are native for two gated systems and nearest-center proxies from deterministic endpoint clustering at the known benchmark count for the catalog systems. They define and score the subset but do not enter family formation or equalize label counts. $H(B | F_{\text{abs}})$ uses natural logarithms and is reported in nats. $|F_{\text{abs}}^{\text{obs}}|$ counts family IDs observed on the scored slice and uses ordinary seed means. Triple-well Duffing’s proxy realizes one label, so its identically zero entropy is excluded from primary aggregates; its raw rows and all-15-system sensitivity are retained. Superscripts mark Wilcoxon/Holm tests against Dense MLP (*, $p < 0.05$); all displayed stars pass on all 14/14 eligible systems. **Bold** marks the best directional entry.

Model	Tie-inclusive high-margin slice, 14 systems	
	$H(B F_{\text{abs}})$ [nats] ↓	$ F_{\text{abs}}^{\text{obs}} $
LISTA	0.138*	3.6
LISTA-BD	0.173*	3.5
LISTA-SB	0.168*	3.5
Sparse MLP, BD	0.386*	3.0
Sparse MLP	0.244*	3.3
Dense MLP	1.37 [baseline]	1.0 [baseline]

nats for Dense MLP. The other sparse rows lie between these values. The scored high-margin slice contains 3.6 LISTA family IDs on average, compared with a mean/median of 4.29/4 represented labels across the 14 systems, whereas Dense MLP exposes one family ID on average. These are ordinary seed means of observed slice counts, not total partition sizes. All-15-system sensitivity values are 0.129 and 1.28 nats for LISTA and Dense MLP, respectively; including the uninformative zero therefore reduces both estimates. A historical 1,345-run sweep is locally stable around Jaccard 0.4–0.6, but 0.2 merges families and 0.9 fragments to about 33.8 families while driving entropy near zero, so family count must accompany entropy. Alignment tests pass after correction on all 14/14 eligible systems. This is evidence of post-hoc alignment to this evaluation label construction; it does not show that the center margin measures true boundary distance, that a frozen family map discovers new basins out of sample, or that every family corresponds one-to-one with a basin.

4.4 Support-family local linearizations

We next test a separate, controlled routing protocol. The staged model and global- K baseline use the same LISTA architecture and total 200,000-step budget. Stage one trains the encoder, decoder, and shared K for 100,000 steps. We then freeze all three and fit $F_{\text{abs}}^{\text{route}}$ on a nominal 512-row packet from the training distribution. The source implementation generated one batch of 256 unique trajectories and repeated it exactly, so this packet contains 256, not 512, unique trajectories. Each trajectory has 192 transitions and 193 states. The greedy 10^{-3} -threshold, Jaccard-0.40 family merge sees all 193 state supports, including terminal states; map counts, source centers, and direct mask keys then use the first 192 source states. This gives 98,816 nominal supports for clustering, 98,304 nominal source transitions, and 49,152 unique source transitions. Families with at least one source transition receive affine maps trained for another 100,000 steps.

Table 3: Staged $F_{\text{abs}}^{\text{route}}$ -routed local affine forecasting on the controlled benchmark. The same-architecture, same-budget global- K LISTA baseline has $d_z = 256$, but checkpoint selection is asymmetric: the staged selector overlaps the first 32/100 reported starts, whereas the global selector uses distinct starts. Both rows also select cadence on the reported starts over the same fixed grid. Pair counts are descriptive nested system–seed outcomes; the last column counts pairs won by the routed model at H100, H500, and H1000 simultaneously. Lower MSEs and ratios are better.

Metric	H100	H500	H1000	All three horizons
Paired local wins	199/225	192/225	189/225	180/225
Mean MSE, $F_{\text{abs}}^{\text{route}}$ -local/global	0.0128/0.0229	0.0347/0.0623	0.0403/0.0706	–
Median local/global MSE ratio	0.355	0.312	0.328	–
Geometric local/global MSE ratio	0.252	0.180	0.174	–

For support family f , let $\bar{z}_f$ be the mean source latent assigned to that family. The route-local update is

$$T_f(z) = d_f + K_f(z - \bar{z}_f), \quad (13)$$

where K_f is initialized from the frozen global matrix and the learned intercept d_f is initialized to $K\bar{z}_f$. The runtime representative of each merged family is its modal source-state support mask. Before every latent transition, the model recomputes the current latent’s absolute support and routes that step against the frozen representatives; an unassigned route falls back to the frozen global K . Scheduled periodic events instead decode and re-encode the predicted state, refreshing the latent before the next step’s route. Thus routing occurs every latent step even when re-encoding is less frequent. Neither construction nor rollout uses basin labels, counts, attractor identities, or trajectory assignments. For each horizon, both staged and global rows are reported at their evaluation-set oracle cadence from $m \in \{1, 2, 5, 10, 20, 25, 50, 100\}$.

Checkpoint selection is not matched. The staged selector scores exactly 200 candidates—steps 100,500, 101,000, $\dots$, 199,500, plus 199,999—on 32 fixed starts at seed offset 12,345. For each candidate, it minimizes the finite-prefix, state-summed squared error independently across the cadence grid at H100, H500, and H1000, averages those three minima, and replaces the incumbent only on a strict improvement. Final reporting uses 100 starts at the same seed offset, so the selector starts are exactly the first 32 reported starts (32% overlap). By contrast, the global baseline uses its standard 16-start, seed-offset-999,999, every-step-re-encoded H200 final-error selector, which has no overlap with the reported starts. Thus architecture and optimization budget are shared, but the staged comparison is descriptive and optimistic rather than a clean held-out checkpoint comparison.

Across the 15 controlled systems and 15 seeds, the routed model has lower MSE on 199/225, 192/225, and 189/225 system–seed pairs at H100, H500, and H1000; median local/global ratios are 0.355, 0.312, and 0.328. Matched H1000 random-count, latent-k-means, and oracle-basin routing packets also improve on global K , with median ratios 0.451/0.341/0.366 and 185/196/195 wins, respectively; local maps help this staged artifact, but support identity is not uniquely implicated. These 225 pairs are nested within 15 systems and are reported descriptively, without treating them as independent units or attaching confirmatory inference. Selector overlap and asymmetry preclude a clean causal or held-out routing comparison.

5 Discussion

The evidence supports three bounded claims. Sparse-latent KAE recipes have lower reported aggregate error than the Dense MLP rows on the controlled benchmark; direct-rollout Lorenz–96

and forecast-optimized Allen–Cahn comparisons extend this to high dimension. The historical Dysts comparison is not evidence because its KAE cache cadence was mislabeled, and its corrected rerun remains pending. The Allen H200 through-horizon advantage persists with the same checkpoints across three new initial-condition datasets, and a subsequently frozen outcome-aware audit of seven correlated state-derived metrics shows broad phase/interface/energy concordance, while independent retraining replication, terminal superiority, PDE-law validation, and cross-condition generalization remain unsupported. On high center-margin controlled-benchmark states, LISTA families have low transductive conditional entropy with evaluation-only native or nearest-center-proxy labels. A different Allen–Cahn temporal-support model transfers a validation-fitted map to unseen PDE trajectories: across all trajectory final states, $T = 20$ supports are informative about the $T = 20$ modal-well fate, and the single-well-dominated final-field slice prospectively frozen for the new holdout after earlier mixed-domain fragmentation has exactly one transferred family per fate for every seed, although the stricter all-trajectory final-state uniqueness gate narrowly fails. In a separate staged artifact, training-distribution support families route local affine maps and have descriptively lower controlled-forecasting error than a same-budget global- K LISTA model, subject to selector leakage and asymmetry. These forecasting, alignment, and routing results arise from different recipes and do not identify a representation-to-forecast causal chain.

These results motivate a hybrid interpretation in which coefficient values carry continuous predictive coordinates and active-coordinate identities provide an inspectable regime key. An exact direct-sum affine lift shows that one matrix can contain several support-selected local laws in principle, without showing that learning discovers them. The held-out controlled diagnostic finds lower support-conditioned one-step leakage than matched coordinate-permutation controls for one unchanged K ; an evaluator-emitted all-current guard gives similar ratios, but adds only 0.96 percentage points on a highly persistent sample. It therefore does not resolve boundary-switching behavior or remove the future-conditioning concern generally. Its operator-differentiation gate still fails, although the triggered exact-dense control establishes complete-recipe specificity relative to cardinality-matched dense coordinates. A final frozen recursive residual-routing diagnostic was invalid at its frozen validity tier and directionally unadjudicated after a nonfinite payload value (`inf`) prevented completion of the exact-ten packet; no partial outcome was inspected. We do not claim an exact global finite-dimensional linearization, a one-to-one support-basin theorem, supervised-quality basin recovery, distinct support-local laws, or that representation quality mediates forecasting. Periodic decode–encode is an autonomous nonlinear composite predictor, not a guaranteed mathematical projection; the best-periodic values in the submitted binary PDF are not uniformly pure K^h and were selected over re-encoding cadences on report trajectories. The separate no-re-encoding comparisons more directly test the KAE objective. The staged result is specific to its fitted partition and affine-map recipe and does not establish deployment-ready routing.

Several protocol limitations constrain the conclusions. First, all reported best-periodic values choose cadence separately at each horizon on the same test trajectories; validation-frozen cadence selection is needed for deployment-style estimates. The MSE reducer also omits nonfinite steps, so finite prefixes can make divergent rollouts appear better than a strict full-horizon failure rule. Every ordinary controlled row and both stages of the staged model also cycle only eight fixed training batches per seed because the maintained child-seed streams are not advanced, substantially limiting training-data diversity. Second, the controlled alignment partition is fitted over the same evaluation trajectory collection that contains the high center-margin states used for scoring. For 13 systems, the same collection also supplies the endpoints used to estimate centers, and the known benchmark count enters that evaluation-only construction. The resulting nearest-center labels and $d_2 - d_1$ margin are proxies and need not recover true basin geometry or boundary distance. The Allen–Cahn transfer and forecasting packets remove this particular fit–score overlap, but remain one

synthetic PDE condition and are not evidence on empirical real-world data. Validation fate metrics selected the transfer packet’s T20 scope, Jaccard-0.40 rule, temporal weight, and advancement of the fixed ten-seed roster; its all-trajectory final-state uniqueness gate misses by 0.009. The original forecast packet fails its four-cell gate and was selected after a three-seed/four-recipe screen; the later H200 same-checkpoint check freezes an outcome-aware endpoint and only changes initial-condition datasets. Neither establishes population generalization over retraining, PDE parameters, grids, or systems.

Third, the 225 staged system–seed pairs are nested within 15 systems and receive no confirmatory system-level inference. Its route-fit packet contains only 256 unique trajectories despite 512 configured rows, and its checkpoint selector reuses 32% of the reported starts while the global baseline uses a different selector. The result should be rerun from a newly versioned 512-unique route fit with disjoint, common checkpoint-validation starts and a validation-frozen cadence before making a general routing claim. Cross-family attribution is also limited because LISTA decoders are atom-normalized whereas MLP decoders are not, and the structured-transition rows use a known controlled-benchmark basin count solely to size their blocks. The corrected Dysts comparison removes the historical LISTA-SB depth confound by using one learned refinement in all three LISTA rows and uses a tanh, zero-sparsity dense baseline. A separate historical 15-seed $d_z = 1,024$ stress test on fixed eight-basin competitive Lotka–Volterra, Hopfield $N = P = 16$, and Kuramoto $N = 16$ favored tanh Dense MLP over every sparse/LISTA recipe at H100/H500/H1000 under evaluation-selected best-periodic cadence; it is a required negative recipe comparison, not a sparsity-only ablation. Finally, standalone controls use three seeds and independently generated trajectories, and the coordinate intervention uses only one Local-linear gates LISTA checkpoint at seed 0. These comparisons remain descriptive.

The immediate priorities are therefore to complete the corrected all-row Dysts rerun; finish the paired Allen–Cahn two-versus-three-refinement study; stop outcome-adaptive work on the opened controlled mechanism condition; freeze forecasting cadence on validation data with strict finiteness accounting; and test frozen recipes prospectively across new physical conditions.

A Additional experimental details

This appendix records the paper-facing protocol. Unstructured LISTA, Sparse MLP, Dense MLP, support-family construction, and staged routing are trained or constructed without basin labels, basin counts, or trajectory-to-basin assignments. For post-hoc alignment evaluation, the two gated systems provide native labels and centers; the other 13 systems use endpoint-estimated centers and nearest-center proxy labels, with the known benchmark count setting the number of estimated centers. Separately, the block-diagonal and soft-block architecture diagnostics use the known controlled count to set the number of transition blocks. No label or count is a feature in support-family formation or routing, and structured rows receive no trajectory label.

Benchmarks. The controlled benchmark contains 15 two-dimensional multibasin systems, with equations and retained metadata in [section B](#). Unstructured rows and runtime prediction observe stored states, not vector fields, integration substeps, native basin annotations, estimated centers, proxy labels, or benchmark counts; the disclosed structured-transition rows use a count only to size their parameterization. A cadence audit found that the historical Dysts KAE caches used each system’s native interval although their metadata claimed $30dt$, while standalone controls used the intended cadence. All historical Dysts forecasts and comparisons are therefore quarantined. The roster in [section C](#) now describes only the corrected campaign.

Model rows and decoder normalization. All six KAE rows have $d_z = 256$. Encoder pre-code or MLP hidden widths are 64, 64. Sparse MLP and LISTA pre-code layers use biased ReLU activations, whereas Dense MLP uses tanh; all decoders are linear. LISTA, LISTA-BD, and LISTA-SB use learned shrinkage; Sparse MLP and Sparse MLP-BD use an L_1 latent penalty; Dense MLP removes the sparse output gate and sets that penalty to zero. BD denotes a block-diagonal K ; SB denotes a dense K with an off-block penalty. The reported LISTA models decode with a normalized linear dictionary. The sparse and dense MLP rows use ordinary learned linear decoders without atom normalization, so decoder scaling is not controlled across encoder families.

The controlled LISTA rows use threshold scale $\alpha = 0.15$, two refinement loops, and a sign-split output that maps a signed 128-dimensional base code to $d_z = 256$ by concatenating its positive and negative parts. The shrinkage threshold is $\alpha/\hat{L}_D$, where initialization estimates $\hat{L}_D$ as 1.05 times a ten-step power-iteration estimate of the initialized dictionary Gram norm.

Controlled multibasin training. Every controlled row uses seeds $0, \dots, 14$, rollout-window length $L = 8$, 200,000 optimization steps, minibatches of 256, and AdamW. Encoder and decoder learning rates are 5×10^{-5} , the transition learning rate is 5×10^{-6} , and non-transition weight decay is 10^{-4} . The objective weights are $\lambda_{\text{pred}} = 1$, $\lambda_{\text{rec}} = 0.03$, $\lambda_{\text{lin}} = 1$, and $\lambda_{\text{sp}} = 3 \times 10^{-3}$ for sparse rows; Dense MLP uses $\lambda_{\text{sp}} = 0$. Soft-block rows add an off-block L_1 penalty with weight 10^{-4} .

All controlled rows use the system-specific observation intervals and base reset boxes in [table 5](#), followed by the same label-free perturbation-sensitive, high-transient training wrapper. Let s be the coordinatewise span of the base reset box. For model seed r , candidate j uses base-reset seed $r + 2,000,000 + j$. A single float32 noise stream seeded $r + 3,000,000$ is consumed in candidate-order chunks of 128. It supplies four Gaussian perturbations per candidate with coordinatewise standard deviation $0.04s$; these are box-clipped and rolled for 32 stored steps. The score is their mean endpoint separation from the unperturbed rollout, normalized by $\|0.04s\|_2$, plus 0.5 times the mean step-24-to-step-32 motion over all five rollouts, normalized by $\|s\|_2$. The top 1,024 of 4,096 candidates form the hard pool. Each training reset chooses that pool with probability 0.5; a selected point receives box-clipped Gaussian jitter with coordinatewise standard deviation $0.01s$, while the other branch uses a base reset. This rule uses no basin labels or boundary geometry and should not be interpreted as sampling measured basin boundaries.

The maintained loader cycles eight fixed batches of 256 length-eight windows per model seed because its eight child-seed streams are not advanced between uses. This fixed-data artifact affects all six ordinary controlled KAE rows and both training stages of the staged model; its global- K comparator is the ordinary LISTA row. Stream $b \in \{0, \dots, 7\}$ supplies child reset seeds $r + 256b + i$, $i = 0, \dots, 255$, to the wrapped reset rule above.

Corrected Dysts campaign status. No historical Dysts result is treated as paper evidence. The corrected campaign sets the installed system integration interval to 30 times its native value before native trajectory generation, verifies the realized observation grid numerically, and uses schema-versioned caches with exact metadata, shape, split, and finiteness gates. Train, validation, policy-selection, and test caches have separate deterministic namespaces and split-specific perturbed initial conditions. All six KAE rows use seeds $0, \dots, 14$, $d_z = 256$, batch size 256, $L = 10$, and 100,000 optimization steps; LISTA, LISTA-BD, and LISTA-SB all use one learned refinement under unified semantics. The exact-dense baseline uses tanh hidden activations, a dense K , and zero sparsity terms. The primary endpoint is direct repeated- K rollout with strict full-horizon finiteness. This campaign is in progress and contributes no result to the current draft.

Checkpoint selection. For the ordinary controlled single-stage KAE runs, including the global- K LISTA baseline used in the staged table, every 500 optimization steps and the final step are candidates. Each candidate is rolled out for 200 stored steps from 16 fixed starts generated at seed offset 999,999, using every-step re-encoding. The checkpoint with the lowest mean final-step Euclidean state error is retained. These starts are distinct from the 100 reported evaluation starts at seed offset 12,345. The staged selector is different and is specified below. Tables aggregate completed seeds and do not choose the best seed.

Forecasting metric and cadence selection. For start i , step t , and state dimension j , the raw step error is $e_{i,t} = \sum_j (\hat{x}_{i,t,j} - x_{i,t,j})^2$. For KAE and local-linear mixture rows, the reported MSE through H is the mean of finite $e_{i,t}$ over $t = 1, \dots, H$ within each start and then over starts. Consequently, a rollout with a finite prefix can contribute even if it becomes nonfinite before H ; strict full-horizon finite coverage is not their ranking rule. The classical DMD, polynomial EDMD, and RBF-dictionary EDMD evaluator instead takes an ordinary mean through H within each start and then omits starts whose resulting mean is nonfinite, so those descriptive rows require full finite-step coverage through H for a start to contribute.

Periodic rollout decodes and re-encodes only the model’s own prediction. For model seed r , controlled forecasting uses 100 base-reset starts with child seeds $r + 12,345 + i$, $i = 0, \dots, 99$; the training-only hard-pool wrapper is not used. The alignment collection analogously uses base-reset seeds $42 + i$, $i = 0, \dots, 127$, and is shared across model seeds. Controlled forecasting uses cadence grid $m \in \{10, 25, 50, 100\}$. For each model, system, seed, and horizon, the reported best-periodic value is the minimum on those same evaluation trajectories over the fixed grid. Thus cadence is selected by an evaluation-set oracle rather than frozen from validation. The staged table uses the wider grid given below.

Standalone controls. DMD, polynomial EDMD, RBF-dictionary EDMD, and the local-linear mixture controls configure seeds 0, 1, 2 and use independently generated trajectories rather than the KAE rows’ matched test samples. The controlled manifest uses 256 trajectories of length 1000, with 60% assigned to fitting. All fits use ridge coefficient 10^{-6} . Polynomial EDMD uses degree 3; RBF-dictionary EDMD uses 128 centers with data-derived gamma and the implementation default as fallback; k -means, GMM, and soft-gated GMM local-linear controls use four components. Evaluation horizons match the columns of [table 1](#). These rows are descriptive references only and do not support paired tests or a causal conclusion about sparsity.

Transductive support-basin alignment. At evaluation seed 42, the evaluator generates 128 trajectories with 128 transitions and therefore 129 states per row, or 16,512 states before subset selection. It constructs $S_{\text{abs}}(x) = \{i : |z_i| > 10^{-3}\}$ for all of those states and greedily merges their masks at Jaccard threshold 0.50 to form $F_{\text{abs}}^{\text{align}}$. The Local-linear gates and Transfer-gated local-linear environments provide native labels and centers. Each of the 13 catalog environments provides neither: the evaluator advances the 128 trajectory endpoints for 5,000 additional steps, uses the known manifest basin count in a deterministic k -means fit (first endpoint followed by farthest-point initialization, then at most 25 Lloyd updates), and labels every state by its nearest estimated center.

For every state, the evaluator computes the distance gap $d_2 - d_1$ between its second-nearest and nearest native or estimated center. Within each observed evaluation label, it retains every state at or above that label’s 75th margin percentile, including ties, and scores $H(B \mid F_{\text{abs}}^{\text{align}})$, using natural logarithms and reporting nats. This operation does not equalize label sample counts or force an

exact 25% retention rate. Most system-seed subsets contain 4,129–7,167 states. Tied margins make Triple-well Duffing retain all 16,512, and its proxy assigns every state to one estimated center. The resulting $H(B | F) = 0$ for all models is uninformative, so the primary alignment aggregate uses the 14 systems with at least two observed labels. Raw Duffing rows remain frozen and an all-15-system sensitivity is retained. A large margin means that the nearest-center assignment is separated from its runner-up; it need not measure distance to a true basin boundary or separatrix. Labels, centers, and the known catalog-system count define and score this subset but do not enter family formation. The result is post-hoc and transductive because the scored states are contained in the same evaluation trajectory collection used to fit the partition and, for catalog systems, to estimate centers; it is not evaluation of a codebook or label rule fitted on disjoint data. The companion family count is the number of fitted family IDs observed on the scored subset, not the total partition size over all 16,512 states.

Representative coordinate intervention. The intervention uses one Local-linear gates LISTA checkpoint at seed 0. It either drops the $k \in \{1, 2, 3, 5, 10\}$ largest-magnitude entries of z_0 or moves active values to randomly selected inactive coordinates, then runs the ordinary transition and decoder. Candidate starts must lie in the tie-inclusive per-native-label high center-margin slice and retain their initial native label throughout the true 21-step future. Sampling balances the 100 starts over sorted native labels, giving 34/33/33; coordinate dropping and the 20-shuffle random-support condition use exactly the same starts. The margin is the same nearest-center separation proxy, not a measured distance to a basin boundary. This intervention is descriptive and has no confirmatory significance test.

Staged support-family local forecasting. The staged and global- K LISTA models use the same architecture and 200,000-step total budget. Stage one trains E_θ , D_ϕ , and a shared K for 100,000 steps. The staged model then freezes them and fits $F_{\text{abs}}^{\text{route}}$ at seed offset 271,828. Its nominal 512-row packet is two bitwise-identical copies of one 256-trajectory batch from the training distribution. A row contains 192 transitions and 193 states. Family formation at absolute threshold 10^{-3} and Jaccard threshold 0.40 consumes all 193 supports per row, so terminal supports affect frequency ordering and the greedy partition. Map counts, source centers, and direct mask-to-family keys use only the first 192 source states. The corresponding nominal counts are 98,816 clustered supports and 98,304 source transitions, of which 49,152 source transitions are unique. A future fit with 512 unique trajectories is a new protocol and requires new checkpoints.

After the greedy merge, the frozen runtime representative for each family is the modal support among its source states, with the historical first-seen tie break; it need not equal the initial frequency-ordered mask that opened the family. Families need at least one source transition. The model trains one affine map $T_f(z) = d_f + K_f(z - \bar{z}_f)$ per retained family for 100,000 further steps. Each K_f starts from the frozen global K , while d_f is a learned intercept initialized to $K\bar{z}_f$.

During rollout, the staged wrapper computes the current latent’s absolute support and selects a family before every latent transition. Unassigned masks use global K . Independently, each scheduled periodic refresh decodes and re-encodes the current prediction; that refreshed latent is routed at the following transition. Period m therefore controls latent refresh, not route lookup: route lookup occurs on all steps. Neither fitting nor routing uses basin labels or counts. Both staged and global rows are reported at their evaluation-set oracle cadence for each horizon from $m \in \{1, 2, 5, 10, 20, 25, 50, 100\}$.

The staged checkpoint selector evaluates exactly 200 candidates: steps 100,500 through 199,500 every 500 steps, plus step 199,999. Its 32 fixed starts use seed offset 12,345. For every candidate

and cadence, it computes state-summed squared error, averages each start’s finite prefix through H100, H500, or H1000, and then averages the finite per-start means. It selects the best cadence independently at each horizon, takes the arithmetic mean of the three minima, and saves only on a strict decrease. Final reporting uses 100 starts generated at the same offset; therefore the selector starts are exactly the first 32 reported starts, a 32% overlap. The global baseline instead uses the standard 16-start offset-999,999, every-step H200 final-error selector described above. Hence the comparison matches architecture and total budget but not checkpoint selection, and is descriptive and optimistic. The 225 system–seed pairs are also nested within 15 systems; win counts and ratios have no confirmatory p -value.

Aggregation and inference. Forecasting and alignment entropy point estimates use SciPy’s `trim_mean` with proportion 0.25 within each system, followed by an arithmetic mean across systems. Thus $\lfloor n/4 \rfloor$ finite values are discarded from each tail and a complete 15-seed cell averages its central nine. Controlled forecasting uses all 15 systems; primary alignment uses the 14 systems with at least two observed labels and retains an all-15 sensitivity. The descriptive family-count column instead uses ordinary seed means within system. Controlled forecasting and alignment stars use paired seed-level Wilcoxon tests within system with Holm correction across eligible systems. The staged and intervention results are descriptive. No Dysts inference is currently reported. Full definitions are in [section F](#).

Hardware. Training used one cluster-assigned CUDA GPU per allocation with four CPU cores. The main controlled KAE rows requested 16 GB host memory, whereas staged support-family training requested 24 GB. Jobs loaded CUDA 12.6.0. Controlled training requested 24 hours per run. Cache construction and paper evaluation used CPU allocations; the retained coordinate-intervention evaluator requests 8 GB. Corrected Dysts compute is not reported until that campaign completes. The scripts do not pin a GPU model, so a specific accelerator model is not claimed.

B Multibasin benchmark inventory and system definitions

[Table 4](#) lists the 15 two-dimensional multibasin systems used in this paper. Every row contributes to controlled forecasting and staged-routing results in [tables 1](#) and [3](#). The primary alignment aggregate in [table 2](#) uses the 14 systems whose frozen evaluation assignment realizes at least two labels. Triple-well Duffing’s one-label proxy rows remain in the evidence packet and in the all-15-system sensitivity sidecar.

The two gated systems expose native labels and centers for evaluation. The remaining 13 catalog systems expose neither; their alignment evaluator uses the known benchmark count to estimate centers from long-rolled endpoints and construct nearest-center proxy labels. Basin counts are also used to set the fixed architecture group count for structured-transition ablations, as described in [section A](#). No native label, proxy label, center, or count is supplied as a runtime feature to support extraction, support-family construction, staged routing, or rollout prediction; structured-transition rows retain the disclosed count-sized parameterization.

For every system except Triple-well Duffing, the frozen seed-42 evaluation assignment realizes the full manifest count shown above. The Duffing endpoint clustering instead assigns all 16,512 evaluation states to one estimated center. Its conditional entropy is therefore zero for every model and cannot measure regime alignment; this evaluation-proxy failure does not remove the system from forecasting or staged-routing experiments.

Table 4: The 15 multibasin benchmark systems. The manifest count sets the number of estimated evaluation centers for catalog systems and the number of transition blocks for structured-transition diagnostic rows.

Display name	Dynamical family	Manifest count (eval./blocks)
Local-linear gates	piecewise local-linear gates	3
Transfer-gated local-linear	piecewise transfer gates	3
Arrested spiral	spiral capture with Gaussian traps	5
Asymmetric three-well	Gaussian wells with independent rotation	3
High-cross three-well	Gaussian wells with stronger rotation	3
Hexagonal six-well	Gaussian wells with independent rotation	6
Octagonal eight-well	Gaussian wells with independent rotation	8
Pentagonal five-well	Gaussian wells with independent rotation	5
Square four-well	Gaussian wells with independent rotation	4
Triple-well Duffing	triple-well Duffing oscillator	3
SNIC multi-attractor	polar SNIC-like phase dynamics	3
Transition-routes four-well	route-augmented Gaussian wells	4
Depth-gradient four-well	Gaussian wells with depth gradient	4
Diamond four-well	Gaussian wells on a diamond layout	4
L-shaped five-well	Gaussian wells on an L-shaped layout	5

Table 5: Stored observation intervals and base reset laws. Each coordinate is sampled independently and uniformly over the listed box; one stored step is one fourth-order Runge–Kutta update.

System(s)	Δt	Base reset box
Local-linear gates	0.04	$[-2.6, 2.6]^2$
Transfer-gated local-linear	0.04	$[-2.8, 2.8]^2$
Arrested spiral	0.02	$[-3, 3]^2$
Asymmetric three-well	0.03	$[-3, 3] \times [-2.5, 3]$
High-cross three-well; Hexagonal six-well; Pentagonal five-well; Square four-well	0.03	$[-3.3, 3.3]^2$
Octagonal eight-well	0.03	$[-4, 4]^2$
Triple-well Duffing; SNIC multi-attractor	0.02	$[-2, 2]^2$
Transition-routes four-well	0.02	$[-3.5, 3.5]^2$
Depth-gradient four-well	0.03	$[-3, 3]^2$
Diamond four-well; L-shaped five-well	0.03	$[-3.5, 3.5]^2$

Stored-step convention. All retained systems are implemented as continuous-time vector fields $\dot{x} = f_s(x)$ and integrated with fourth-order Runge–Kutta at the system-specific intervals in [table 5](#) to produce the stored observation map $F_{\Delta t}$ used in [equation \(1\)](#). The training windows contain only stored states x_k , not f_s , integration substeps, or basin assignments.

Ground-truth vector-field visualizations. [Figure 7](#) shows the continuous-time ground-truth vector fields for all 15 retained two-dimensional multibasin systems. Streamlines show the direction of $f_s(x)$, and black crosses mark the generator attractor centers or analytic equilibrium references used for benchmark annotation. The streamline colors use a panel-local $\log_{10} \|f_s(x)\|_2$ scale to reveal structure within each system; they should not be read as comparable speed scales across panels. Detailed panels with per-system colorbars are provided in [figures 8 to 12](#).

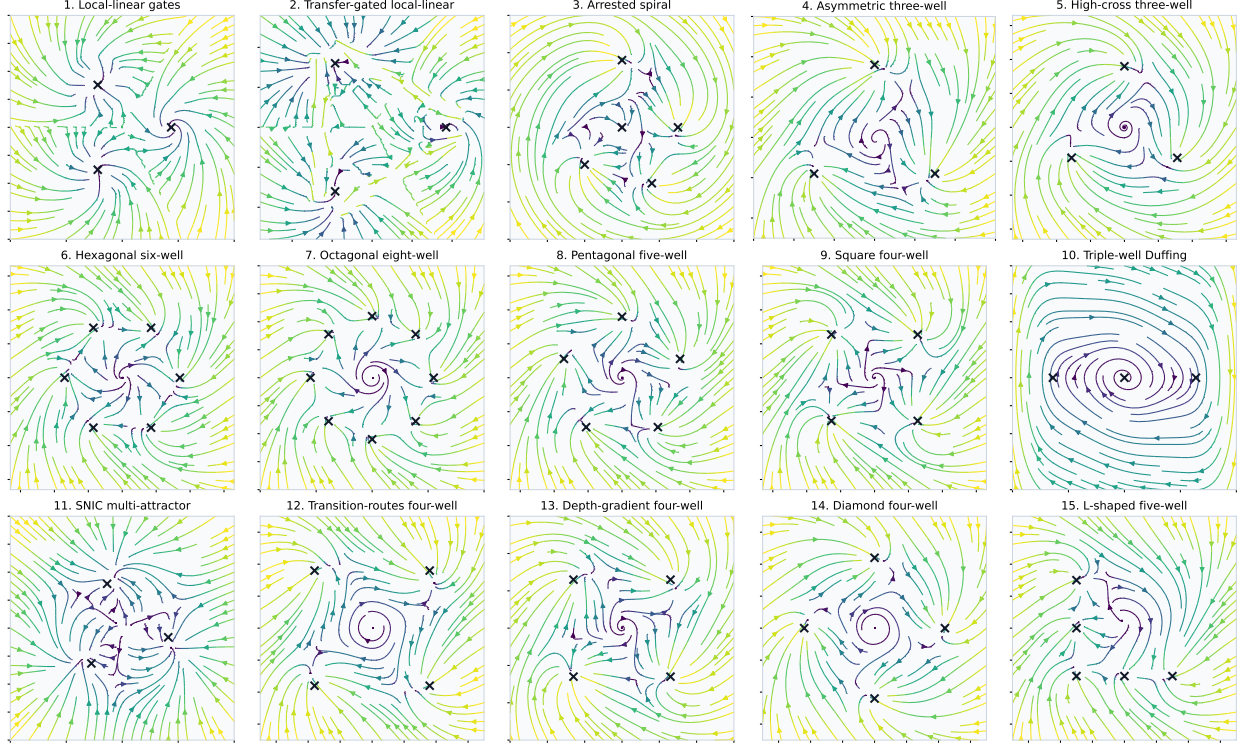


Figure 7: Ground-truth vector fields for the retained 15-system multibasin benchmark. The models are trained only on stored state trajectories; these continuous-time vector fields and generator reference markers are shown only to document the benchmark and are not provided to the learner. For the 13 catalog systems, these markers are not native alignment labels or centers; the evaluation-only proxies are estimated from long-rolled endpoints.

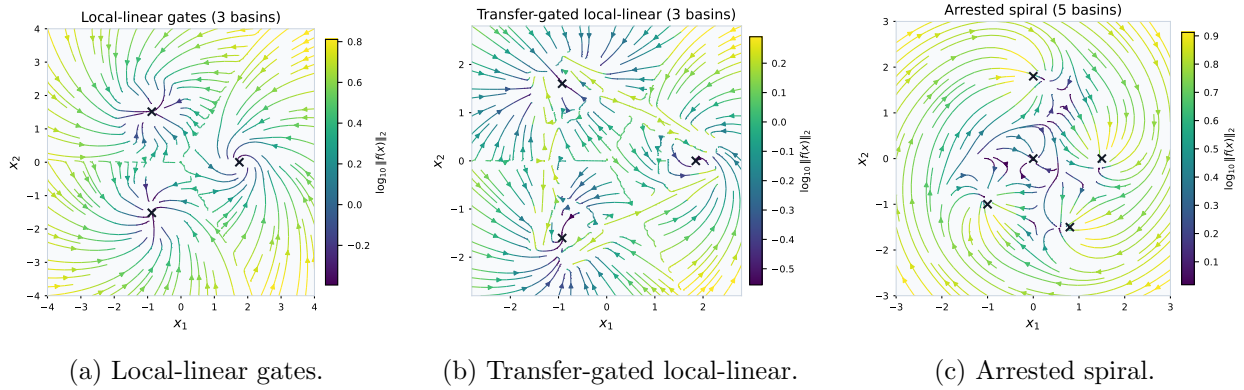


Figure 8: Detailed ground-truth vector-field panels for the first three retained multibasin systems. Colorbars report the panel-local $\log_{10} \|f_s(x)\|_2$ scale.

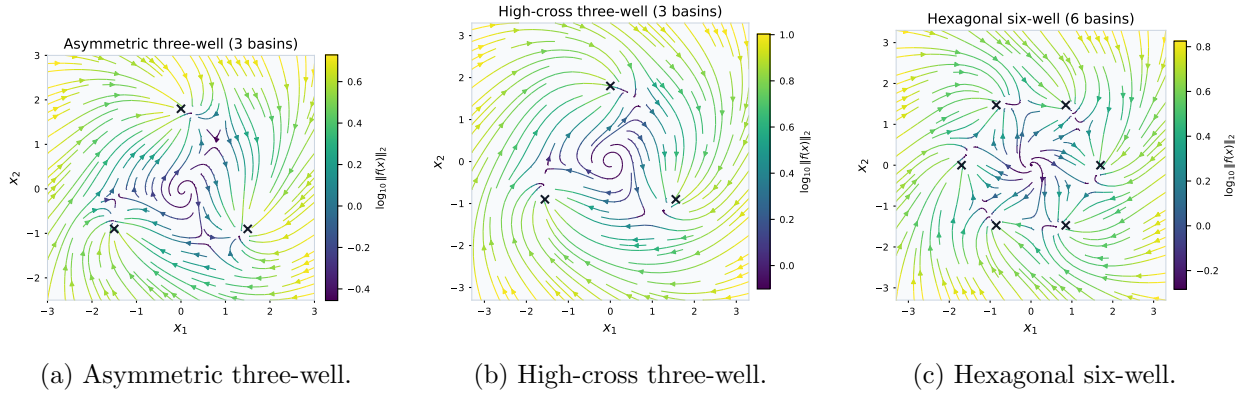


Figure 9: Detailed ground-truth vector-field panels for retained Gaussian-well systems. Colorbars report the panel-local $\log_{10} \|f_s(x)\|_2$ scale.

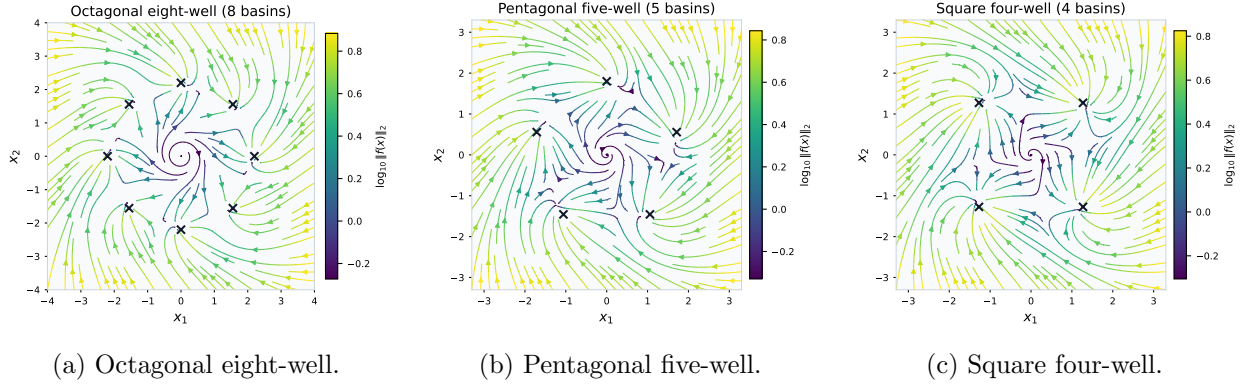


Figure 10: Detailed ground-truth vector-field panels for retained polygonal Gaussian-well systems. Colorbars report the panel-local $\log_{10} \|f_s(x)\|_2$ scale.

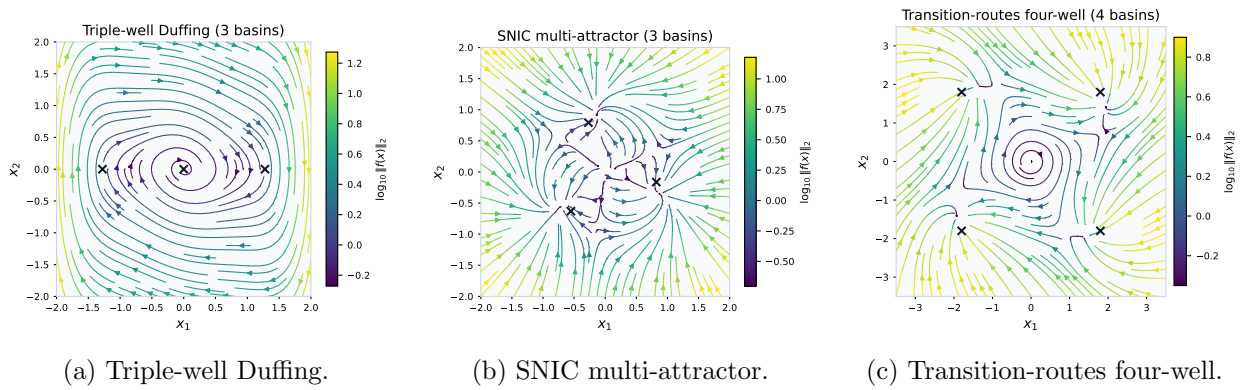


Figure 11: Detailed ground-truth vector-field panels for specialized retained systems. Colorbars report the panel-local $\log_{10} \|f_s(x)\|_2$ scale.

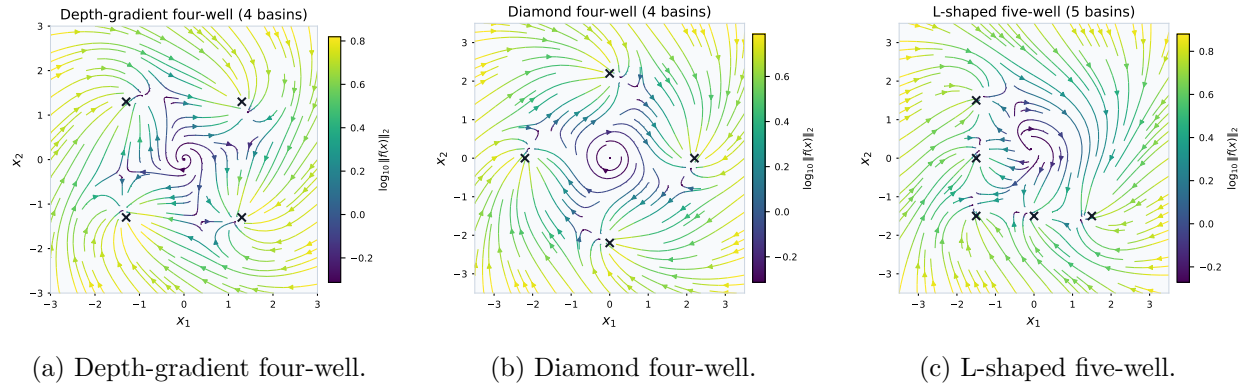


Figure 12: Detailed ground-truth vector-field panels for retained geometry variants. Colorbars report the panel-local $\log_{10} \|f_s(x)\|_2$ scale.

Table 6: Parameters for retained Gaussian-well systems. Repeated pairs in the (a_i, σ_i) column apply to every listed center.

System	$\{c_i\}$	(a_i, σ_i)	(ω, γ)
High-cross three-well	$\{p_i^{(3)}(1.8, \pi/2)\}$	(3.0, 0.5)	(2.0, 0.03)
Hexagonal six-well	$\{p_i^{(6)}(1.7, 0)\}$	(2.0, 0.5)	(1.0, 0.03)
Octagonal eight-well	$\{p_i^{(8)}(2.2, 0)\}$	(3.0, 0.5)	(0.90, 0.02)
Pentagonal five-well	$\{p_i^{(5)}(1.8, \pi/2)\}$	(2.0, 0.5)	(1.1, 0.03)
Square four-well	$\{p_i^{(4)}(1.8, \pi/4)\}$	(3.0, 0.5)	(1.0, 0.03)
Diamond four-well	$\{(0, 2.2), (2.2, 0), (0, -2.2), (-2.2, 0)\}$	(2.5, 0.5)	(1.0, 0.02)
L-shaped five-well	$\{(-1.5, 1.5), (-1.5, 0), (-1.5, -1.5), (0, -1.5), (1.5, -1.5)\}$	(2.5, 0.5)	(1.0, 0.03)
Asymmetric three-well	$\{(0, 1.8), (-1.5, -0.9), (1.5, -0.9)\}$	$\{(2.5, 0.55), (1.5, 0.4), (2.0, 0.5)\}$	(1.0, 0.03)
Depth-gradient four-well	$\{(-1.3, 1.3), (1.3, 1.3), (1.3, -1.3), (-1.3, -1.3)\}$	$\{(2.2, 0.55), (2.5, 0.5), (3.0, 0.5), (3.5, 0.5)\}$	(1.3, 0.03)

Gaussian-well family. Most retained catalog systems use a common two-dimensional Gaussian potential with an independent rotational drift. For $x = (x_1, x_2) \in \mathbb{R}^2$, let $Rx = (x_2, -x_1)$ and

$$V_s(x) = \sum_{i=1}^{M_s} -a_{s,i} \exp\left(-\frac{\|x - c_{s,i}\|_2^2}{2\sigma_{s,i}^2}\right) + \gamma_s(x_1^4 + x_2^4).$$

The implemented vector field is

$$\dot{x} = -\nabla V_s(x) + \omega_s Rx. \quad (14)$$

Define $p_i^{(M)}(r, \varphi) = r(\cos(2\pi i/M + \varphi), \sin(2\pi i/M + \varphi))$, $i = 0, \dots, M-1$. The retained Gaussian-well systems instantiate [equation \(14\)](#) with the parameters in [table 6](#).

The transition-routes system uses the same Gaussian-well form with centers

$$\{c_i\} = \{(-1.8, 1.8), (1.8, 1.8), (-1.8, -1.8), (1.8, -1.8)\},$$

$a_i = 3.0$, $\sigma_i = 0.6$, $\omega = 1.0$, and $\gamma = 0.03$, and adds a corridor-dependent rotation term:

$$\dot{x} = -\nabla V_s(x) + (1.0 + 0.3 C_{\text{route}}(x_2)) Rx, \quad C_{\text{route}}(x_2) = e^{-(x_2-1.8)^2/0.3} + e^{-(x_2+1.8)^2/0.3}. \quad (15)$$

Native gated local-linear systems. Both native gated systems place three centers on a circular layout, but with different radii and region definitions. Let $Q(\alpha) = \begin{pmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{pmatrix}$ and $\alpha_b = 2\pi b/3$, $b \in \{0, 1, 2\}$.

For Local-linear gates, $c_b = 1.75(\cos \alpha_b, \sin \alpha_b)$. Inside the radius-1.05 core of basin b ,

$$\dot{x} = A_b(x - c_b), \quad A_b = Q(\alpha_b) \tilde{A}_b Q(\alpha_b)^\top,$$

with

$$\tilde{A}_0 = \begin{pmatrix} -0.9 & -1.2 \\ 1.2 & -0.9 \end{pmatrix}, \quad \tilde{A}_1 = \begin{pmatrix} -1.35 & 0.2 \\ -0.3 & -0.7 \end{pmatrix}, \quad \tilde{A}_2 = \begin{pmatrix} -0.7 & -0.1 \\ 0.5 & -1.2 \end{pmatrix}.$$

Outside the cores, the active sector is the nearest angular sector $s(x)$ and the shared gate matrix

$$G = \begin{pmatrix} -1.35 & -0.9 \\ 0.9 & -1.35 \end{pmatrix}$$

drives the state as $\dot{x} = G(x - c_{s(x)})$.

For Transfer-gated local-linear, $c_b = 1.85(\cos \alpha_b, \sin \alpha_b)$. The implemented radii and widths are

$$\begin{aligned} \rho_{\text{core}} &= 0.30, & \rho_{\text{source}} &= 0.80, & \rho_{\text{exit}} &= 0.60, & \beta_{\text{exit}} &= 0.72, \\ w_{\text{chan}} &= 0.22, & \delta_{\text{lane}} &= 0.28, & \rho_{\text{hand}} &= 0.45. \end{aligned}$$

The core matrices use the same rotated form with templates

$$\begin{pmatrix} -1.0 & -1.1 \\ 1.1 & -1.0 \end{pmatrix}, \quad \begin{pmatrix} -1.4 & 0.2 \\ -0.2 & -0.7 \end{pmatrix}, \quad \begin{pmatrix} -0.8 & -0.3 \\ 0.5 & -1.3 \end{pmatrix}.$$

For each ordered pair $p = (s, t)$, $s \neq t$, define

$$d_{s \rightarrow t} = \frac{c_t - c_s}{\|c_t - c_s\|_2}, \quad n_{s \rightarrow t} = (-d_{s \rightarrow t, 2}, d_{s \rightarrow t, 1}), \quad o_t = \frac{c_t}{\|c_t\|_2},$$

and $\chi_{s,t} = 1$ when $\sin(\alpha_s - \alpha_t) \geq 0$, otherwise $\chi_{s,t} = -1$. The implemented channel entry and handoff points are

$$\begin{aligned} e_p &= c_s + \rho_{\text{source}} d_{s \rightarrow t} + \chi_{s,t} \delta_{\text{lane}} n_{s \rightarrow t}, \\ h_p &= c_t + \rho_{\text{hand}} o_t + \chi_{s,t} \delta_{\text{lane}} (-o_{t, 2}, o_{t, 1}), \end{aligned}$$

with channel direction $d_p = (h_p - e_p) / \|h_p - e_p\|_2$, channel normal $n_p = (-d_{p, 2}, d_{p, 1})$, and channel length $\ell_p = (h_p - e_p) \cdot d_p$.

The vector field is piecewise analytic. Core regions $\|x - c_b\|_2 \leq \rho_{\text{core}}$ use $A_b(x - c_b)$. Source annuli use $0.65A_b(x - c_b)$, and other non-channel background regions use $0.50A_b(x - c_b)$ for the nearest center c_b . Exit wedges for $p = (s, t)$ are source-neighborhood states outside the core with $\|x - c_s\|_2 \geq \rho_{\text{exit}}$ and $(x - c_s) \cdot d_{s \rightarrow t} \geq \|x - c_s\|_2 \cos \beta_{\text{exit}}$; they use

$$\dot{x} = v_{\text{exit}} d_{s \rightarrow t} - \lambda_{\text{exit}} ((x - c_s) \cdot n_{s \rightarrow t}) n_{s \rightarrow t}, \quad v_{\text{exit}} = 1.0, \quad \lambda_{\text{exit}} = 1.8;$$

and channel rectangles satisfying

$$0 \leq (x - e_p) \cdot d_p \leq \ell_p, \quad |(x - e_p) \cdot n_p| \leq w_{\text{chan}}$$

use

$$\dot{x} = v_{\text{chan}} d_p - \lambda_{\text{chan}} ((x - e_p) \cdot n_p) n_p, \quad v_{\text{chan}} = 1.55, \quad \lambda_{\text{chan}} = 2.8.$$

Other specialized systems. The arrested spiral system has a background spiral flow plus four Gaussian traps and an origin trap:

$$\dot{x} = -0.3x + 2.0Rx - 4.0 \sum_{i=1}^4 e^{-\|x - c_i\|_2^2 / (2 \cdot 0.25)} \frac{x - c_i}{0.25} - 1.5e^{-\|x\|_2^2 / 0.3} \frac{x}{0.15} - 0.02(x_1^3, x_2^3),$$

with $c_i \in \{(1.5, 0), (0, 1.8), (-1, -1), (0.8, -1.5)\}$. Its fifth basin is the origin trap, matching the benchmark manifest count.

The triple-well Duffing system uses $x = (q, p)$ and

$$V(q) = q^6/6 - q^4/2 + aq^2, \quad V'(q) = q^5 - 2q^3 + 2aq,$$

with $a = 0.3$, damping $\delta = 0.5$, rotation strength $\omega = 1.0$, and soft cubic confinement $-\epsilon u^3/s^3$ with $\epsilon = 0.003$ and $s = 4.0$:

$$\begin{aligned} \dot{q} &= (1 + \omega)p - \epsilon q^3/s^3, \\ \dot{p} &= -V'(q) - \delta p - \omega q - \epsilon p^3/s^3. \end{aligned} \tag{16}$$

The SNIC system is implemented in polar form and then converted to Cartesian coordinates. With $\varepsilon_{\text{num}} = 10^{-8}$, $r^2 = x_1^2 + x_2^2 + \varepsilon_{\text{num}}$, $r = \sqrt{r^2}$, $\theta = \text{atan2}(x_2, x_1)$, $c_\theta = x_1/(r + \varepsilon_{\text{num}})$, and $s_\theta = x_2/(r + \varepsilon_{\text{num}})$,

$$\dot{r} = r(1 - r^2) - 0.3 \cos(3\theta), \quad \dot{\theta} = 1 - 1.2 \cos(3\theta).$$

Table 7: The ten-system roster and target observation grid for the corrected Dysts $dt \times 30$ campaign. The “target stored step” is 30 times the native Dysts integration interval and does not describe the quarantined historical KAE caches. The final column gives the hand-specified diagnostic structure count used only to size the BD and SB transition parameterizations; it is a heuristic lobe, scroll, or equilibrium count, not basin truth.

Display name	Dimension	Native dt	Target stored step $30dt$	Diagnostic count
Chua	3	$2.8474745791 \times 10^{-4}$	$8.5424237373 \times 10^{-3}$	2
Dadras	3	$6.5782963827 \times 10^{-4}$	$1.9734889148 \times 10^{-2}$	2
Dequan Li	3	$1.6763993999 \times 10^{-5}$	$5.0291981997 \times 10^{-4}$	3
Hadley	3	$2.9086847808 \times 10^{-4}$	$8.7260543424 \times 10^{-3}$	3
Lu–Chen–Cheng	3	$1.8469678280 \times 10^{-4}$	$5.5409034839 \times 10^{-3}$	4
Qi–Chen	3	$7.8371061844 \times 10^{-5}$	$2.3511318553 \times 10^{-3}$	2
Sakarya	3	$9.9704617436 \times 10^{-4}$	$2.9911385231 \times 10^{-2}$	2
San–Um–Srisuchinwong	3	$1.4933288881 \times 10^{-3}$	$4.4799866644 \times 10^{-2}$	2
Shimizu–Morioka	3	$2.4080013336 \times 10^{-3}$	$7.2240040007 \times 10^{-2}$	2
Wang–Sun	3	$5.3924987498 \times 10^{-3}$	$1.6177496249 \times 10^{-1}$	4

The Cartesian vector field is

$$\begin{aligned}\dot{x}_1 &= \dot{r}c_\theta - r\dot{\theta}s_\theta - 0.01x_1(x_1^2 + x_2^2) + 0.5x_2, \\ \dot{x}_2 &= \dot{r}s_\theta + r\dot{\theta}c_\theta - 0.01x_2(x_1^2 + x_2^2) - 0.5x_1.\end{aligned}$$

These explicit generators define the benchmark trajectories. Evaluation annotations use native labels for the two gated systems and nearest-estimated- center proxy labels after endpoint rollout for the catalog systems; the learned models observe only stored states.

C Dysts benchmark inventory and system definitions

This appendix lists the ten Dysts systems frozen for the corrected $dt \times 30$ forecasting campaign. The systems are drawn from Dysts (Gilpin, 2021, 2023). We used the continuous-time right-hand sides and metadata from the pinned package version resolved in the project environment, `dysts` 0.96. All ten systems are three-dimensional autonomous flows. The equations below are written in unstandardized native Dysts coordinates (x, y, z) ; the training cache standardizes coordinates only after trajectories are generated. This roster does not validate or describe the quarantined historical KAE caches.

Closed-form convention. For the corrected campaign, stored observations are generated from $\dot{x} = f_s(x)$ with a numerically verified observation interval $\Delta t = 30 dt_{\text{native}}$, where dt_{native} is the system-specific Dysts integration interval in table 7. The learner observes only the resulting stored states, not the vector field, continuous-time solver, or substeps. All ten systems have explicit closed-form right-hand sides in Dysts. Most are polynomial vector fields. The Chua and San–Um–Srisuchinwong systems include absolute-value terms, so they are piecewise smooth rather than globally analytic at the corresponding switching surfaces.

Chua. The Chua system uses the piecewise-linear diode nonlinearity

$$h(x) = m_1x + \frac{1}{2}(m_0 - m_1)(|x + 1| - |x - 1|),$$

with parameters

$$\alpha = 15.6, \quad \beta = 28, \quad m_0 = -1.142857, \quad m_1 = -0.71429.$$

The vector field is

$$\begin{aligned}\dot{x} &= \alpha(y - x - h(x)), \\ \dot{y} &= x - y + z, \\ \dot{z} &= -\beta y.\end{aligned}\tag{17}$$

Dadras. With parameters

$$c = 2, \quad e = 9, \quad o = 2.7, \quad p = 3, \quad r = 1.7,$$

the Dadras system is

$$\begin{aligned}\dot{x} &= y - px + oyz, \\ \dot{y} &= ry - xz + z, \\ \dot{z} &= cxy - ez.\end{aligned}\tag{18}$$

Dequan Li. With parameters

$$a = 40, \quad c = 1.833, \quad d = 0.16, \quad \varepsilon = 0.65, \quad f = 20, \quad k = 55,$$

the Dequan Li system is

$$\begin{aligned}\dot{x} &= a(y - x) + dxz, \\ \dot{y} &= kx + fy - xz, \\ \dot{z} &= cz + xy - \varepsilon x^2.\end{aligned}\tag{19}$$

Hadley. With parameters

$$a = 0.2, \quad b = 4, \quad f = 9, \quad g = 1,$$

the Hadley system is

$$\begin{aligned}\dot{x} &= -y^2 - z^2 - ax + af, \\ \dot{y} &= xy - bxz - y + g, \\ \dot{z} &= bxy + xz - z.\end{aligned}\tag{20}$$

Lu–Chen–Cheng. With parameters

$$a = -10, \quad b = -4, \quad c = 18.1,$$

the Lu–Chen–Cheng system is

$$\begin{aligned}\dot{x} &= -\frac{ab}{a+b}x - yz + c, \\ \dot{y} &= ay + xz, \\ \dot{z} &= bz + xy.\end{aligned}\tag{21}$$

Qi–Chen. With parameters

$$a = 38, \quad b = 2.666, \quad c = 80,$$

the Qi–Chen system is

$$\begin{aligned}\dot{x} &= a(y - x) + yz, \\ \dot{y} &= cx + y - xz, \\ \dot{z} &= xy - bz.\end{aligned}\tag{22}$$

Sakarya. With parameters

$$a = -1, \quad b = 1, \quad c = 1, \quad h = 1, \quad p = 1, \quad q = 0.4, \quad r = 0.3, \quad s = 1,$$

the Sakarya system is

$$\begin{aligned}\dot{x} &= ax + hy + syz, \\ \dot{y} &= -by - px + qxz, \\ \dot{z} &= cz - rxy.\end{aligned}\tag{23}$$

San–Um–Srisuchinwong. With parameter $a = 2$, the San–Um–Srisuchinwong system is

$$\begin{aligned}\dot{x} &= y - x, \\ \dot{y} &= -z \tanh(x), \\ \dot{z} &= -a + xy + |y|.\end{aligned}\tag{24}$$

Shimizu–Morioka. With parameters

$$a = 0.85, \quad b = 0.5,$$

the Shimizu–Morioka system is

$$\begin{aligned}\dot{x} &= y, \\ \dot{y} &= x - ay - xz, \\ \dot{z} &= -bz + x^2.\end{aligned}\tag{25}$$

Wang–Sun. With parameters

$$a = 0.2, \quad b = -0.01, \quad d = -0.4, \quad e = -1, \quad f = -1, \quad q = 1,$$

the Wang–Sun system is

$$\begin{aligned}\dot{x} &= ax + qyz, \\ \dot{y} &= bx + dy - xz, \\ \dot{z} &= ez + fxy.\end{aligned}\tag{26}$$

D Support definitions and family construction

All support objects are computed after or between training stages from $z = E_\theta(x)$. Basin labels, basin counts, and trajectory assignments do not define an exact support, merge masks, or route a rollout. The known count used to estimate catalog-system evaluation centers is therefore not an input to $F_{\text{abs}}^{\text{align}}$, although it is an input to the label and scoring-slice construction applied afterward. The paper uses two protocol-specific absolute-threshold families and does not assume that they are the same partition.

Absolute support. For every state,

$$m_{\text{abs},i}(x) = \mathbf{1}\{|z_i| > 10^{-3}\}, \quad S_{\text{abs}}(x) = \{i : m_{\text{abs},i}(x) = 1\}.$$

The 10^{-3} threshold is fixed in both experiments.

Deterministic greedy family merge. For a specified state collection and Jaccard threshold τ , distinct Boolean masks are counted and ordered from most to least frequent; equal frequencies use ascending byte order after `numpy.packbits` with its default big-bit ordering. During the greedy merge, each family is compared against the first mask that opened it. For a new mask m , compute

$$J(m, r_f) = \frac{|S(m) \cap S(r_f)|}{|S(m) \cup S(r_f)|}, \quad J(0, 0) = 1.$$

The mask joins the representative with largest J when that score is at least τ ; equal Jaccard scores choose the earliest-created family. Otherwise it starts a new family. These merge-time representatives are not averaged or updated, so family identifiers are meaningful only through their induced partition for that fitted collection. The alignment diagnostic retains these representatives. The staged route subsequently replaces each runtime representative with that family’s modal source-state support, as specified below.

Input: masks and counts from collection $\mathcal{X}$; threshold τ .

Initialize: ordered representative list $R \leftarrow []$.

For each distinct mask m : assign m to the best existing $r \in R$ if $J(m, r) \geq \tau$; otherwise append m to R .

Output: the fixed representatives and mask-to-family assignment.

Alignment family. $F_{\text{abs}}^{\text{align}}$ uses $\tau = 0.50$. At evaluation seed 42, it is fitted on all $128 \times 129 = 16,512$ states from 128 trajectories with 128 transitions each. The two gated environments supply native labels and centers. The 13 catalog environments supply neither, so the evaluator advances the 128 trajectory endpoints for 5,000 additional steps and estimates the known manifest number of centers by deterministic k -means. Initialization starts with the first converged endpoint and adds the endpoint farthest from its nearest current center; a farthest-point tie chooses the first endpoint index. Lloyd assignment ties choose the first center index, an empty cluster retains its prior center, and at most 25 updates follow; iteration stops early under PyTorch’s default `allclose` test. Every evaluation state is then assigned the nearest estimated center as a proxy label, again with first-center tie breaking.

Only after family fitting, each state’s center-margin proxy $d_2 - d_1$ is computed from its second-nearest and nearest native or estimated center. The evaluator uses NumPy’s default linear 75th-percentile rule within each observed evaluation label and retains all ties at the cutoff; labels with fewer than four states retain all their states. It does not equalize label counts or enforce an exact one-quarter subset. Most system-seed subsets contain 4,129–7,167 states. Triple-well Duffing retains all 16,512 because of tied margins, but all states also receive one proxy label, so its $H(B | F) = 0$ for every model is uninformative. Primary alignment aggregates use the 14 systems with at least two observed labels. Duffing’s raw rows are retained for an all-15-system sensitivity. The historical entropy reducer computes joint entropy minus family entropy with natural logarithms and $-p \log(p + 10^{-12})$ terms; values are reported in nats on this high-margin subset. Labels and centers do not enter the mask merge as features. The result is transductive because the fitting

collection contains the scored states and, for catalog systems, the endpoints used to estimate centers. In the main alignment table, the compact notation F_{abs} refers specifically to $F_{\text{abs}}^{\text{align}}$.

Low $H(B | F_{\text{abs}}^{\text{align}})$ means that the fitted partition leaves little uncertainty about the native or proxy label on that slice. The margin expresses separation under a nearest-center construction; it need not equal distance from a true basin boundary or separatrix. The result does not imply a frozen classifier for new states, one family per basin, or recovery near basin boundaries. The reported family count is the number of family IDs observed on the scored high-margin slice, not the total number formed over all evaluation states; it is compared descriptively with the number of represented evaluation labels.

Staged routing family. $F_{\text{abs}}^{\text{route}}$ uses $\tau = 0.40$ after stage one. The source artifact’s nominal 512-row route-fit packet is two exactly identical copies of 256 unique training-distribution trajectories, generated at seed offset 271,828. Each row contains 193 states and 192 transitions. The greedy family merge consumes all 193 supports, including the terminal state, for 98,816 nominal support observations. Only afterward are family counts, source centers, and exact mask-to-family keys restricted to the first 192 source states, for 98,304 nominal and 49,152 unique source transitions.

For runtime routing, each merged family’s representative is replaced by the modal support among that family’s source states; ties follow the first-seen source-key order. A family is retained for a local affine map when it has at least one source transition. During rollout, an exact source mask observed during fitting maps directly to its frozen family. An unseen mask is assigned to the modal source representative with greatest Jaccard overlap only if the overlap reaches 0.40; otherwise the rollout uses the frozen global K . The current latent’s support is recomputed and routed before every latent transition. A scheduled decode–re-encode event only refreshes that latent before the next transition; it is not the routing cadence. This modal-source replacement is part of the frozen source protocol; replacing it with the merge-opening mask or fitting 512 unique trajectories would define a new protocol.

The 0.40 routing merge and 0.50 alignment merge answer different questions and were applied to different state collections. Alignment statistics cannot be used to infer the route partition’s entropy, and the staged forecasting result cannot be read as an out-of-sample validation of the transductive alignment partition.

E Support-coordinate intervention table

[Figure 5](#) plots the full curves for one Local-linear gates LISTA checkpoint at seed 0. [Table 8](#) reports accumulated MSE at $H = 21$, defined here as the sum over rollout steps of the squared error averaged across state coordinates. This differs from the state-summed, through-horizon mean used by the headline forecasting table, so their numerical scales are not comparable. Candidate starts lie in the tie-inclusive per-native-label high center-margin subset and must retain their initial native label throughout the true 21-step future. The evaluator balances 100 starts across the three labels (34/33/33). Coordinate dropping and random support use exactly these same starts; random support uses 20 inactive-coordinate reassignments per start. Random-support summaries pool the resulting 2,000 dependent point–shuffle outcomes at each horizon. The margin is not a true basin-boundary distance. This one-system, one-model, one-seed experiment is descriptive and has no confirmatory significance test.

Table 8: Accumulated MSE (sum of per-coordinate step MSE) at $H = 21$ for descriptive support-coordinate interventions on the Local-linear gates LISTA checkpoint at seed 0. Drop rows summarize 100 starts; the random row pools 2,000 dependent point-shuffle outcomes. Lower is better; no confirmatory inference is attached.

Rollout	Mean $\pm$ SD	Median [IQR]
Standard	0.0158 $\pm$ 0.0127	0.0140 [0.0055, 0.0224]
Drop top-1	0.508 $\pm$ 0.647	0.218 [0.148, 0.677]
Drop top-2	1.37 $\pm$ 0.938	1.06 [0.802, 1.63]
Drop top-3	2.08 $\pm$ 1.10	1.80 [1.34, 2.10]
Drop top-5	3.26 $\pm$ 2.44	1.95 [1.61, 4.17]
Drop top-10	8.51 $\pm$ 6.83	5.16 [4.58, 8.11]
Random support	1.69 $\times 10^3 \pm 2.19 \times 10^3$	874 [377, 2.06 $\times 10^3$]

F Statistical testing protocol

Point estimates and significance annotations answer different questions. Forecasting and alignment entropy tables first summarize finite completed seeds within each system by `scipy.stats.trim_mean` with proportion 0.25, then take an arithmetic mean across systems. This sorts n values and discards $\lfloor n/4 \rfloor$ from each tail; complete 15-seed cells therefore average their central nine. We refer to this implementation as the interquartile mean (IQM). Forecasting averages all 15 controlled systems; primary alignment averages the 14 systems with at least two observed labels and retains an all-15 sensitivity. The descriptive family-count column instead averages per-system seed means. We do not pool trajectories, seeds, and systems into one inferential sample.

Forecasting metric and missingness. For start i , rollout step t , and state coordinate j , let

$$e_{i,t} = \sum_{j=1}^{d_x} (\hat{x}_{i,t,j} - x_{i,t,j})^2.$$

For the KAE rows used in formal comparisons, the reported MSE through horizon H takes the mean of finite $e_{i,t}$ over $t = 1, \dots, H$ within each start, then averages finite per-start means. It is therefore a state-summed squared error, not per-coordinate MSE. A rollout can contribute through its finite prefix even if it is nonfinite before H , so the KAE ranking does not enforce full-horizon finiteness. Local-linear mixture controls use the same finite-step convention; the unmatched classical DMD/EDMD evaluator requires a finite ordinary through- H mean for each contributing start, as detailed in [section A](#).

The error $E_{s,r,h}$ used below is also best-periodic: for every system s , seed r , model, and horizon h , the lowest MSE is selected on the same test trajectories over the fixed cadence grid. This test-set oracle is shared by the point estimates and formal KAE comparisons and should not be interpreted as validation-frozen deployment selection.

Controlled forecasting effects. For candidate model a , Dense MLP baseline, system s , seed r , and horizon h , the paired effect is

$$\Delta_{s,r,h}^a = \log_{10} E_{s,r,h}^a - \log_{10} E_{s,r,h}^{\text{Dense}}.$$

Only mutually finite positive MSEs are eligible. Within each system and candidate–horizon cell, a one-sided paired Wilcoxon signed-rank test uses the common seeds with alternative $\Delta < 0$. At least four finite pairs are required. Raw per-system p -values are Holm-corrected across eligible systems in that candidate–horizon family. A main-table star denotes corrected $p < 0.05$; every displayed controlled sparse-row forecasting cell passes on all 15/15 systems.

The per-system tables in [section G](#) display the arithmetic mean of the paired seed effects $\Delta_{s,r,h}^a$, together with the Wilcoxon/Holm p -value. The displayed effect is a mean, not the median used by an older forest-plot artifact. Because the signed-rank test depends on ranks whereas the mean does not, an extreme finite outlier can make their directions appear inconsistent; the Transfer-gated local-linear LISTA–BD H500 cell is the one such case, with 12/15 negative paired effects but a positive arithmetic mean.

Transductive alignment effects. The alignment partition $F_{\text{abs}}^{\text{align}}$ is fit on all states in an evaluation trajectory collection and scored on states at or above the within-label 75th center-margin percentile contained in that collection; ties are retained, so the subset need not contain exactly one quarter of a label, and no between-label count balancing is performed. Labels and centers are native for the two gated systems. For the 13 catalog systems they are nearest-center proxies obtained by deterministic clustering of long-rolled endpoints at the known benchmark count. The margin $d_2 - d_1$ is nearest-center separation, not a true boundary-distance measurement. $H(B \mid F_{\text{abs}}^{\text{align}})$ uses natural logarithms and is reported in nats. The seed is the paired unit within each system and the one-sided alternative is candidate $<$ Dense MLP. Holm correction is across eligible systems within each model–metric cell. Triple-well Duffing’s frozen proxy realizes one observed label, so its zero entropy for every model is uninformative and it is excluded from primary aggregates; the all-zero paired differences and undefined signed-rank statistic are consequences. Every displayed alignment star passes on all 14/14 eligible systems. The observed high-margin-slice family count has no directional test because closeness to the number of represented evaluation labels, rather than monotone increase, is the relevant descriptive comparison. These tests quantify model differences in the transductive diagnostic; they do not turn it into an out-of-sample classification experiment.

Dysts forecasting effects. The historical Dysts KAE rows and their associated point estimates, sign tests, Holm corrections, and significance markers are quarantined because the KAE and standalone-control trajectories used mismatched physical cadences. No Dysts inferential result is reported until the corrected all-six-row campaign is complete.

Descriptive comparisons. The standalone DMD, EDMD, and local-linear controls configure three seeds and use independently generated evaluation trajectories; their per-system summaries omit nonfinite seed values, so they are not included in paired KAE tests. The staged support-family result reports 225 system–seed pairs nested within 15 systems, plus MSE ratios, without a confirmatory p -value. Its checkpoint score applies the same finite-prefix state-summed error reducer as the reported table, minimizes separately over the eight cadences at H100, H500, and H1000, and averages those three minima. The 32 selector starts are the first 32 of the 100 reported starts, whereas the global baseline selects its checkpoint on 16 distinct starts with an H200 final-error proxy. This 32% staged overlap and selector asymmetry make the comparison optimistic and preclude interpreting the paired win counts as held-out causal evidence. The Local-linear gates coordinate intervention uses one LISTA checkpoint at seed 0; bootstrap intervals summarize its 100 starts, while random-support medians and interquartile ranges pool 2,000 dependent point–shuffle outcomes per horizon. These describe within-checkpoint variability only.

High-dimensional stress tests. The Lorenz-96, Allen-Cahn temporal-support, and forecast-optimized Allen-Cahn packets each average trajectories within model seed and use ten paired model-initialization seeds as inferential units; trajectories are not independent replicates. Lorenz-96 and the temporal-support packet use 100,000 percentile bootstrap resamples of seed-level outcomes and exact two-sided sign-flip tests over all 2^{10} assignments. Lorenz-96 applies Holm correction over its two declared long physical horizons, 2.5 and 5; temporal-support forecasting corrects over physical times 8, 12, 16, 20, and its final-state support gate combines frozen absolute thresholds with a paired sparse-minus-dense ARI interval. The separate forecast-optimized Allen-Cahn packet uses 50,000 paired-seed bootstrap resamples of each ratio-of-arm-means reduction. Those four intervals are marginal; the frozen primary decision is their four-cell intersection-union gate. An exact one-sided paired sign-flip max- t calculation over the four H160/H200 mean/terminal cells is reported only as a multiplicity sensitivity. H160 is a prefix of H200, so these cells are dependent. Every interval conditions on its packet’s fixed generated data and does not cover new dataset draws.

The same-checkpoint Allen-Cahn new-initial-condition check crosses three newly generated datasets with each of the same ten paired sparse/dense checkpoint seeds. It first averages the three datasets within checkpoint seed, then uses the ten paired seeds for a 100,000-resample paired bootstrap interval and exact one-sided sign-flip test over all 2^{10} assignments. Thus 30 seed-dataset cells are not treated as independent replicates, and the three dataset-specific effects are descriptive. The H200 through-horizon endpoint was frozen only after the original four-cell gate failed, so this inference is an outcome-aware same-checkpoint robustness check rather than independent retraining replication or a repair of that gate.

The secondary Allen-Cahn physical-metric packet uses the identical forecasts and the same within-checkpoint averaging over three datasets. Its sole statistical family contains all seven H200 cumulative phase, interface, and energy metrics. For each metric, the one-sided paired sign-flip test enumerates all 2^{10} assignments of the oriented seed effects; Holm correction is over all seven raw values. The 100,000-resample paired-seed interval is for the oriented absolute arm-mean difference. Relative error reductions divide that absolute difference by the observed dense arm mean; modal accuracy is instead reported in percentage points. Instantaneous curves, terminal values, H160, and the late half are mandatory descriptives and receive no separate inference. Because the metrics translate an outcome-selected H200 advantage on unchanged checkpoints, favorable results remain secondary and cannot reclassify the original gate.

Holm correction and assumptions. For each formal family, eligible raw p -values are sorted, multiplied by the number of remaining hypotheses, made monotone by a running maximum, and clipped at one. Paired Wilcoxon tests assume the common seed differences are the relevant exchangeable units within a controlled system. All conclusions condition on the fixed benchmark roster. Evaluation-set period selection, finite-prefix omission, staged checkpoint-selection overlap, selector asymmetry, and transductive family fitting are protocol limitations independent of the multiple-comparison correction.

G Full per-system controlled-benchmark tables

Tables 9–11 report the per-system forecasting effects and Holm-corrected Wilcoxon p -values underlying the controlled columns of table 1. Each displayed Δ is the arithmetic mean over common completed seeds of the paired $\log_{10}$ -MSE difference, candidate minus Dense MLP. The associated test is the one-sided paired Wilcoxon test on those seed effects. Bold entries pass Holm-corrected $p < 0.05$ in favor of the candidate. The rank test and arithmetic mean need not have the same sign

Table 9: Per-system forecasting effects at $H = 100$ on the 15-system controlled benchmark. Δ is the arithmetic mean paired $\log_{10}$ -MSE difference over common seeds (candidate minus Dense MLP), not a median; p_H is the one-sided paired Wilcoxon p -value Holm-corrected across retained systems within each candidate column.

System	LISTA		LISTA-BD		LISTA-SB		Sparse MLP, BD		Sparse MLP	
	Δ	p_H	Δ	p_H	Δ	p_H	Δ	p_H	Δ	p_H
Local-linear gates	-2.16	$<10^{-3}$	-2.19	$<10^{-3}$	-2.20	$<10^{-3}$	-2.33	$<10^{-3}$	-2.52	$<10^{-3}$
Transfer-gated local-linear	-0.27	0.001	-0.21	0.008	-0.31	0.007	-0.37	$<10^{-3}$	-0.27	0.004
Arrested spiral	-1.85	$<10^{-3}$	-2.09	$<10^{-3}$	-2.03	$<10^{-3}$	-1.43	$<10^{-3}$	-1.90	$<10^{-3}$
Asymmetric three-well	-2.00	$<10^{-3}$	-1.92	$<10^{-3}$	-1.75	$<10^{-3}$	-2.43	$<10^{-3}$	-2.44	$<10^{-3}$
High-cross three-well	-1.05	$<10^{-3}$	-1.12	$<10^{-3}$	-1.19	$<10^{-3}$	-0.95	$<10^{-3}$	-1.17	$<10^{-3}$
Hexagonal six-well	-1.76	$<10^{-3}$	-1.73	$<10^{-3}$	-1.85	$<10^{-3}$	-1.70	$<10^{-3}$	-1.83	$<10^{-3}$
Octagonal eight-well	-1.85	$<10^{-3}$	-1.90	$<10^{-3}$	-1.80	$<10^{-3}$	-1.66	$<10^{-3}$	-1.63	$<10^{-3}$
Pentagonal five-well	-1.86	$<10^{-3}$	-1.58	$<10^{-3}$	-1.20	$<10^{-3}$	-1.53	$<10^{-3}$	-2.16	$<10^{-3}$
Square four-well	-1.44	$<10^{-3}$	-1.62	$<10^{-3}$	-1.58	$<10^{-3}$	-1.26	$<10^{-3}$	-1.02	$<10^{-3}$
Triple-well Duffing	-0.54	$<10^{-3}$	-0.51	0.008	-0.35	0.007	-0.90	$<10^{-3}$	-0.78	$<10^{-3}$
SNIC multi-attractor	-2.01	$<10^{-3}$	-2.03	$<10^{-3}$	-1.91	$<10^{-3}$	-2.21	$<10^{-3}$	-2.13	$<10^{-3}$
Transition-routes four-well	-2.21	$<10^{-3}$	-1.77	$<10^{-3}$	-1.90	$<10^{-3}$	-2.14	$<10^{-3}$	-2.47	$<10^{-3}$
Depth-gradient four-well	-1.39	$<10^{-3}$	-1.33	$<10^{-3}$	-1.37	$<10^{-3}$	-0.94	$<10^{-3}$	-0.82	$<10^{-3}$
Diamond four-well	-2.00	$<10^{-3}$	-1.80	$<10^{-3}$	-1.84	$<10^{-3}$	-1.51	$<10^{-3}$	-1.67	$<10^{-3}$
L-shaped five-well	-2.23	$<10^{-3}$	-1.94	$<10^{-3}$	-1.90	$<10^{-3}$	-1.85	$<10^{-3}$	-2.23	$<10^{-3}$
<i>Significant/eligible</i>	$K=15 / N=15$		$K=15 / N=15$		$K=15 / N=15$		$K=15 / N=15$		$K=15 / N=15$	

under an extreme outlier: at H500 for Transfer-gated local-linear LISTA-BD, 12/15 paired seed effects are negative, but one divergent finite candidate seed makes the displayed mean positive.

H Dysts forecasting campaign status

A cadence audit found that the historical KAE caches retained the native Dysts integration interval rather than the requested $30dt$ interval, while the standalone controls used the intended cadence. Consequently, every historical Dysts result, ratio table, horizon curve, and significance annotation is quarantined and is not evidence in this manuscript. A corrected matched campaign retraining all six KAE rows at a numerically verified $30dt$ cadence is in progress; no result from that campaign is reported here.

I High-dimensional stress-test protocols

Lorenz-96. The fixed-data chaotic forecasting comparison uses the standard Lorenz-96 system with forcing 8, $d_x = 128$, and all coordinates observed. One generated dataset (seed 20,260,719) is fixed across ten model-initialization seeds and contains 128/32/64 train/validation/test trajectories with 201 stored states each. The stored interval is 0.05, training observations receive noise at 5% of the training scale, and the declared forecast horizons 1, 5, 10, 25, 50, 100 therefore extend through physical time 5. Both KAEs use a four-times overcomplete lift, $d_z = 512$, 20-step training windows, 3,000 updates, batch size 1,024, and a full trainable latent matrix. Forecasts encode only the initial state and then repeatedly apply the learned K without periodic decode-encode refresh. The deterministic references are persistence, ordinary DMD, and rank-32 truncated DMD fitted to

Table 10: Per-system forecasting effects at $H = 500$. Effect and test definitions match [table 9](#).

System	LISTA		LISTA-BD		LISTA-SB		Sparse MLP, BD		Sparse MLP	
	Δ	p_H	Δ	p_H	Δ	p_H	Δ	p_H	Δ	p_H
Local-linear gates	-2.01	$<10^{-3}$	-2.24	$<10^{-3}$	-2.27	$<10^{-3}$	-2.29	$<10^{-3}$	-2.52	$<10^{-3}$
Transfer-gated local-linear	-0.08	0.013	+1.19	0.008	-0.21	$<10^{-3}$	-0.17	$<10^{-3}$	-0.18	$<10^{-3}$
Arrested spiral	-2.26	$<10^{-3}$	-2.57	$<10^{-3}$	-2.57	$<10^{-3}$	-1.72	$<10^{-3}$	-2.33	$<10^{-3}$
Asymmetric three-well	-2.27	$<10^{-3}$	-2.12	$<10^{-3}$	-1.94	$<10^{-3}$	-3.11	$<10^{-3}$	-3.15	$<10^{-3}$
High-cross three-well	-0.93	$<10^{-3}$	-0.97	$<10^{-3}$	-1.03	$<10^{-3}$	-0.86	$<10^{-3}$	-1.26	$<10^{-3}$
Hexagonal six-well	-2.01	$<10^{-3}$	-2.01	$<10^{-3}$	-2.18	$<10^{-3}$	-1.95	$<10^{-3}$	-2.16	$<10^{-3}$
Octagonal eight-well	-1.94	$<10^{-3}$	-2.09	$<10^{-3}$	-1.91	$<10^{-3}$	-1.78	$<10^{-3}$	-1.74	$<10^{-3}$
Pentagonal five-well	-2.48	$<10^{-3}$	-2.10	$<10^{-3}$	-1.61	$<10^{-3}$	-2.02	$<10^{-3}$	-2.77	$<10^{-3}$
Square four-well	-1.60	$<10^{-3}$	-1.89	$<10^{-3}$	-1.81	$<10^{-3}$	-1.55	$<10^{-3}$	-1.17	$<10^{-3}$
Triple-well Duffing	-0.83	$<10^{-3}$	-0.81	$<10^{-3}$	-0.65	$<10^{-3}$	-1.10	$<10^{-3}$	-1.05	$<10^{-3}$
SNIC multi-attractor	-2.30	$<10^{-3}$	-2.45	$<10^{-3}$	-2.31	$<10^{-3}$	-2.56	$<10^{-3}$	-2.49	$<10^{-3}$
Transition-routes four-well	-1.73	$<10^{-3}$	-1.52	$<10^{-3}$	-1.49	$<10^{-3}$	-1.55	$<10^{-3}$	-1.86	$<10^{-3}$
Depth-gradient four-well	-1.63	$<10^{-3}$	-1.45	$<10^{-3}$	-1.47	$<10^{-3}$	-1.17	$<10^{-3}$	-0.92	$<10^{-3}$
Diamond four-well	-1.77	$<10^{-3}$	-1.58	$<10^{-3}$	-1.40	$<10^{-3}$	-1.28	$<10^{-3}$	-1.75	$<10^{-3}$
L-shaped five-well	-2.98	$<10^{-3}$	-2.04	$<10^{-3}$	-2.03	$<10^{-3}$	-2.27	$<10^{-3}$	-2.96	$<10^{-3}$
<i>Significant/eligible</i>	$K=15 / N=15$		$K=15 / N=15$		$K=15 / N=15$		$K=15 / N=15$		$K=15 / N=15$	

Table 11: Per-system forecasting effects at $H = 1000$. Effect and test definitions match [table 9](#).

System	LISTA		LISTA-BD		LISTA-SB		Sparse MLP, BD		Sparse MLP	
	Δ	p_H	Δ	p_H	Δ	p_H	Δ	p_H	Δ	p_H
Local-linear gates	-1.98	$<10^{-3}$	-2.24	$<10^{-3}$	-2.29	$<10^{-3}$	-2.28	$<10^{-3}$	-2.52	$<10^{-3}$
Transfer-gated local-linear	-0.08	0.013	-0.15	$<10^{-3}$	-0.20	$<10^{-3}$	-0.16	$<10^{-3}$	-0.17	$<10^{-3}$
Arrested spiral	-2.32	$<10^{-3}$	-2.65	$<10^{-3}$	-2.66	$<10^{-3}$	-1.76	$<10^{-3}$	-2.39	$<10^{-3}$
Asymmetric three-well	-2.27	$<10^{-3}$	-2.12	$<10^{-3}$	-1.93	$<10^{-3}$	-3.16	$<10^{-3}$	-3.22	$<10^{-3}$
High-cross three-well	-0.92	$<10^{-3}$	-0.96	$<10^{-3}$	-0.99	$<10^{-3}$	-0.85	$<10^{-3}$	-1.26	$<10^{-3}$
Hexagonal six-well	-2.02	$<10^{-3}$	-2.02	$<10^{-3}$	-2.19	$<10^{-3}$	-1.96	$<10^{-3}$	-2.18	$<10^{-3}$
Octagonal eight-well	-1.89	$<10^{-3}$	-2.05	$<10^{-3}$	-1.86	$<10^{-3}$	-1.73	$<10^{-3}$	-1.70	$<10^{-3}$
Pentagonal five-well	-2.55	$<10^{-3}$	-2.15	$<10^{-3}$	-1.64	$<10^{-3}$	-2.06	$<10^{-3}$	-2.84	$<10^{-3}$
Square four-well	-1.62	$<10^{-3}$	-1.92	$<10^{-3}$	-1.84	$<10^{-3}$	-1.60	$<10^{-3}$	-1.21	$<10^{-3}$
Triple-well Duffing	-0.80	$<10^{-3}$	-0.73	$<10^{-3}$	-0.64	$<10^{-3}$	-1.05	$<10^{-3}$	-1.02	$<10^{-3}$
SNIC multi-attractor	-2.31	$<10^{-3}$	-2.49	$<10^{-3}$	-2.35	$<10^{-3}$	-2.59	$<10^{-3}$	-2.53	$<10^{-3}$
Transition-routes four-well	-1.54	$<10^{-3}$	-1.38	$<10^{-3}$	-1.36	$<10^{-3}$	-1.40	$<10^{-3}$	-1.62	$<10^{-3}$
Depth-gradient four-well	-1.67	$<10^{-3}$	-1.48	$<10^{-3}$	-1.50	$<10^{-3}$	-1.20	$<10^{-3}$	-0.94	$<10^{-3}$
Diamond four-well	-1.76	$<10^{-3}$	-1.53	$<10^{-3}$	-1.33	$<10^{-3}$	-1.20	$<10^{-3}$	-1.74	$<10^{-3}$
L-shaped five-well	-3.08	$<10^{-3}$	-2.03	$<10^{-3}$	-2.07	$<10^{-3}$	-2.26	$<10^{-3}$	-3.05	$<10^{-3}$
<i>Significant/eligible</i>	$K=15 / N=15$		$K=15 / N=15$		$K=15 / N=15$		$K=15 / N=15$		$K=15 / N=15$	

Table 12: Per-system transductive $H(B | F_{\text{abs}}^{\text{align}})$ effects, in nats, on the within-label high center-margin slice. Families are fitted on all states in each evaluation trajectory collection and scored at or above each label’s 75th margin percentile, retaining ties; native or nearest-estimated-center proxy labels select only that subset, without equalizing label counts. The known benchmark count is used to estimate catalog-system centers but does not enter family formation. The family-count companion is the number of family IDs observed on the scored subset. The one-sided paired Wilcoxon alternative is candidate entropy < Dense MLP entropy, with Holm correction across eligible systems. Triple-well Duffing’s proxy realizes one label and is excluded from primary alignment aggregation; its zero paired differences are a consequence. Displayed alignment stars pass on all 14/14 eligible systems.

System	LISTA		LISTA-BD		LISTA-SB		Sparse MLP, BD		Sparse MLP	
	Δ	p_H	Δ	p_H	Δ	p_H	Δ	p_H	Δ	p_H
Local-linear gates	-1.09	<10 ⁻³	-1.09	<10 ⁻³	-1.09	<10 ⁻³	-1.09	<10 ⁻³	-1.09	<10 ⁻³
Transfer-gated local-linear	-1.10	<10 ⁻³	-1.10	<10 ⁻³	-1.10	<10 ⁻³	-1.10	<10 ⁻³	-1.10	<10 ⁻³
Arrested spiral	-1.04	0.001	-0.92	0.001	-0.98	0.001	-0.27	0.026	-0.98	0.001
Asymmetric three-well	-1.09	<10 ⁻³	-1.09	<10 ⁻³	-1.09	<10 ⁻³	-1.09	<10 ⁻³	-1.09	<10 ⁻³
High-cross three-well	-0.95	<10 ⁻³	-0.79	0.001	-1.01	<10 ⁻³	-0.20	0.026	-0.70	0.001
Hexagonal six-well	-1.40	0.001	-1.31	0.001	-1.17	0.001	-0.77	0.001	-0.82	0.001
Octagonal eight-well	-1.34	0.001	-1.25	0.001	-1.34	0.001	-1.08	0.001	-1.09	0.001
Pentagonal five-well	-1.57	<10 ⁻³	-1.51	<10 ⁻³	-1.41	0.001	-0.95	0.001	-1.34	0.001
Square four-well	-1.34	<10 ⁻³	-1.36	<10 ⁻³	-1.34	<10 ⁻³	-1.32	<10 ⁻³	-1.36	<10 ⁻³
Triple-well Duffing	+0.00	–	+0.00	–	+0.00	–	+0.00	–	+0.00	–
SNIC multi-attractor	-1.09	<10 ⁻³	-1.09	<10 ⁻³	-1.09	<10 ⁻³	-1.09	<10 ⁻³	-1.09	<10 ⁻³
Transition-routes four-well	-1.39	<10 ⁻³	-1.39	<10 ⁻³	-1.39	<10 ⁻³	-1.39	<10 ⁻³	-1.39	<10 ⁻³
Depth-gradient four-well	-1.30	<10 ⁻³	-1.34	<10 ⁻³	-1.36	<10 ⁻³	-1.24	0.001	-1.36	<10 ⁻³
Diamond four-well	-1.33	<10 ⁻³	-1.33	<10 ⁻³	-1.36	<10 ⁻³	-1.36	<10 ⁻³	-1.36	<10 ⁻³
L-shaped five-well	-1.01	0.001	-0.96	0.001	-0.99	0.001	-0.92	<10 ⁻³	-0.92	<10 ⁻³
<i>Significant/eligible</i>	<i>K=14 / N=14</i>		<i>K=14 / N=14</i>		<i>K=14 / N=14</i>		<i>K=14 / N=14</i>		<i>K=14 / N=14</i>	

the same fixed dataset.

The exact-dense control uses tanh in every hidden encoder and decoder layer and a linear latent output. It contains no ReLU, GELU, soft-shrinkage, or dropout module; its sparsity coefficient, weight decay, transition penalty, and soft-block penalty are zero; and it uses Adam rather than AdamW. All ten fail-closed audits pass. At an absolute activation threshold of 10^{-4} , its empirical latent density is 0.9996, compared with 0.4884 for the signed LISTA KAE. Thus this comparison does not obtain a nominally dense row through an activation that itself creates zeros. It remains a complete-recipe comparison: sign splitting mechanically permits at most one active coefficient per pair, LISTA and MLP encoders and decoder normalization differ, and the signed/dense models have 459,264/443,136 parameters. Mean training times were approximately 85.5/34.9 seconds under the common update budget.

At physical times 2.5 and 5, sparse mean NRMSE is 0.8987 and 0.9669, versus 0.9595 and 1.2889 for exact dense. Paired mean relative reductions are 6.30% (95% model-seed bootstrap interval [5.09%, 7.32%]) and 24.87% ([22.98%, 26.51%]); sparse wins all ten paired seeds. Exact two-sided sign-flip tests give $p = 0.001953$ at each horizon and Holm adjustment over the two declared long horizons gives $p = 0.003906$. Sparse also has lower mean NRMSE than persistence, DMD, and truncated DMD at both horizons. Since the dataset is fixed, this inference quantifies model-initialization variability conditional on one generated dataset, not variability over new Lorenz-96 datasets.

Four-well vector Allen-Cahn representation packet. The synthetic physics-motivated multibasin study evolves two fields on a periodic 16×16 grid under diffusion coefficient 0.005 and a four-well reaction potential. This gives $d_x = 2(16)^2 = 512$; both KAEs use the required four-times overcomplete $d_z = 2,048$ lift. RK4 integration uses step 0.005, every twentieth state is stored, and the 512/128/256 train/validation/test trajectories extend to 200 stored transitions, or physical time 20. Models train without basin labels or counts on fields through physical time 12; the longer fields measure whether forecasts and supports remain meaningful well beyond a short observation interval. The evaluation-only fate label is the modal final well at time 20.

The comparison includes a direct tanh circular-convolutional forecaster, an exact-dense convolutional KAE, and a signed sparse convolutional KAE. The dense KAE has tanh hidden activations, a linear latent output, a full $2,048 \times 2,048$ transition, Adam, and zero coefficients for ordinary sparsity, temporal group sparsity, weight decay, transition stability, and soft-block regularization. No zero-inducing activation or dropout is present; all ten dense audits pass. Its holdout active density at threshold 10^{-4} is 0.998, versus 0.499 for the selected sign-split temporal-support model.

The first sealed test showed finite-time fate information but fragmented within-fate families: sparse transfer coverage was 0.988, ARI was 0.704, and normalized $H(F | B)/H(F) = 0.396$ exceeded the frozen 0.35 ceiling. Once opened, that test became development evidence. A label-free temporal-group objective had already been implemented and frozen before test access as the sole architectural fallback; the failed result triggered its validation screen rather than motivating a new post-test objective. For a sampled 20-state training window, it penalizes the sum over latent coordinates of each coordinate’s root-mean-square activation across time. The screened weights were $10^{-3}, 3 \times 10^{-3}, 10^{-2}, 3 \times 10^{-2}$; the weakest weight was the only row selected because it passed all six frozen checks on the original validation data. More broadly, validation fate metrics had selected physical-time-20 scoring and family Jaccard 0.40, and three screen checks—ARI and the two normalized conditional entropies—selected the temporal weight. The recipe is therefore label-assisted. No fate label, basin count, or trajectory assignment enters the objective, encoding, representative fitting, or family assignment conditional on these selected hyperparameters.

The selected weight was then evaluated as one fixed roster of ten fresh initialization seeds using the same label-assisted validation fate gates; no seed was omitted, and the entire ten-seed packet had to pass. Mean validation transfer coverage was 0.961, normalized $H(B | F)/H(B) = 0.110$, normalized $H(F | B)/H(F) = 0.299$, ARI was 0.756 (95% model-seed bootstrap interval [0.702, 0.798]), NMI was 0.784, and purity was 0.936. Sparse beat exact dense in all ten paired ARI comparisons ($p = 0.001953$, exact two-sided sign flip), satisfying every frozen promotion check. Only then was a new dataset generated with a different simulation seed. Support representatives were formed without labels from absolute-threshold masks on even-indexed trajectories of the original validation set and transferred at fixed Jaccard 0.40; they were never refit on the new data. Thus map formation and new-simulation-seed holdout assignment remain label-free after validation-selected recipe freezing; the new 256 test initial conditions and their labels are used only for scoring.

On this new-initial-condition dataset, sign-split temporal-support transfer coverage is 0.982. Its all-256 ARI treats unassigned family -1 as a category rather than dropping uncovered trajectories; this all-trajectory statistic is 0.767 (95% interval [0.742, 0.787]). NMI is 0.730, purity is 0.923, and normalized $H(B | F)/H(B) = 0.151$. Exact dense transfers one threshold-support family and has ARI and NMI zero; this shows that the support extractor finds no dense partition, not that dense latent values contain no fate information. Sparse wins all ten paired support-ARI comparisons ($p = 0.001953$). Across all 256 trajectory final states, the normalized $H(F | B)/H(F) = 0.359$ ([0.339, 0.382]) misses its 0.350 ceiling by 0.009 and the map has a mean 10.8 families. Both the support family and modal-well fate label are computed from the $T = 20$ final field: this is not x_0 fate prediction or alignment throughout each trajectory. This prespecified primary gate therefore fails and was not moved after test access.

After the earlier opened test exposed mixed-domain fragmentation but before opening the new holdout, we prospectively froze a single-well-dominated final-field slice: a trajectory is retained when at least 90% of its final grid occupies its modal Allen–Cahn well. The criterion uses the new-simulation-seed holdout fate label only to stratify evaluation. It retains 130/256 trajectories and all four modal-well fates. On this slice, every one of the ten sparse model seeds transfers exactly four families with ARI, NMI, and purity 1.000 and both normalized conditional entropies zero. This is exactly one transferred support family per finite-time fate on the declared slice, not proof of geometric basin-interior membership. Residual family fragmentation is associated with spatially mixed, interface-rich final fields.

Forecasts are evaluated at physical times 0.1, 0.5, 1, 2, 4, 8, 12, 16, 20, so the display does not rely on a nominally multi-step but physically negligible horizon. Temporal sparse has lower error than dense through time 4, but the ordering reverses at time 8 and remains reversed. At time 20, direct, dense, and sign-support final field MSE are 0.0293, 0.1184, and 0.2991; their final RMSE ratios to persistence are 0.218, 0.442, and 0.649. All three learned rows beat persistence, but the direct control is best. This separates the high-dimensional sparse forecasting claim, which comes from Lorenz–96 and the separate packet below, from this physics-based multibasin representation claim. The temporal-support model’s pre-split group activity is nearly complete, so it is not evidence of intrinsic coefficient sparsity before its sign split.

Separate forecast-optimized global- K packet. This experiment shares the $d_x = 512$, $d_z = 2,048$, grid, stored interval, and four-well system, but it is a different trained model and dataset split. It uses 512 training trajectories, full 200-step sequences, batch size 8, 2,000 autoencoder-pretraining updates, and 3,500 forecast-training updates. Adam has learning rate 3×10^{-4} for the network and 10^{-6} for K , with zero weight decay. The selected soft-thresholded recipe uses shrinkage 0.15, ordinary elementwise sparsity weight 0.01, and zero temporal-group penalty. The dense arm uses

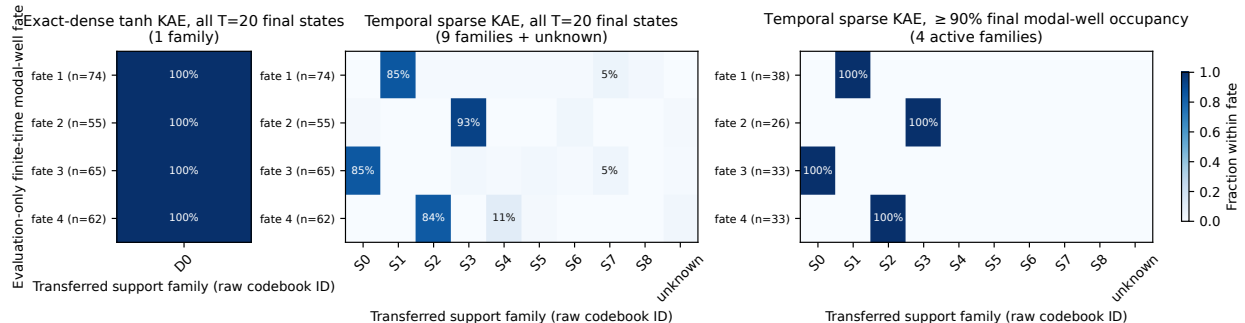


Figure 13: Row-normalized final-state fate–support contingency for the fixed display seed on the new-simulation-seed Allen–Cahn holdout. Exact dense assigns every fate to one family (left). Temporal sparse has one dominant family per fate on all trajectories but additional families in mixed-domain fields (center). On the single-well-dominated final-field slice prospectively frozen for this new holdout after earlier mixed-domain fragmentation, it transfers exactly four families in one-to-one correspondence with the four finite-time modal-well fates (right). Aggregate claims use all ten model seeds, not this display seed alone. Columns preserve the raw training-codebook family IDs in numeric order, independent of evaluation fate; row labels give panel-specific trajectory counts; the center and right panels share one family axis, so blank right-panel columns are families absent from the declared slice.

tanh hidden activations, a linear latent output, a full trainable K , and zero sparsity, shrinkage, dropout, weight decay, transition, or block penalties; all ten fail-closed audits pass. An exact checkpoint audit finds the same trainable tensor shapes/count and shared convolutional, decoder, and K backbone, with 12,698,690 trainable parameter entries in both arms. The nominal $2,048^2$ LISTA recurrence matrix stays exactly zero and is inert with one LISTA loop, leaving 8,504,386 effective forward-path parameters per arm. Thus the treatment is jointly soft-thresholding at 0.15 and an L_1 weight of 0.01 under matched parameter count and backbone, not equal function-class capacity or an isolated component test.

Selection used a three-seed, four-recipe development screen. The selected recipe was then trained for seeds 64–73 and scored on 256 fresh initial conditions from simulation seed 20,260,724, using no basin label or count. Checkpointing used the even development partition and joint H160/H200 endpoints; held-out scoring uses all fresh validation trajectories and no checkpoint reselection. Forecasting is direct repeated- K rollout without re-encoding. Mean near-zero fraction at threshold 10^{-3} is 0.434 for sparse and 0.0025 for dense. All 20 runs are finite; mean active A100L utilization is 95.55% (range 90.56–96.78%). On the opened development screen, the one-pass soft-thresholded sparse arm had lower through-horizon mean and terminal MSE than its same-recipe dense arm in all 16 comparisons across the four forecast-weighting/latent-consistency recipes. This is descriptive screen robustness, not confirmation or a threshold-versus- L_1 ablation.

At H160, sparse/dense through-horizon mean field MSE is 0.04220/0.04503; at H200 it is 0.04472/0.04731. These are reductions of 6.30% and 5.48%, each with 8/10 paired wins. Terminal reductions are 2.38% and 3.22%, each with 7/10 wins and an interval crossing zero. The corresponding through-horizon MSE ratios to persistence are 0.0958/0.1023 at H160 and 0.0938/0.0993 at H200 for sparse/dense; terminal ratios are 0.0824/0.0844 and 0.0937/0.0968. Thus the physical times 16–20 are nontrivial relative to persistence even though the terminal sparse–dense effect remains uncertain. The rule frozen before fresh confirmation-data generation required all four mean and terminal cells to pass; it failed. In addition to both terminal intervals crossing zero, H200 terminal

Table 13: Separate forecast-optimized Allen–Cahn comparison. “Mean” averages field MSE over every forecast state through the horizon; it is not an unnormalized cumulative sum. Intervals resample paired model seeds. The one-sided max- t values are a multiplicity sensitivity, not a replacement for the failed internally frozen four-cell gate.

Horizon	Endpoint	Dense	Sparse	Reduction	Wins	Marginal 95% CI; max- t p
H160	Mean through H	0.04503	0.04220	6.3%	8/10	[2.6, 9.5]%; 0.014
H160	Terminal	0.05189	0.05065	2.4%	7/10	[-1.6, 6.0]%; 0.197
H200	Mean through H	0.04731	0.04472	5.5%	8/10	[1.7, 8.8]%; 0.019
H200	Terminal	0.06086	0.05890	3.2%	7/10	[-0.8, 6.7]%; 0.112

Matched trainable tensor shapes and effective forward-path parameter count; sparse jointly applies soft-thresholding 0.15 and L_1 weight 0.01.
Internally frozen confirmation gate: failed; sealed holdout not generated or opened.

misses the minimum 5% reduction and 8/10-win criteria. The conditional seed-20,260,725 holdout was never generated or opened.

Same-checkpoint new-initial-condition robustness check. After the preceding four-cell decision failed, we prospectively froze the already positive H200 through-horizon endpoint. This endpoint choice was outcome-aware. We crossed the same ten sparse and ten dense checkpoints with three newly generated datasets (simulation seeds 1,775,404,171, 74,732,421, and 293,789,188), each containing 256 trajectories. There was no retraining, checkpoint reselection, periodic re-encoding, label use, or basin-count use. The rollout encodes once and applies the unchanged global K for 200 steps, reaching physical time 20. Datasets are averaged within each paired model seed before inference, so the inferential unit remains the ten paired checkpoint seeds rather than the 30 crossed seed–dataset cells.

Sparse/dense through-horizon field MSE is 0.04616/0.04903, a 5.86% reduction with 8/10 paired-seed wins. A 100,000-resample paired-seed bootstrap interval is [2.40%, 9.21%] and the exact one-sided paired sign-flip value is $9/1,024 = 0.00879$. Dataset-specific reductions are 4.68%, 5.88%, and 7.03%, all in the same direction; these three values are descriptive rather than independent hypothesis tests. H200 terminal error is descriptive only: 0.06476/0.06683, a 3.09% reduction with 7/10 wins. A separate authenticated recomputation exactly reproduced all primary values and authenticated the complete 60-cell roster, all 20 checkpoints, and all three datasets. The result establishes same-checkpoint robustness to new initial conditions in this fixed PDE condition. It does not retroactively repair the original four-cell gate or establish terminal, parameter, resolution, physics, or system generalization. Descriptively, across the full physical-time trace, the sparse arm-mean instantaneous MSE is lower at all 200 times, but the running-mean reduction declines from about 14% early to 5.86% at $T = 20$. Over the late half, $T = 10.1$ –20, sparse/dense instantaneous means are 0.05216/0.05380, only a 3.06% reduction; no simultaneous curve-wide inference is claimed. The result remains a trajectory-average rather than terminal or asymptotic advantage. H200 equals the trained temporal horizon and is not extrapolation beyond it.

Outcome-aware physical concordance on the same forecasts. On the identical 20 checkpoints, three datasets, 256 trajectories per dataset, and 200-step direct rollouts, sparse is directionally better at the H200 cumulative endpoint on all seven frozen secondary metrics. Pixel nearest-well disagreement, well-area total-variation error, interface-edge disagreement, full-free-energy error, and gradient-energy error fall by 4.61%, 6.07%, 2.25%, 6.20%, and 8.58%, respectively, and survive Holm correction across all seven metrics. Modal-well accuracy rises by 0.236 percentage points and potential-energy error falls by 3.77%, but their Holm-adjusted values are 0.303 and 0.119 and their

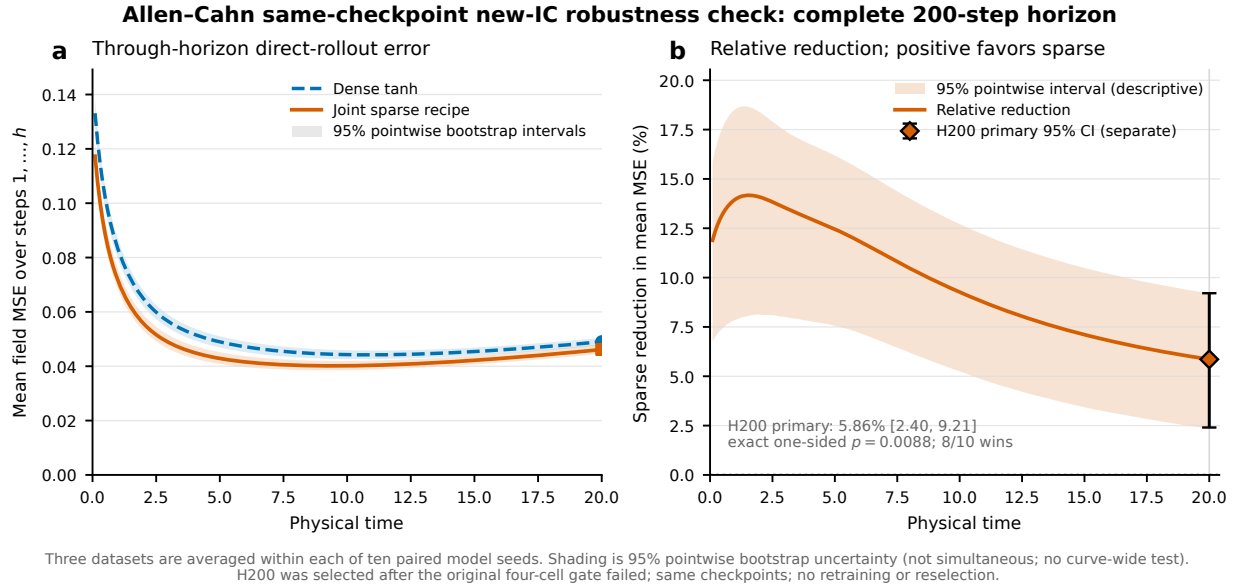


Figure 14: Complete 200-step Allen–Cahn same-checkpoint new-initial-condition robustness trace. H200 was selected after the original four-cell gate failed; the checkpoints were neither retrained nor reselected. The left panel reports mean field MSE over steps $1, \dots, h$ against physical time. The right panel reports the ratio-of-arm-means reduction. Shading is a descriptive pointwise 95% paired-seed bootstrap band and is neither simultaneous nor a curve-wide test; the black H200 interval is the separate 100,000-resample primary interval. Datasets are averaged within each of ten paired model seeds.

Table 14: Outcome-aware, same-checkpoint secondary Allen–Cahn analysis showing all seven frozen metrics at the H200 cumulative endpoint. Effects are relative reductions for the six lower-is-better errors and percentage-point differences for modal accuracy. Confidence intervals are paired-seed 95% intervals for the oriented absolute effect; q is Holm-adjusted across all seven metrics. Bold q values are at most 0.05.

Metric	Dense	Sparse	Oriented effect	Wins	Holm q / abs. 95% CI
Pixel phase error	0.09424	0.08990	4.61%	9/10	0.0146 / [0.00229, 0.00610]
Modal-well accuracy	0.90742	0.90977	+0.236 pp	5/10	0.3027 / [-0.00551, 0.01061]
Well-area TV error	0.05402	0.05074	6.07%	8/10	0.0352 / [0.00112, 0.00535]
Interface-edge error	0.09966	0.09742	2.25%	9/10	0.0146 / [0.00118, 0.00311]
Free-energy error	0.02843	0.02667	6.20%	10/10	0.0068 / [0.00112, 0.00243]
Potential-energy error	0.03597	0.03462	3.77%	8/10	0.1191 / [-0.00006, 0.00311]
Gradient-energy error	0.02897	0.02649	8.58%	10/10	0.0068 / [0.00164, 0.00332]

intervals include zero. Thus the result is broad phase-assignment, interface, and thermodynamic concordance, not seven statistically significant endpoints.

Every metric was frozen and computed for every trajectory and time. Pixel and modal-well scores use the four known wells only for evaluation; well-area error compares the four phase fractions; interface error compares periodic nearest- well boundary indicators; and the energy terms use the discrete Lyapunov functional whose derivative matches the pinned reaction–diffusion generator. Datasets are first averaged within model seed, leaving ten paired checkpoint seeds as the inferential units. Exact one-sided paired sign flips are Holm- adjusted as one seven-metric family, and intervals resample paired seeds. No periodic re-encoding, fate label, basin count, finite-prefix filtering, or trajectory exclusion alters a forecast.

This translation addresses whether lower field MSE merely reflects benign amplitude changes: the favorable interface and separated potential/gradient energy components argue against that narrow explanation in this condition. It does not identify a sparse component as causal, repair the original gate, establish terminal superiority, or generalize over retraining, PDE parameters, grid resolution, or another physical system. The appropriate next experiment is therefore a prospectively frozen recipe across new physical conditions, not additional tuning on the already opened condition.

Historical hard-system negative. A separate 15-seed stress test used $d_z = 1,024$, sequence length 8, batch size 256, and 100,000 updates on 15-dimensional fixed eight-basin competitive Lotka–Volterra, Hopfield $N = P = 16$, and identical-frequency ring Kuramoto $N = 16$. Under each row’s evaluation-selected best periodic-reencoding cadence, the tanh/no-output-gate Dense MLP beat every sparse/LISTA recipe on all three systems at H100, H500, and H1000. At H1000 its system-median MSEs were 0.2999/2.8034/7.7872, versus 0.3210/8.3277/15.5167 for the closest LISTA-SB row. All arms used AdamW weight decay 10^{-4} ; activation, iterative-encoder, block- K , capacity, and learning-rate differences preclude a sparsity-only attribution. This is a mandatory negative stress test and rules out a universal sparse forecasting advantage.

Requested half-global/half-local diagnostic. At one separate seed (61), a sequence-80 sign-split model received 2,000 autoencoder and 1,750 shared- K updates, followed by 1,750 updates to either full route-local $2,048 \times 2,048$ matrices or a matched continued-global branch. Every support and cosine-cluster local recipe selected its zero-update initialization. At H200, local/global through-horizon mean MSE is 0.07871/0.06463 (local 21.8% worse), terminal MSE is 0.12048/0.10168 (18.5%

Allen-Cahn direct-rollout physics
all seven frozen metrics at the H200 cumulative endpoint

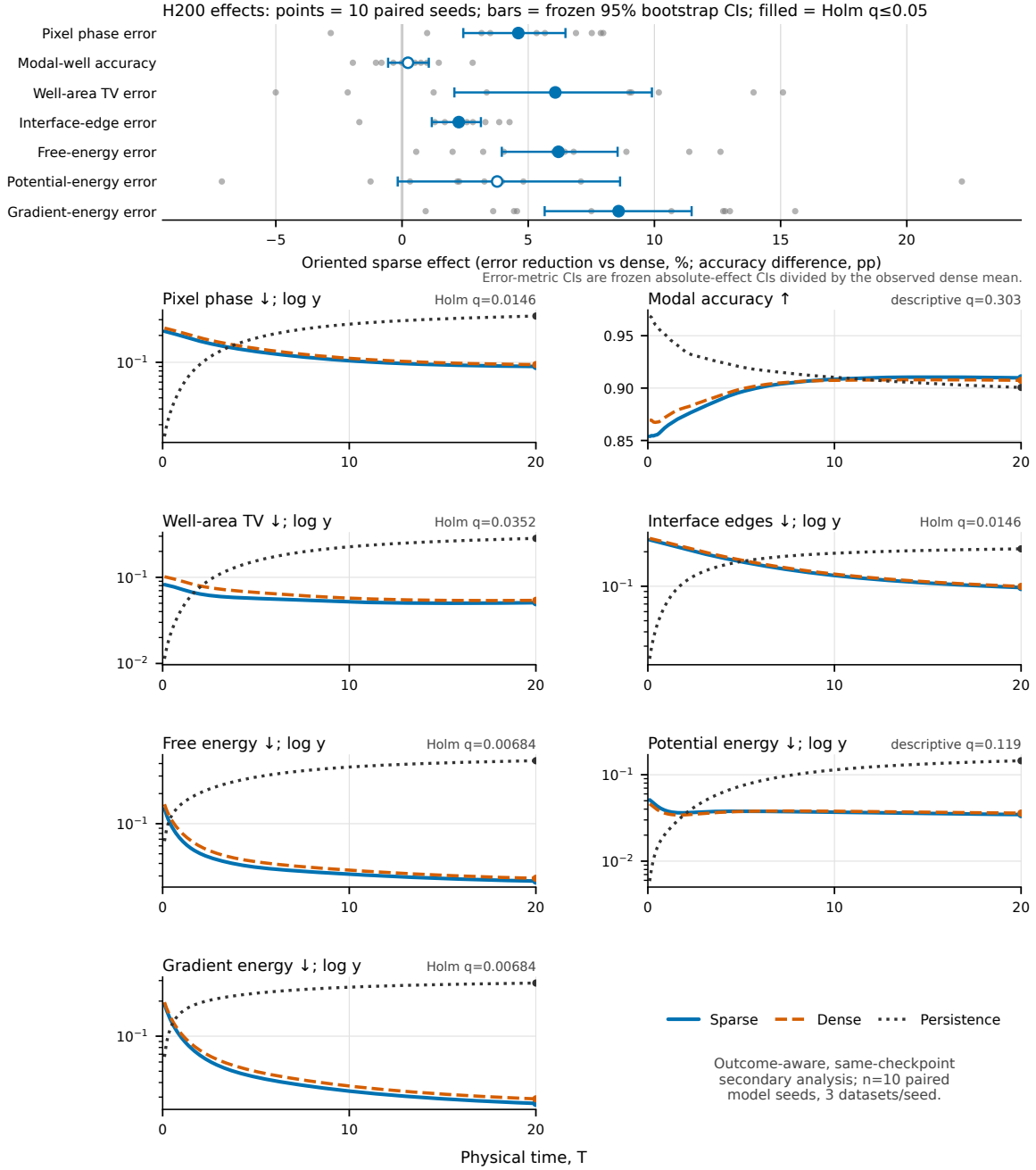


Figure 15: All seven frozen Allen–Cahn physical metrics on the same-checkpoint new-initial-condition forecasts. The top panel shows the H200 cumulative oriented effect, all paired-seed effects, and paired-seed intervals; filled marks survive seven-way Holm correction. Lower panels show every arm-mean curve through physical time 20, including persistence. Curves are descriptive; the statistical family consists of the seven H200 cumulative endpoints. The analysis is outcome-aware because H200 and these checkpoints were retained after the original four-cell forecasting gate failed.

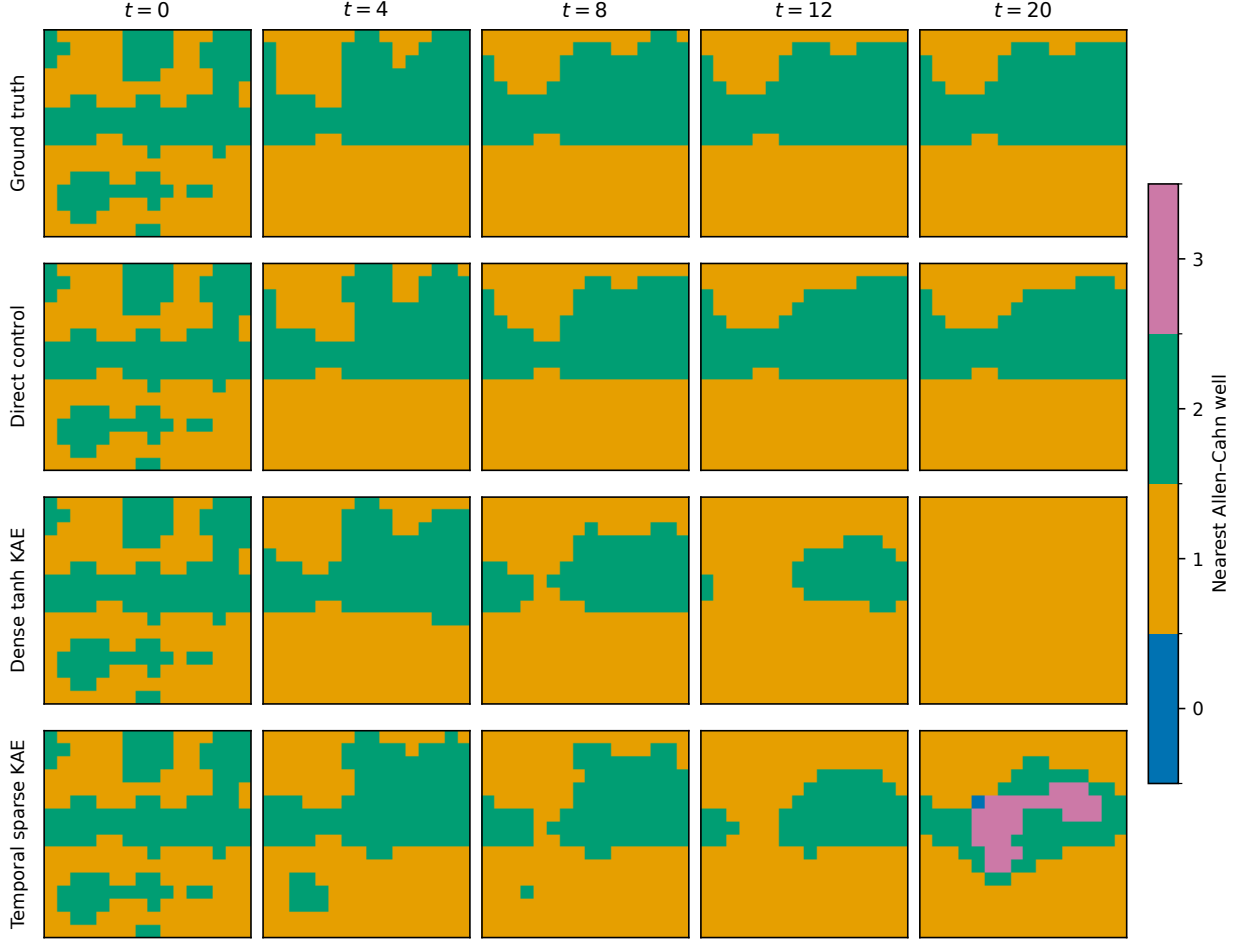


Figure 16: Predeclared qualitative long-horizon Allen–Cahn trajectory on the new-simulation-seed holdout, shown as nearest-well maps at physical times through 20. The direct control preserves the evolving interface most faithfully; the sign-split temporal-support KAE has the weakest long-horizon forecast. The trajectory and model seed were fixed before the new holdout was opened and are not used for quantitative selection.

worse), and route coverage is $0.840 < 0.90$. Full matrices rule out insufficient rank as the sole explanation, but this one-seed near-dense ancestor (pre-split activity 0.9982) differs from the positive global recipe; the result falsifies this routing recipe, not all local- K methods.

Inference. The Lorenz–96, Allen–Cahn temporal-support, forecast-optimized Allen–Cahn, same-checkpoint new-IC, and secondary physical-metric packets average trajectories within model seed and use ten paired model-initialization seeds as inferential units. Lorenz–96 and temporal-support intervals use 100,000 percentile bootstrap resamples; their paired tests enumerate all sign flips, with long-horizon p -values Holm-adjusted within each declared family. The forecast-optimized packet instead uses 50,000 paired-seed resamples of each ratio-of-arm-means reduction. Its four intervals are marginal and its frozen primary decision is an intersection-union gate; the exact one-sided 1,024-swap max- t calculation is a multiplicity sensitivity over the dependent H160/H200 mean/terminal cells (H160 is a prefix of H200), not a replacement decision. Lorenz–96 and the original Allen–Cahn intervals condition on one fixed evaluation dataset. The new-IC interval averages three fixed datasets

within each paired checkpoint seed. They quantify variation across the ten trained seed pairs, but not an independent repeat of the training/selection pipeline, a future retraining packet, PDE parameters, resolutions, or physical systems. The new-IC check crosses the three datasets with every checkpoint, averages them within paired model seed, uses 100,000 paired-seed resamples, and enumerates the same 1,024 paired sign flips. Neither benchmark uses fate labels, basin counts, or trajectory assignments for training, encoding, representative fitting, or family transfer. In the temporal-support packet, validation fate metrics select the T20 scope, Jaccard-0.40 rule, temporal weight, and advancement of the whole fixed ten-seed packet; labels on the new-simulation-seed holdout only score the frozen assignments and the declared single-well-dominated final-field slice. The physical-metric packet uses the same within-seed dataset averaging and 100,000 paired-seed resamples, one-sided exact sign flips, and Holm correction over all seven H200 cumulative endpoints. Its two nonsignificant metrics and all terminal and late-window summaries remain visible rather than being removed from the family.

J Support-projected one-step leakage of one global operator

This diagnostic tests a narrow mechanism hypothesis: a single learned global operator can act approximately within several support-selected coordinate sets on observed projected codes even when no basin label, basin count, or trajectory-to-basin assignment is used. It does not train local operators. In the implementation’s row-vector convention, a family mask defines a diagonal projector P_f and the learned prediction is $z_{t+1} = z_t K$. Cross-support output is therefore $z_t P_f K (I - P_f)$. This is the transpose-equivalent of $(I - P_f) K P_f z_t$ in the paper’s column-vector convention.

Existence construction. One matrix is not mathematically incompatible with several local laws. For row-form affine dynamics $x^+ = x A_b + c_b$, let $\psi_b(x) = [x, 1]$ and $K_b = \begin{bmatrix} A_b & 0 \\ c_b & 1 \end{bmatrix}$. In an overcomplete direct-sum lift, set $K = \text{diag}(K_1, \dots, K_B)$ and let a routed basin-interior encoder place $\psi_b(x)$ only in block b . If P_b selects that block, then $P_b K (I - P_b) = 0$ and $z P_b K^h = z K^h$, while different restrictions contain different laws; a block-aware decoder recovers the physical state. Unused blocks mean the architecture need not be told the realized basin count. This proves an algebraic interior-chart possibility with a block-aware decoder, not a continuous global encoder across basin boundaries, learned discovery, label-free routing, or realization by our checkpoints.

Protocol frozen before evaluation. The roster contains unstructured signed sparse encoders with full 256×256 learned matrices: three initialization seeds on each of the 15 two-dimensional controlled systems. Their sign-split ReLU output mechanically permits at most one active coefficient per positive/negative pair, so the result concerns the learned sign-split coordinate basis rather than generic sparse encoders. For every checkpoint, 128 newly generated trajectories are split 64/64 by whole trajectory. The first half alone forms support families using $|z_i| > 10^{-3}$, greedy Jaccard-0.50 merging, and a minimum of 128 fit transitions per retained family. The second half alone is scored. The primary slice contains transitions whose current and next support masks both map to the same retained family. This future-conditioned persistence criterion makes the closure test easier and is an important scope restriction. The authenticated evaluator also emitted an all-current regime before execution, without conditioning on the next support. Its reduction was performed post hoc as a no-next-state guard and is not a second frozen decision. Eligibility requires at least two retained families, current/persistent coverage of at least 0.80/0.70, and at least 64 scored transitions per reported family.

The activity leakage diagnostic is $\|z P_f K (I - P_f)\| / \|z P_f K\|$; its matrix-wide analogue is $\|P_f K (I - P_f)\|_F / \|P_f K\|_F$. To exclude the trivial case where $K \approx I$, change-normalized versions instead use $\|z P_f (K - I)\|$ and $\|P_f (K - I)\|_F$ in their denominators. We also report the inside-support residual of the post-hoc restricted predictor $P_f K P_f$, the encoded next-state energy outside the current support, and the residual of the unmodified global K relative to identity. The restricted predictor is an evaluation diagnostic, not a retrained model. Finally, pairwise symmetric normalized Frobenius distance between restricted operators tests whether the selected coordinate sets have more differentiated restrictions than a matched null.

The null uses 16 coordinate permutations frozen in advance. Each permutation moves positive/negative sign partners together and is applied jointly to codes, next codes, masks, and projectors while leaving K fixed. It therefore preserves norms, support cardinality, activity, and sign-split semantics while breaking alignment with the learned operator. Metrics reduce first within a run, then by the median across eligible seeds within each system, and finally by the median across eligible systems; transitions are not treated as independent replicates.

All 45 runs and all 15 systems are eligible. Mean current and persistent coverage are 0.963 and

Table 15: Held-out one-step support-closure diagnostics for one unchanged global K . The table reports both the post-hoc reduction of the pre-execution evaluator-emitted all-current guard and the internally frozen persistent-family primary. The matched null permutes sign pairs while holding K fixed. The frozen branch is labeled partial closure, but its failed differentiation guard permits only a lower-leakage claim.

Slice	Diagnostic	Observed	Matched null	Obs./null	Wins	Gate
All-current guard [†]	Raw- K activity leakage	0.0047	0.2065	0.022	15/15	Ref. pass
	Raw- K matrix Frobenius leakage	0.0538	0.2080	0.240	15/15	Diagnostic
	$(K - I)$ -normalized leakage	0.3382	0.7830	0.433	15/15	Ref. pass
	$(K - I)$ matrix Frobenius leakage	0.5541	0.7834	0.699	15/15	Diagnostic
	Post-hoc PKP inside residual	0.0100	0.1646	0.053	15/15	Ref. pass
	Restricted-operator distance	0.9741	0.9830	0.993	0/15	Ref. fail
Persistent primary	Raw- K activity leakage	0.0047	0.2065	0.022	15/15	Pass
	Raw- K matrix Frobenius leakage	0.0538	0.2080	0.240	15/15	Diagnostic
	$(K - I)$ -normalized leakage	0.3394	0.7831	0.435	15/15	Pass
	$(K - I)$ matrix Frobenius leakage	0.5539	0.7834	0.699	15/15	Diagnostic
	Post-hoc PKP inside residual	0.0100	0.1646	0.053	15/15	Pass
	Restricted-operator distance	0.9741	0.9830	0.993	–	Fail

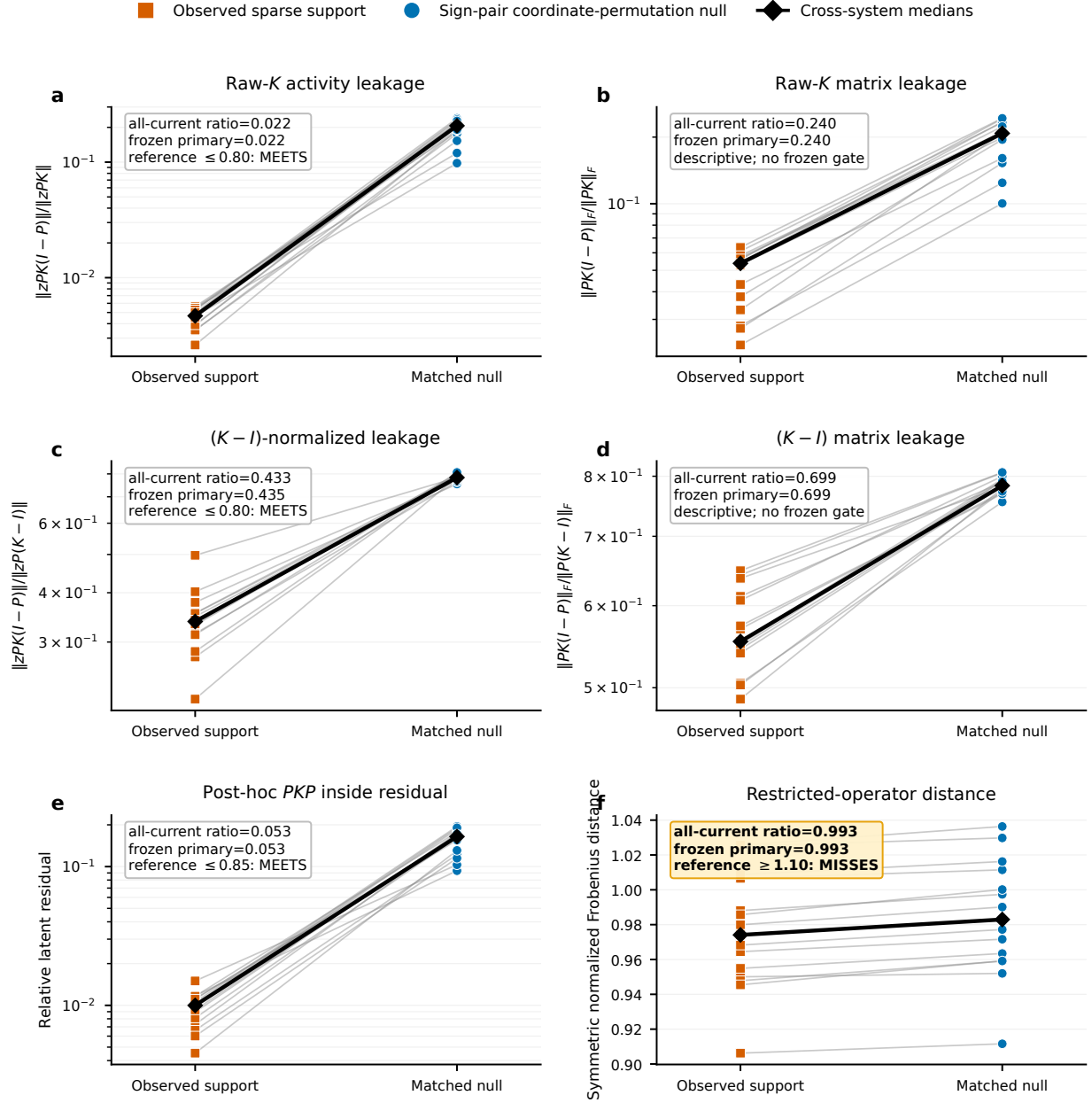
[†]Post-hoc reduction of a pre-execution evaluator-emitted no-next-state guard; not a second frozen decision.

The guard adds only 3,527 transitions (0.96 percentage points); medians are over system-level seed medians.

Frozen primary decision: partial closure; the differentiation guard fails; PKP is a post-hoc restriction.

Table 16: Triggered exact-dense specificity control. Dense tanh models use zero weight decay and no sparsity-producing mechanism. At every matched physical state, the dense projector selects the top-magnitude coordinates with exactly the corresponding sparse support cardinality. Values are observed/permutation-null ratios; the last column is sparse/dense. The source-locked reducer retains 41/45 dense runs and 14 systems, versus 45/45 runs and 15 systems for sparse; its pre-outcome two-eligible-seed rule was omitted from the card. Sparse-to-dense ratios of observed/null activity and residual are 0.039/0.108 on the 14 common systems, 0.040/0.109 under the literal-card all-15 rule, and 0.041/0.110 when all three seeds are required (12 systems).

Metric	Sparse / null	Dense / null	Sparse / dense
Raw- K activity leakage / null	0.022	0.567	0.039
Post-hoc restricted residual / null	0.053	0.486	0.108



Plotted: post-hoc all-current guard, 15 systems $\times$ 3 seeds; frozen persistent-primary ratios appear in annotations.
The guard adds 0.96 percentage points of transitions. PKP is post-hoc; failed differentiation precludes invariant-subspace or distinct-law claims.

Figure 17: System-level seed medians for the no-next-state all-current guard; annotations give its ratio and the internally frozen persistent-primary ratio. Lower is better in the first five panels. The two matrix-wide panels are descriptive diagnostics without retrofitted decision thresholds. Restricted-operator differentiation is null-like and fails its reference threshold.

0.953. Retained families average 5.84 per run, their representatives average 95.0 of 256 coordinates, and the projected source captures 0.997 of source-code RMS energy. At the system-median level, the all-current observed/null activity leakage is 0.0047/0.2065 (ratio 0.022); change-normalized leakage is 0.338/0.783 (ratio 0.433); the corresponding matrix-wide values are 0.0538/0.2080 (ratio 0.240) and 0.554/0.783 (ratio 0.699); and restricted inside-support residual is 0.0100/0.1646 (ratio 0.053). All 15 systems favor the observed masks on each of these five diagnostics. Their ratios are numerically close to the frozen persistent-primary values 0.022/0.240/0.435/0.699/0.053. Removing the next-family condition therefore leaves the aggregate essentially unchanged on this high-persistence sample. Because the guard adds only 3,527 transitions, or 0.96 percentage points, it does not determine behavior on boundary-switching transitions. The unmodified global- K residual is 0.583 times identity on the all-current guard.

The frozen strong decision nevertheless fails: restricted-operator distance is 0.974 for observed supports and 0.983 under the null, a ratio of 0.993 rather than the required 1.10. Thus support-projected one-step leakage is lower than under matched permutations in this learned sign-split basis, but the result does not establish distinct local laws, exact invariant subspaces, a direct-sum decomposition, decoded forecasting, or Allen–Cahn behavior. Because the result is basis-dependent, the frozen protocol triggered a cardinality-matched exact-dense tanh specificity control.

That control is complete. Across the 45-run dense roster, 41 runs and 14 systems meet the source-locked reducer’s eligibility rule. Sparse/dense observed-to-null ratios are 0.039 for raw- K activity leakage and 0.108 for the post-hoc $P_f K P_f$ residual, far below both frozen 0.90 gates; all 14 common eligible systems favor sparse on both metrics. The reducer’s requirement of at least two eligible seeds per system existed before outcomes but was omitted from the prediction card, so it is not described as card-preregistered; the frozen headline also aggregates 15 sparse versus 14 dense systems. Common-14 sensitivities are 0.039/0.108, literal-card minimum-one-seed all-15 sensitivities are 0.040/0.109, and an all-three-seed 12-system sensitivity is 0.041/0.110, so the conclusion is nearly unchanged under these population choices. Thus the low support-projected one-step leakage is specific to the complete sparse recipe relative to cardinality-matched dense tanh top- k coordinate masks. Dense masks have no natural dense-support semantics, and encoder/decoder parameterization and optimization differ; this is a complete-recipe specificity control, not a sparsity-only causal ablation, basis-invariant result, distinct-law test, or forecasting comparison.

Prospective physical-law test. A separate frozen test trained ten new paired sparse/dense models on a controlled two-dimensional system with three basin-specific laws. Training and support-family discovery used neither basin labels nor the basin count; evaluation labels only matched discovered families to the benchmark laws. The authenticated result is negative. Only 24/30 sparse seed–basin rows passed the prespecified differentiability gate, leaving 6/10 complete sparse seeds rather than the required 8/10. On those rows, the Jacobian of the actual restricted predictor selected the correct law in 9/24 nearest-law comparisons, passed its ratio gate in 4/24, its affine gate in 2/24, and its finite-radius gate in 0/24. The isolated K -induced update component was descriptively nearest to the correct law in 24/24 rows and met projected-code closure in 23/24, but its aggregate matched-null guard failed (null won 6/10 seeds; $p = 0.377$). Dense specificity was invalid because only six pairs were jointly differentiable. The isolated-component diagnostic is therefore stronger than the actual-predictor diagnostic, but it also fails its aggregate null and the packet fails validity. This contrast motivates testing reconstruction geometry without identifying it as the cause and supports no multiple-law, invariant-subspace, or sparse-superiority claim.

A final frozen recursive residual-routing diagnostic used one unchanged K but did not yield a scientific decision: five of ten seed shards completed, a sixth encountered a nonfinite payload

value (`inf`) during strict serialization, four were not run, and the required telemetry assessment and summary were absent. Under its exact-ten fail-closed protocol, every forecasting claim tier is invalid at the frozen validity tier and directionally unadjudicated; no partial shard was inspected and the execution is not evidence for or against the hypothesis.

K Additional background

K.1 Koopman operators and finite-dimensional approximations

Consider the stored-step map used in [section 2](#),

$$x_{k+1} = F_{\Delta t}(x_k), \quad x_k \in \mathcal{X} \subseteq \mathbb{R}^{d_x}. \quad (27)$$

Koopman operator theory studies the evolution of observables rather than the evolution of the state directly. An observable is a measurement function $g : \mathcal{X} \rightarrow \mathbb{R}$, and the Koopman operator $\mathcal{K}$ acts by composition:

$$(\mathcal{K}g)(x) = g(F_{\Delta t}(x)). \quad (28)$$

The map $F_{\Delta t}$ may be nonlinear, but $\mathcal{K}$ is linear on the function space of observables. If a finite vector of observables

$$\Phi(x) = [g_1(x), \dots, g_{d_z}(x)]^\top \quad (29)$$

spans a Koopman-invariant subspace, then there is a matrix $K \in \mathbb{R}^{d_z \times d_z}$ such that

$$\Phi(x_{k+1}) = \Phi(F_{\Delta t}(x_k)) = \mathcal{K}\Phi(x_k) = K\Phi(x_k). \quad (30)$$

In that case the nonlinear dynamics are exactly linear in the observable coordinates $z_k = \Phi(x_k)$. Learning Koopman models can therefore be viewed as searching for observables whose span is invariant enough to support useful finite-dimensional linear prediction.

In practice, the infinite-dimensional operator is approximated from data. Given snapshot pairs $\{(x_k, x_{k+1})\}$ and a chosen feature map Φ , dynamic mode decomposition and extended dynamic mode decomposition (eDMD) estimate a finite transition by least squares ([Schmid, 2010](#); [Rowley et al., 2009](#); [Tu et al., 2014](#); [Williams et al., 2015, 2016](#)):

$$K = \arg \min_{\tilde{K} \in \mathbb{R}^{d_z \times d_z}} \sum_k \left\| \Phi(x_{k+1}) - \tilde{K}\Phi(x_k) \right\|_2^2. \quad (31)$$

DMD is the special case $\Phi(x) = x$, so it fits a linear state-space surrogate without a learned nonlinear lift. eDMD extends this by choosing a richer observable dictionary. Koopman autoencoders replace the fixed observable dictionary with a learned encoder and decoder, which is the setting used in this paper.

K.2 Why multibasin systems may benefit from local structure

Near appropriate attracting sets, classical linearization and Koopman eigenfunction constructions can produce useful local or basin-restricted coordinates. Global results are more restrictive: multi-attractor systems should not in general be assumed to admit one exact finite-dimensional linearization with a continuous, state-reconstructing encoder–decoder pair ([Lan and Mezić, 2013](#); [Brunton et al., 2016](#); [Pan and Duraisamy, 2024](#); [Liu et al., 2025](#)). These results motivate local structure, but they do not prove that every learned global KAE must fail or that sparse supports recover the theoretical local coordinates.

For the present paper, a learned finite-dimensional KAE is treated as an approximation. Sparse supports provide one possible discrete and inspectable key: a support may index a basin part or local predictive chart while coefficient values carry within-chart information. Whether that interpretation holds is an empirical question tested by the transductive alignment and staged forecasting experiments.

K.3 Koopman autoencoders and periodic re-encoding

A standard Koopman autoencoder learns an encoder $E_\theta : \mathbb{R}^{d_x} \rightarrow \mathbb{R}^{d_z}$, decoder $D_\phi : \mathbb{R}^{d_z} \rightarrow \mathbb{R}^{d_x}$, and latent transition K so that

$$z_k = E_\theta(x_k), \quad z_{k+1} \approx K z_k, \quad x_k \approx D_\phi(z_k). \quad (32)$$

A schematic one-step objective has the form

$$\min_{E_\theta, D_\phi, K} \sum_k \|x_k - D_\phi(E_\theta(x_k))\|_2^2 + \lambda_{\text{lin}} \|E_\theta(x_{k+1}) - K E_\theta(x_k)\|_2^2. \quad (33)$$

This objective is useful as a reference point, but it is not sufficient for the present study. It does not directly train decoded multi-step rollouts, does not impose a sparse support object, and does not test whether the support itself selects a useful local predictive rule. The training objective in [section 3.3](#) therefore keeps the autoencoding and latent-linearity terms but adds decoded rollout prediction over windows, sparsity on the rolled latent states, and optional transition-structure penalties.

Periodic re-encoding interrupts a latent rollout after a fixed number of stored observation steps. For period m ,

$$\tilde{z}_{k+m} = K^m z_k, \quad \tilde{x}_{k+m} = D_\phi(\tilde{z}_{k+m}), \quad z_{k+m} = E_\theta(\tilde{x}_{k+m}). \quad (34)$$

This is the same mechanism defined in [equation \(12\)](#). It uses only the model’s own predicted state and is therefore not a teacher-forced correction. The point is to refresh the latent embedding and, in sparse models, to refresh the support as the predicted state moves through regions where a different local linear description may be more appropriate ([Fathi et al., 2024](#)).

K.4 Sparse coding and LISTA

Classical sparse coding seeks a code z that reconstructs a signal from a small set of active dictionary atoms:

$$z^\star(x) = \arg \min_{z \in \mathbb{R}^{d_z}} \frac{1}{2} \|x - Dz\|_2^2 + \lambda_{\text{sc}} \|z\|_1, \quad \lambda_{\text{sc}} > 0, \quad (35)$$

where D is a dictionary and λ_{sc} controls sparsity ([Olshausen and Field, 1996](#); [Chen et al., 1998](#); [Donoho and Elad, 2003](#)). For a fixed dictionary, and under the usual uniqueness and non-boundary conditions for the Lasso solution, the active set and signs are locally constant over regions of the input space, while the coefficient values vary continuously within a fixed active set. The decoder reconstruction then lies in the span of the selected dictionary columns, giving a union-of-subspaces geometry. Each exact support can index a local chart, and the greedy Jaccard procedure in [section 3.4](#) groups nearby Boolean support vectors into support families when threshold noise or boundary effects fragment the exact masks.

LISTA replaces an iterative sparse-coding solver with a learned finite-depth shrinkage network ([Gregor and LeCun, 2010](#)). In the notation of [equation \(6\)](#), the encoder forms a pre-code from the input and repeatedly applies learned affine refinement followed by soft thresholding. In this

paper, the encoder is trained jointly with the Koopman rollout objective rather than solving the sparse-coding problem at evaluation time. Thresholding turns the latent into continuous coefficients plus an explicit support. The experiments ask whether absolute-threshold support families align transductively with evaluation-only native or nearest-center-proxy labels and whether a separately fitted training-distribution family partition can route useful local affine predictors.

L Preliminary coauthor checklist

Status. This is a current working record for coauthor review of the revised manuscript and rebuttal, not a record of the binary PDF submitted at commit `fd68bc5`. Application of the final venue style determines the official question set, numbering, and response format; these answers must then be reconciled with that form. Nothing here asserts that a public artifact, anonymous URL, DOI, or release has been completed.

Claims and evidence. The manuscript makes three bounded empirical claims: sparse KAE rows have lower reported aggregate error than Dense MLP on the controlled and separate high-dimensional comparisons; sparse support families align with evaluation-only labels in the transductive controlled diagnostic and under frozen transfer on the independent high-dimensional PDE holdout; and a staged training-distribution support-family route has descriptively lower controlled-forecasting error than a same-budget global- K LISTA baseline under an optimistic asymmetric selector. The paper states no new theorem or proof, no exact global linearization, and no out-of-sample basin-discovery guarantee. Benchmark definitions and training/evaluation protocols are in [sections A, B, I and J](#); support construction and inference are in [sections D and F](#).

Limits on interpretation.

- Best-periodic cadence is chosen on the test trajectories, and the MSE reducer omits nonfinite steps, so finite prefixes of divergent rollouts can contribute.
- Alignment fits families on all states in each evaluation trajectory collection. Two gated systems use native labels and centers; the other 13 use deterministic endpoint-estimated centers at the known benchmark count and nearest-center proxy labels. Scoring retains states at or above each label’s 75th center-margin percentile, including ties, so it need not retain exactly one quarter; Triple-well Duffing retains all 16,512 states. This margin need not equal true basin-boundary distance. Labels and centers select only the scored subset, label sample counts are not equalized, and the reported family count includes only family IDs observed on that subset. Conditional entropy uses natural logarithms and is reported in nats. The result is transductive and two-dimensional; alignment uses Jaccard threshold 0.50, while the separate staged route uses 0.40.
- The staged route-fit packet has 512 configured rows but only 256 unique trajectories because one batch is duplicated exactly. Its checkpoint selector uses the first 32/100 reported starts, while the global baseline uses a distinct 16-start, H200 final-error selector. Thus the 225 nested system–seed comparisons are optimistic and descriptive, not a clean held-out or causal comparison.
- Standalone controls use three seeds and unmatched trajectories; the intervention covers one checkpoint.
- LISTA decoders are atom-normalized whereas MLP decoders are not. Structured-transition rows use a known controlled-benchmark basin count only to set the number of blocks. Historical Dysts evidence is quarantined; the corrected campaign matches all LISTA rows at one learned refinement.
- Lorenz–96 uses one fixed generated dataset and is chaotic rather than multibasin. Allen–Cahn is one synthetic PDE: its primary all-trajectory final-state uniqueness score misses the frozen gate by 0.009, its exact one-family-per-fate result is on an evaluation-only single-well-dominated finite-time slice rather than proven basin interiors, and both support and fate are measured at $T = 20$, not from x_0 or throughout a trajectory. Conditional on the recipe, its objective,

representative fitting, and holdout assignment use no labels, but validation fate metrics selected T20 scoring, Jaccard 0.40, the temporal weight, and whole-packet advancement. Temporal sparse is worse than dense at long forecast horizons. A separate direct-rollout sparse model has lower through-horizon mean error, but its terminal-inclusive gate fails. Neither experiment establishes empirical real-world generalization.

- The global- K closure diagnostic is support-projected, one-step, and restricted to two-dimensional controlled systems. Its frozen primary is future-conditioned. A pre-execution evaluator-emitted all-current guard gives similar leakage, but it adds only 0.96 percentage points because persistence is high and therefore does not resolve boundary-heavy switching. The operator-differentiation gate fails, so neither slice establishes invariant subspaces or distinct local laws.

Reproducibility and access status. The working repository contains explicit controlled-system vector fields, benchmark adapters, training and evaluation code, seeds, locked dependencies, tests, manuscript sources, and paper-facing aggregate artifacts. Controlled trajectories are regenerable from the explicit analytic or piecewise-analytic systems. The quarantined historical Dysts trajectories are not treated as regenerable paper evidence; the corrected campaign records a schema-versioned, numerically verified cache protocol. Exact model, metric, split, cadence, support, and standalone-control recipes are recorded in [sections A and D](#). Bitwise equality across GPU models and numerical-library versions is not claimed. No verified public anonymous archive or DOI is identified in the manuscript, so open code/data availability is currently answered *No*.

Compute accounting. The following are conservative SLURM allocation ceilings, not measured consumption or energy estimates:

- Controlled training comprises $15 \times 6 \times 15 = 1350$ runs, each requesting one cluster-assigned GPU, four CPU cores, 16 GB host memory, and at most 24 hours: at most 32,400 GPU-hours and 129,600 allocated CPU-core-hours.
- The staged routing packet comprises $15 \times 15 = 225$ runs, each requesting one GPU, four CPU cores, 24 GB, and at most three hours: at most 675 GPU-hours and 2,700 allocated CPU-core-hours.
- Controlled support diagnostics add approximately 290 CPU-core-hours; the one-checkpoint intervention allocation adds at most 3 GPU-hours.
- The final Lorenz-96 comparison uses one two-hour GPU allocation. The Allen-Cahn temporal screen, promotion, holdout evaluation, and new-data generation request at most 64.2 GPU-hours in total. Measured mean GPU utilization is 81% for Lorenz-96 and 84.7%–92.6% for the promoted dense and sparse Allen-Cahn training runs.
- The separate 20-run Allen-Cahn direct-rollout comparison used A100L GPUs with 95.55% mean active utilization. The 45-checkpoint raw global- K closure evaluator is CPU-only because its host-routed 256-dimensional algebra would not use an accelerator efficiently.

Except where a study records its assigned hardware above, the GPU model was scheduler-assigned and is unspecified. These ceilings exclude superseded historical packets and can substantially exceed actual use because jobs may finish early and packed runs share allocations.

Assets, attribution, and licenses. The controlled benchmark contains 15 author-retained analytic or piecewise-analytic systems. Claude Opus 4.5 assisted system ideation; human authors audited the retained definitions and metadata before their experimental use. The LLM is not part of model training, rollout, routing, metric computation, or statistical testing. LLM assistance also

contributed to manuscript and checklist editing; the authors remain responsible for the equations, protocols, and numerical claims. The corrected external campaign uses right-hand sides and metadata from `dysts` 0.96, with project-generated standardized trajectories, and the manuscript cites the associated Gilpin papers. Local package metadata is not sufficient to state a definitive redistribution license: it contains an MIT license classifier but a bundled Apache-2.0 license text. The authoritative `Dysts` license and any system-level attribution requirements therefore remain unresolved. The repository itself has a top-level MIT license, but no public artifact or separate license for generated benchmark definitions, figures, tables, caches, or checkpoints is claimed here.

Ethics, human subjects, and broader impact. The experiments use simulated dynamical-system states and public benchmark definitions, with no participants, crowdworkers, personal data, private records, or human-generated labels. Human-subject review, consent, compensation, and IRB approval are therefore not applicable. This is methodological forecasting research, not a validated safety-critical control system. Learned support families can be misread as physical or causal regimes, and failures near boundaries, in additional high-dimensional systems, or on empirical data could be consequential; downstream use requires independent validation and safeguards.

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